

August 20, 2026 01:52 AM GMT

IT Hardware | North America

# Monthly Data Tracker: C3Q App Store Weakness; Minor iPad Build Cuts; Incremental Cloud Capex Upside

App Store net revs now tracking -0.6% Y/Y MTD through mid-Aug, the first Y/Y decline in 4 yrs. C3Q iPhone builds are unchanged, but iPad build cuts suggest some demand elasticity. PC builds down a material 24% Y/Y in July. Cloud Capex Tracker now +38% Y/Y in CY27 (9 pts higher prior).

AlphaWise 

## Key Takeaways

- Aug App Store net revs are tracking -0.6% Y/Y MTD, the first Y/Y decline in 4 yrs, and +0.5% Y/Y QTD, implying 20bps downside to our Sept Q Services growth est.
- We see limited Services reacceleration as App Store faces lower take rates, link-outs, weaker gaming demand and FX headwinds; AppleCare pricing helps offset.
- C3Q iPhone builds unchanged at 54M, supporting ~58M Sept Q shipments; iPad builds were cut by 1M, implying expects for minor demand elasticity post price hikes.
- July NB build declines accelerate to -24% Y/Y, w/ C3Q builds now forecasted at -20% Y/Y (vs. -8% Y/Y in C2Q) - further evidence the PC mkt is rolling over.
- Our cloud capex tracker now points to +38% CY27 growth, with Consensus at \$1.39T and MSe at \$1.61T, showing there is room for further Consensus Capex revisions.

**August App Store net revenue is tracking -0.6% Y/Y MTD, the first time of Y/Y decline over the past 4 years despite a 200bps easier Y/Y compare, translating to +0.5% Y/Y growth QTD and running below our September quarter forecast.**

Based on the latest Sensor Tower disclosures, we estimate App Store net revenue is declining 0.6% Y/Y MTD through August 17th, representing an 170bps deceleration vs. June against a 200bps easier Y/Y compare ( [Exhibit 25](#) ). If the quarter ended on August 17, C3Q App Store revenue growth would have been just 0.5% Y/Y, 50bps below our September quarter forecast of +1.0% Y/Y. That would translate to roughly 20bps (or ~\$45M) of downside to our September quarter Services revenue growth estimate of +9.5% (vs. +10.7% Y/Y consensus).

Digging into the details, growth in the top 3 App Store markets in aggregate is decelerating in August MTD, driven by the US (tracking -10% Y/Y vs. -7% Y/Y in June;

|                                 |  |                 |
|---------------------------------|--|-----------------|
| MORGAN STANLEY & CO. LLC        |  |                 |
| Erik W Woodring                 |  |                 |
| Equity Analyst                  |  |                 |
| Erik.Woodring@morganstanley.com |  | +1 212 296-8083 |
| Dylan Liu                       |  |                 |
| Research Associate              |  |                 |
| Dylan.Liu@morganstanley.com     |  | +1 212 761-4519 |
| Maya C Neuman                   |  |                 |
| Research Associate              |  |                 |
| Maya.Neuman@morganstanley.com   |  | +1 212 761-1946 |
| Rauf Ural                       |  |                 |
| Research Associate              |  |                 |
| Rauf.Ural@morganstanley.com     |  | +1 212 761-5958 |

|               |         |
|---------------|---------|
| IT HARDWARE   |         |
| North America |         |
| Industry View | In-Line |

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

**For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.**

[Exhibit 30](#) ) and China (tracking +8% Y/Y, vs. +9% Y/Y in July; [Exhibit 28](#) ). Conversely, Japan App Store gross developer revenue declined 4% Y/Y (vs. -5% Y/Y in July) despite being the first major market that allowed alternative app marketplace ( [Exhibit 30](#) ). The Rest of World App Store growth in aggregate is +6% Y/Y this MTD, a 160bps deceleration vs. 7% Y/Y growth in July ( [Exhibit 31](#) ) off a 230bps easier Y/Y compare. **Notably for the US App Store**, net revenue per download is declining 2.1% Y/Y MTD, a 90bps acceleration from -1.2% Y/Y in July, outperforming the ROW, where revenue per download declines are decelerating 150bps vs. July. Apple recently proposed a 15% take rate for link-out transactions in the US, and while the ultimate commission structure will still depend on the Supreme Court's ruling, there is likely upside to Apple's App Store monetization in the US given Apple does not collect any fees on link-out transactions in the US today.

Given that the App Store accounts for ~28% of total FY27 Services revenue on our estimates, App Store monetization remains a key topic in our investor conversations, particularly now that overall Services growth has fallen below a double-digit rate.

**Looking ahead, we do not expect Services growth to reaccelerate meaningfully anytime soon**, driven by (1) ongoing uncertainty around App Store monetization as blended take rates decline and alternative app marketplaces emerge across key regions including the EU, Japan, and Brazil, (2) weaker mobile gaming activity, as highlighted by Apple management, and (3) challenging FX Y/Y compares that lap as early as the June '27 quarter. We've already embedded the Apple Music price increases in our model, but the AppleCare pricing actions should still provide a partial positive offset. For context, we are currently modeling total Services revenue growth of +9.3%/+8.6%/+12.3% for Dec '26/Mar '27/Jun '27 quarters (vs. +11.8%/+11.6%/+12.5% Consensus). For what it's worth, Apple recently announced a lower commission structure for link-out transactions in the EU App Store ([link](#)), which represents an incremental headwind to App Store revenue growth, although we believe investors have largely anticipated this change. For context, Sensor Tower data indicates that EU countries account for ~10% of App Store gross developer revenue. While it remains difficult to precisely quantify the impact of the revised commission structure, the downside is likely limited. Illustratively, every 10% decline in EU App Store revenue would equate to a 1 point headwind to App Store revenue growth, a ~30bps headwind to total Services revenue growth, and a \$0.02 EPS headwind, all else equal.

**C3Q26 iPhone builds unchanged, but iPad builds very modestly lower.** Our Greater China Hardware team [last week](#) maintained its C3Q26 iPhone build forecast of 54M units (-2% Y/Y, +4% Q/Q), including 1M foldable units, which remains broadly consistent with our supply chain checks pointing to CY26 shipments of 265-270M units (+5-7% Y/Y). We would remind investors that Apple has historically communicated its "bull-case" forecasts to the supply chain and typically has not revised those plans until mid-to-late October at the earliest. Based on the historical builds-to-shipments relationship, 54M builds in C3Q would imply September quarter shipments of roughly 58-58.5M units, or 2-2.5M above our 56.0M forecast but still 1-1.5M below the current Consensus estimate of 59.5M. For now, we are maintaining our CY26 shipment forecast of 247M units (-2% Y/Y). Separately, September quarter iPad builds have been reduced by 1M units to 12M (-14% Y/Y, -8% Q/Q), below the

typical seasonality of +19% Q/Q, likely reflecting modestly weaker demand following recent price hikes. As a reminder, our [Sector Quant team's work](#) suggests iPhones (0.2-0.4) and Macs (0.8) have historically exhibited relatively inelastic demand, while iPads (1.0) have shown roughly unitary price elasticity, making them more susceptible to volume pressure following price increases.

**July monthly Notebook ODM build declines of 24% Y/Y signals accelerating demand erosion in the PC market.** Based on the [latest projections from our Greater China Tech Hardware team](#), notebook ODM builds in the month of July declined 24% Y/Y, the worst Y/Y notebook ODM declines in 40 months, and 4% below MSe, due to weak demand and supply constraints. C3Q26 notebook ODM builds are now expected to be 27.3M units, down 20% Y/Y and 6% (or 1.6M units) below prior expectations. We estimate this translates into 41.3M notebook shipments in C3Q, 2% below our current 43.3M unit forecast ([Exhibit 54](#)). This implies Y/Y PC shipment declines deteriorate from -4% Y/Y in C2Q, to -22% Y/Y in C3Q, the worst quarter of Y/Y unit declines in over 3 years, as 1) OEM pull-ins abate following a stronger than expected 1H, 2) consumer PC demand deteriorates further, 3) Windows 11 upgrade demand slows, eliminating a key demand tailwind from the last 18 months, and 4) component inventory gets allocated away from PCs given relative strength in other memory-intensive markets such as Servers and Storage. Taking 1H reported PC shipments alongside the updated 3Q builds implies PC shipments declines of 8% Y/Y in C1Q-C3Q26, with C4Q PC shipments likely to deteriorate further as demand deteriorates further, off tougher C4Q Y/Y compares, consistent with recent commentary from Intel and Lenovo pointing to double digit Y/Y PC unit declines in C2H26 ([Exhibit 55](#)). 2027 PC shipment/build visibility remains limited at the ODMs, but we see a high probability of Y/Y unit declines next year.

**Data we're watching closely over the coming weeks:**

- **Our cloud capex tracker is now pointing to +38% Y/Y growth in CY27, up 9 points from earlier this month, following +104% Y/Y growth in CY26 ([Exhibit 115](#)).** After incorporating the latest Consensus updates, the largest upward revisions to CY27 capex forecasts have come from GOOGL (+\$17.8B), META (+\$8.8B), MSFT (+\$7.5B), AMZN (+\$6.0B), and NBIS (+\$5.0B). We are also adding SPCX to our cloud capex tracker, contributing an incremental \$120.9B of CY27 capex (up \$62.0B Y/Y). Taken together, Consensus CY27 cash cloud capex now stands at \$1.39T, implying capital intensity (i.e. capex as a % of revenue) of 38.6% in aggregate across the top 15 cloud players, 380bps higher than just earlier this month ([Exhibit 118](#)). Even so, our latest MSe points to CY27 cash cloud capex of \$1.61T (+58% Y/Y), compared with buy-side expectations of \$1.8T+ (including financial and operating leases), suggesting there could still be meaningful room for further upward Consensus revisions.
- **What's the latest on PC pricing?** According to IDC's full 2Q26 PC data release, PC industry ASPs rose 19% Y/Y (+8% Q/Q) in C2Q26, an acceleration from +11% Y/Y in C1Q and the strongest growth in PC market history ex-COVID, with DELL/HPQ ASPs up 26% and 23% Y/Y, respectively ([Exhibit 58](#)). For the full 2Q IDC PC data, please reach out.

- **Our IBM Consulting jobs tracker downticks vs. the end of 2Q26.** Based on data as of August 10th, on a daily basis, our IBM Consulting Job Posting tracker is down 53% from the end of C2Q, furthering a steady decline since the beginning of April ([Exhibit 50](#)). We'd also highlight that on a rolling 90-day average basis, our tracker is down 45% vs. the end of C2Q ([Exhibit 51](#)). The declining trend in the daily/monthly job posting data is a clear negative read-through to IBM's 3Q Consulting growth, contrasting with management's constructive tone at 2Q earnings.

# Table of Contents

## [Apple Supply Chain](#) - pg. 5

- Includes 1) Taiwan Suppliers Monthly Sales Tracker, and 2) Morgan Stanley iPhone and iPad Build estimates

## [Smartphone Demand](#) - pg. 10

- Includes 1) Apple Retail Store Status Tracker, 2) iPhone 14 Lead Time Tracker, 3) CAICT China Smartphone Shipment data, 4) IDC Monthly China Smartphone data, and 5) Google Search iPhone Demand Tracker

## [Apple Services](#) - pg. 19

- Includes 1) App Store Net Revenue and Download data, and 2) Apple Services Google Search Trackers

## [Apple AI Job Postings](#) - pg. 25

- Includes the number of active Apple job postings related to AI including NLP, Computer Vision and Deep Learning

## [Apple Vision Pro Demand](#) - pg. 26

- Includes our Vision Pro Lead Time Tracker

## [IBM Related Data](#) - pg. 27

- Includes 1) Active IBM Consulting job postings, and 2) IBM WatsonX Google Trends

## [PC Supply and Demand](#) - pg. 30

- Includes 1) Monthly Notebook ODM Builds, 2) Weekly PC Lead Time Tracker, and 3) AlphaWise US Survey Data on Computer & Accessories Purchasing Intentions

## [Consumer Electronics](#) - pg. 37

- Includes 1) App MAU data, 2) AlphaWise Wearables Demand Trackers, 3) Weekly Lead Time Trackers, 4) Consumer Electronics Web Traffic Data, 5) Morning Consult Data on Apple, Garmin and Sonos, and 6) AlphaWise US Survey Data on Consumer Electronics Purchasing Intentions

## [Hard Disk Drives](#) - pg. 53

- Includes 1) Monthly TSR HDD Production data, 2) Quarterly TSR Takeaways, 3) TSR HDD Metrics

## [Cloud and Enterprise Servers](#) - pg. 56

- Includes 1) Monthly Taiwan Server ODM Sales, 2) Weekly Public Cloud Use / Hyperscale Data Center Commentary

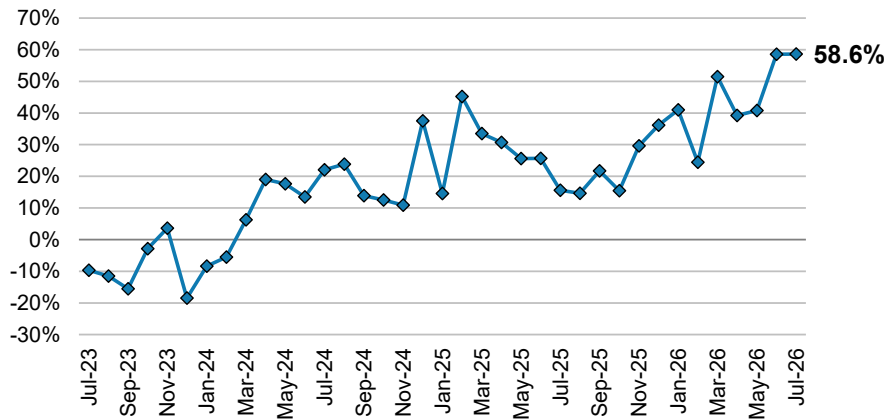
# Apple Supply Chain

## Taiwan Suppliers Monthly Sales Tracker

- Publicly traded Taiwanese companies are required to report monthly sales. This tracker aggregates and analyzes the results from over 50 Taiwan-based Apple supply chain partners across the semiconductor, assembly, optoelectronics and electrical component end markets.

**Exhibit 1:** UPDATED: Sales at Apple's Taiwanese supply chain partners grew 59% Y/Y in July

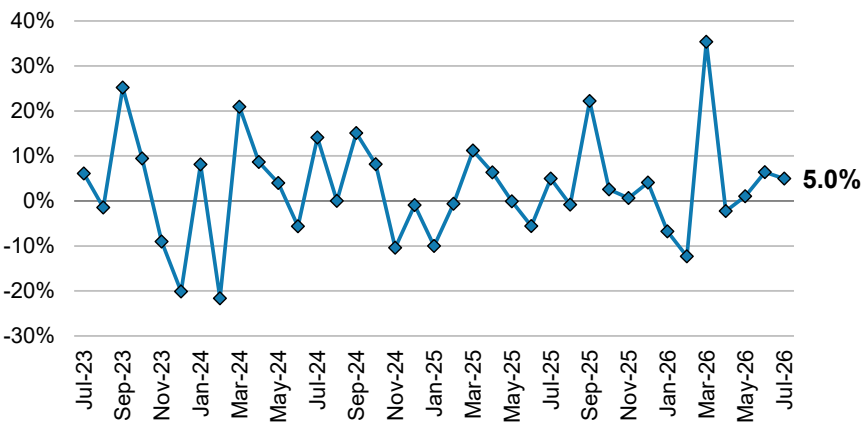
Y/Y Change in Taiwan Suppliers Monthly Sales



Source: AlphaWise, Morgan Stanley Research, Company data

**Exhibit 2:** UPDATED: On a seasonal basis, revenue at Apple's Taiwan-based iPhone suppliers increased ~5% M/M in July...

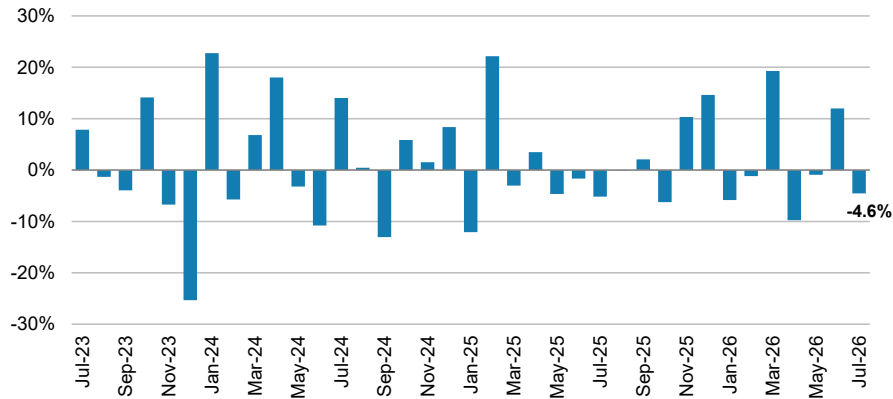
M/M Change in Taiwan Suppliers Monthly Sales



Source: AlphaWise, Morgan Stanley Research, Company data

**Exhibit 3:** UPDATED: ... which is above the trailing 2-year historical seasonality for the month of July

Diff Btw M/M Change in Taiwan Monthly Sales & Trailing 2 Year Seasonality



Source: AlphaWise, Morgan Stanley Research, Company data

## iPhone and iPad Builds

Every month, our colleagues on the Greater China Hardware equity research team (led by Andy Meng) publish their latest iPhone and iPad builds estimates (see their latest Monthly Databook [here](#)), which represents the total number of iPhones and/or iPads that are being manufactured by the supply chain each quarter.

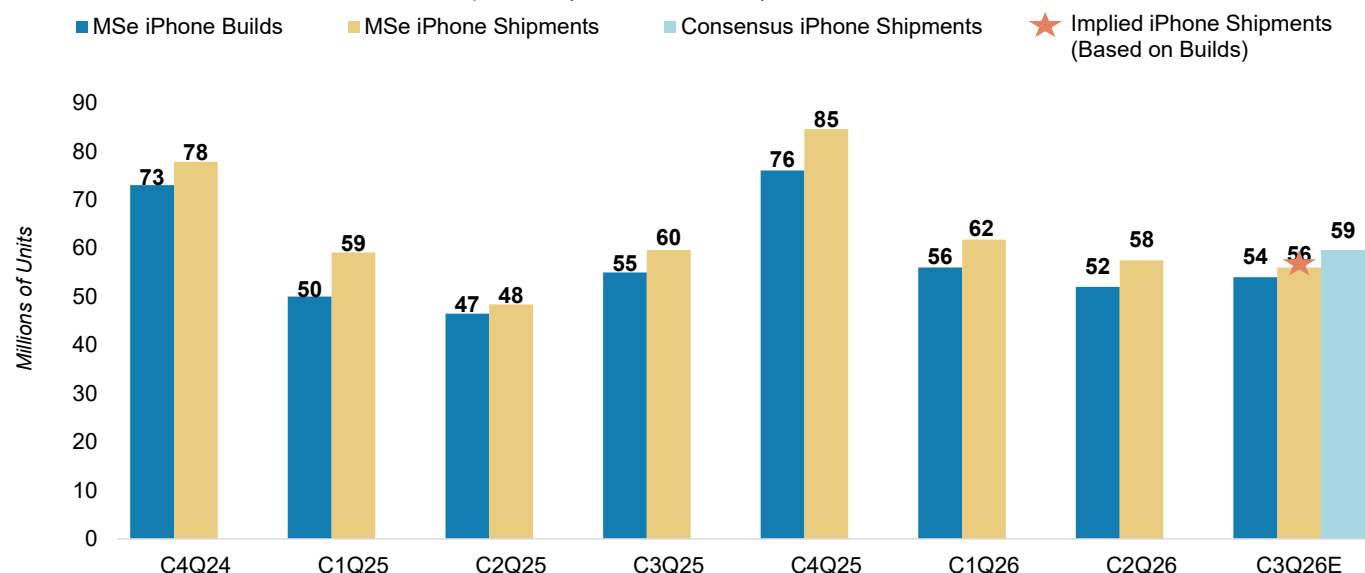
**Exhibit 4:** UPDATED: Our Greater China Tech Hardware equity research colleagues kept their Sept Q builds unchanged at 54M (-2% Y/Y or +4% Q/Q), below seasonality of +26% Q/Q due to a staggered iPhone 18 family launch

| iPhone Builds            | 1Q23        | 2Q23        | 3Q23        | 4Q23        | 2023         | 1Q24        | 2Q24        | 3Q24        | 4Q24        | 2024         | 1Q25        | 2Q25        | 3Q25        | 4Q25        | 2025         | 1Q26        | 2Q26        | 3Q26e       |
|--------------------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|
| iPhone SE3 / SE 5G / 16e | 1.0         | 2.5         | 5.0         | 3.0         | 11.5         | 1.0         | 2.0         | 5.0         | 3.0         | 11.0         | 4.0         | 9.0         | 4.0         | 1.0         | 18.0         | 2.0         | 10.0        | 8.0         |
| iPhone X                 | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone XR                | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 12 mini (5.4")    | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 12 (6.1")         | 0.5         | 0.5         | -           | -           | 1.0          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 12 Pro (6.1")     | 0.5         | -           | -           | -           | 0.5          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 12 Pro Max (6.7") | 0.5         | -           | -           | -           | 0.5          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 13 mini (5.4")    | 1.0         | 0.5         | -           | -           | 1.5          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 13 (6.1")         | 1.0         | 0.5         | -           | -           | 1.5          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 13 Pro (6.1")     | 1.0         | 2.0         | 1.0         | -           | 4.0          | 0.5         | 0.5         | -           | -           | 1.0          | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 13 Pro Max (6.7") | 2.0         | 2.0         | 1.0         | -           | 5.0          | 0.5         | 0.5         | -           | -           | 1.0          | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 14 (6.1")         | 7.0         | 4.0         | 5.0         | 2.0         | 18.0         | 1.0         | 0.5         | -           | -           | 1.5          | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 14 Plus (6.7")    | 5.0         | 3.0         | 1.0         | 1.0         | 10.0         | 1.0         | 0.5         | -           | -           | 1.5          | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 14 Pro (6.1")     | 15.0        | 12.0        | 7.0         | 3.0         | 37.0         | 2.0         | 2.0         | 1.0         | 1.0         | 6.0          | 1.0         | 1.0         | 1.0         | -           | 3.0          | -           | -           | -           |
| iPhone 14 Pro Max (6.7") | 19.5        | 14.0        | 9.0         | 4.0         | 46.5         | 2.0         | 2.0         | 1.0         | 1.0         | 6.0          | 1.0         | 1.0         | 1.0         | -           | 3.0          | -           | -           | -           |
| iPhone 15 (6.1")         | -           | -           | 6.0         | 12.0        | 18.0         | 6.0         | 5.0         | 4.0         | 1.0         | 16.0         | 1.0         | 0.5         | 0.5         | 0.5         | 2.5          | -           | -           | -           |
| iPhone 15 Plus (6.7")    | -           | -           | 5.0         | 10.0        | 15.0         | 4.0         | 3.0         | 1.0         | 1.0         | 9.0          | 1.0         | 0.5         | 0.5         | 0.5         | 2.5          | -           | -           | -           |
| iPhone 15 Pro (6.1")     | -           | -           | 8.0         | 16.0        | 24.0         | 12.0        | 10.0        | 8.5         | 2.0         | 32.5         | 2.0         | 2.0         | 2.0         | 1.0         | 7.0          | 1.0         | 1.0         | -           |
| iPhone 15 Pro Max (6.7") | -           | -           | 2.0         | 24.0        | 26.0         | 18.0        | 13.0        | 10.5        | 3.0         | 44.5         | 3.0         | 3.0         | 2.0         | 1.0         | 9.0          | 1.0         | 1.0         | -           |
| iPhone 16 (6.1")         | -           | -           | -           | -           | -            | -           | -           | 6.0         | 12.0        | 18.0         | 5.0         | 4.0         | 4.0         | 1.0         | 14.0         | 2.0         | 2.0         | 1.0         |
| iPhone 16 Plus (6.7")    | -           | -           | -           | -           | -            | -           | -           | 5.0         | 6.0         | 11.0         | 4.0         | 3.0         | 1.0         | -           | 8.0          | -           | -           | -           |
| iPhone 16 Pro (6.3")     | -           | -           | -           | -           | -            | -           | -           | 8.0         | 17.0        | 25.0         | 11.5        | 9.5         | 4.0         | 2.0         | 27.0         | 2.0         | 2.0         | 2.0         |
| iPhone 16 Pro Max (6.9") | -           | -           | -           | -           | -            | -           | -           | 4.0         | 26.0        | 30.0         | 16.5        | 13.0        | 10.0        | 4.0         | 43.5         | 5.0         | 3.0         | 2.0         |
| iPhone 17 (6.1")         | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | 6.0         | 16.0        | 22.0         | 10.0        | 8.0         | 6.0         |
| iPhone 17 Air (6.7")     | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | 4.0         | 4.0         | 8.0          | -           | -           | -           |
| iPhone 17 Pro (6.3")     | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | 8.0         | 22.0        | 30.0         | 18.0        | 13.0        | 9.0         |
| iPhone 17 Pro Max (6.9") | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | 7.0         | 23.0        | 30.0         | 15.0        | 12.0        | 9.0         |
| iPhone 18 Pro (6.3")     | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | 8.0         |
| iPhone 18 Pro Max (6.9") | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | 8.0         |
| Foldable iPhone          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | 1.0         |
| <b>Total Builds</b>      | <b>54.0</b> | <b>41.0</b> | <b>50.0</b> | <b>75.0</b> | <b>220.0</b> | <b>48.0</b> | <b>39.0</b> | <b>54.0</b> | <b>73.0</b> | <b>214.0</b> | <b>50.0</b> | <b>46.5</b> | <b>55.0</b> | <b>76.0</b> | <b>227.5</b> | <b>56.0</b> | <b>52.0</b> | <b>54.0</b> |
| Y/Y Growth               | 0%          | -9%         | 0%          | -3%         | -3%          | -11%        | -5%         | 8%          | -3%         | -3%          | 4%          | 19%         | 2%          | 4%          | 6%           | 12%         | 12%         | -2%         |
| Q/Q Growth               | -30%        | -24%        | 22%         | 50%         | -            | -36%        | -19%        | 38%         | 35%         | -            | -32%        | -7%         | 18%         | 38%         | -            | -26%        | -7%         | 4%          |

Source: Company data, Morgan Stanley Research estimates

**Exhibit 5:** UPDATED: Our Sept Q shipment estimate of 56.0M is below both builds-implied shipments of 58.2M and 59.5M consensus

### MSe iPhone Builds vs. MSe iPhone Shipments (with Consensus)

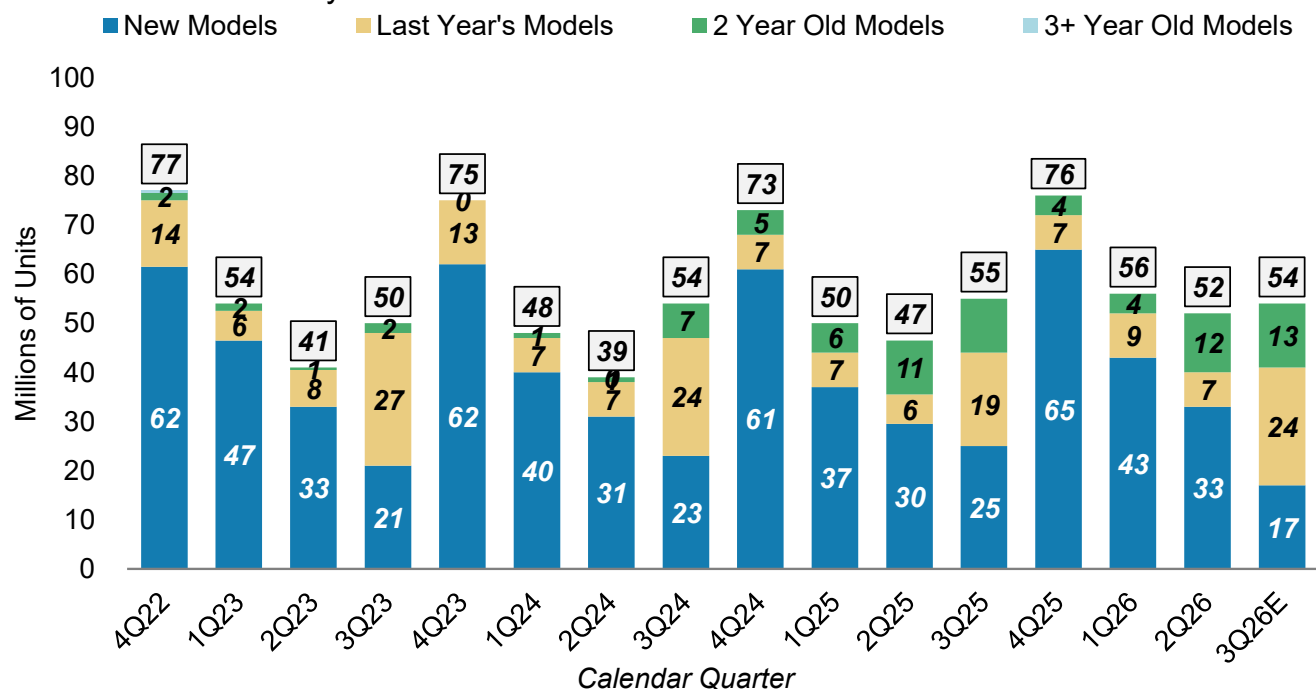


Source: Company data, Morgan Stanley Research estimates



**Exhibit 6:** UPDATED: For C3Q, the team's 54.0M builds include 17.0M iPhone 18 models (31% of mix)

### MSe iPhone Builds By Model Generation



Source: Company data, Morgan Stanley Research estimates

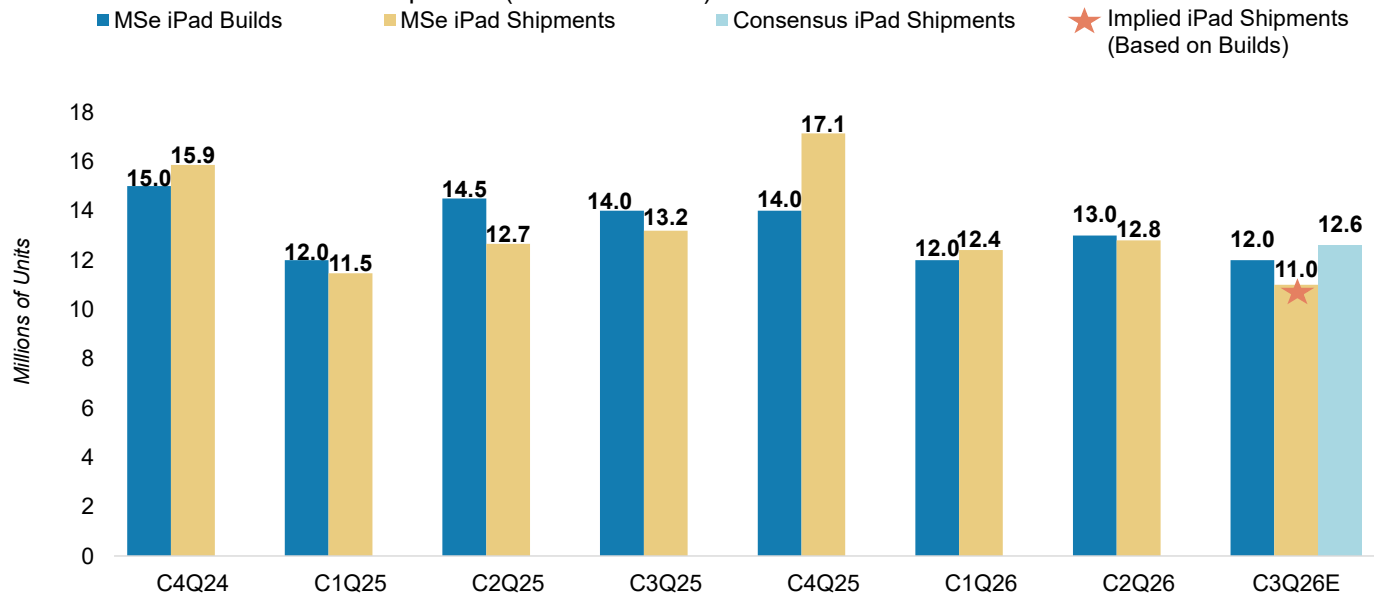
**Exhibit 7:** UPDATED: Our Greater China Tech Hardware equity research colleagues reduced their Sept Q iPad builds by 1.0M to 12.0M (-14% Y/Y or -8% Q/Q), below seasonality of +19% Q/Q likely due to builds pull-in in C1H and demand elasticity amongst price increases

| Millions Units                   | 1Q23        | 2Q23        | 3Q23        | 4Q23        | 2023        | 1Q24        | 2Q24        | 3Q24        | 4Q24        | 2024        | 1Q25        | 2Q25        | 3Q25        | 4Q25        | 2025        | 1Q26        | 2Q26        | 3Q26e       |
|----------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>iPad Builds</b>               |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |             |
| iPad Mini 4                      | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Mini 5                      | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Mini 6                      | 0.5         | -           | -           | -           | 0.5         | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Mini 7                      | 1.0         | 2.5         | 3.0         | 2.0         | 8.5         | 1.0         | 1.0         | 3.0         | 4.5         | 9.5         | 2.5         | 2.0         | 1.5         | 1.0         | 7.0         | 0.5         | 0.5         | 0.5         |
| The iPad (9.7")                  | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| The iPad (10.2")                 | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| The iPad (10.9")                 | 4.0         | 4.5         | 4.5         | 6.0         | 19.0        | 3.5         | 2.5         | 5.5         | 5.5         | 17.0        | 5.5         | 6.0         | 6.0         | 6.0         | 23.5        | 5.5         | 6.0         | 5.5         |
| iPad Air (10.5"/10.9"/11")       | 3.0         | 2.0         | 3.0         | 4.0         | 12.0        | 2.0         | 2.0         | 3.0         | 2.5         | 9.5         | 2.0         | 3.0         | 3.0         | 3.5         | 11.5        | 3.0         | 3.5         | 3.0         |
| iPad Air (12.9"/13.0")           | -           | -           | -           | -           | -           | 1.0         | 2.0         | 2.0         | 1.5         | 6.5         | 1.0         | 1.5         | 1.5         | 1.5         | 5.5         | 2.0         | 2.0         | 2.0         |
| iPad Pro 12.9"                   | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 10.5"                   | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 12.9" w/ Face ID (2018) | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 11" w/ Face ID (2018)   | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 12.9" w/ Face ID (2020) | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 11" w/ Face ID (2020)   | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 12.9" w/ Face ID (2021) | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 11" w/ Face ID (2021)   | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 12.9" w/ Face ID (2022) | 1.0         | 1.0         | 1.0         | 1.5         | 4.5         | 1.0         | -           | -           | -           | 1.0         | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 11" w/ Face ID (2022)   | 1.0         | 1.5         | 1.5         | 1.5         | 5.5         | 1.0         | -           | -           | -           | 1.0         | -           | -           | -           | -           | -           | -           | -           | -           |
| iPad Pro 11" OLED (2024)         | -           | -           | -           | -           | -           | 1.0         | 1.5         | 1.0         | 0.5         | 4.0         | 0.5         | 1.0         | 1.0         | 1.0         | 3.5         | 0.5         | 0.5         | 0.5         |
| iPad Pro 12.9" OLED (2024)       | -           | -           | -           | -           | -           | 1.0         | 1.5         | 1.0         | 0.5         | 4.0         | 0.5         | 1.0         | 1.0         | 1.0         | 3.5         | 0.5         | 0.5         | 0.5         |
| <b>Total Builds</b>              | <b>10.5</b> | <b>11.5</b> | <b>13.0</b> | <b>15.0</b> | <b>50.0</b> | <b>11.5</b> | <b>10.5</b> | <b>15.5</b> | <b>15.0</b> | <b>52.5</b> | <b>12.0</b> | <b>14.5</b> | <b>14.0</b> | <b>14.0</b> | <b>54.5</b> | <b>12.0</b> | <b>13.0</b> | <b>12.0</b> |
| Y/Y Growth                       | -13%        | -12%        | -28%        | -17%        | -18%        | 10%         | -9%         | 19%         | 0%          | 5%          | 4%          | 38%         | -10%        | -7%         | 4%          | 0%          | -10%        | -14%        |
| Q/Q Growth                       | -42%        | 10%         | 13%         | 15%         | -           | -23%        | -9%         | 48%         | -3%         | -           | -20%        | 21%         | -3%         | 0%          | -           | -14%        | 8%          | -8%         |

Source: Company data, Morgan Stanley Research estimates

**Exhibit 8:** UPDATED: Our Sept Q shipments estimate of 11.0M is 2.6M below consensus estimate but largely in line with builds-implied shipments of 10.8M

MSe iPad Builds vs. MSe iPad Shipments (with Consensus)



Source: Company data, Morgan Stanley Research estimates

# Smartphone Demand

## Apple Retail Store Status Tracker (Updated Weekly; Available Daily)

- l Apple currently operates 525 retail stores across 26 different nations, with 271 of those stores located in the US and 254 located in international markets. We track store status (open/closed) on a daily basis using publicly available information from the Apple retail store webpage.

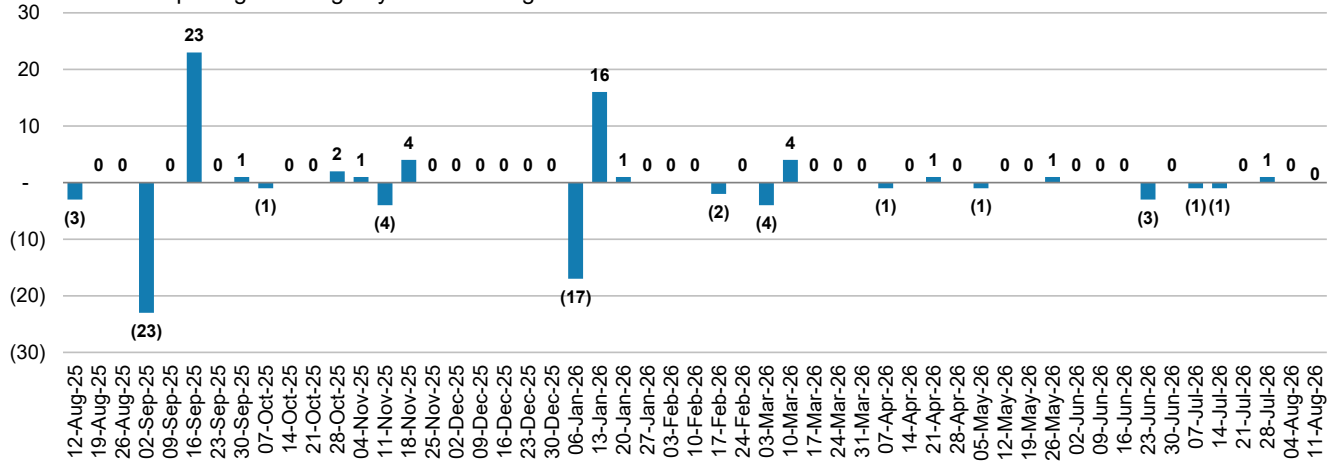
**Exhibit 9:** UPDATED: As of Tuesday, August 11th, 516 of Apple's 525 global retail stores are open (98% of stores)

|                      |              |             |               | As of: 08/11/2026 |
|----------------------|--------------|-------------|---------------|-------------------|
|                      | Total Stores | Stores Open | Stores Closed | % of Stores Open  |
| United States        | 271          | 268         | 3             | 99%               |
| Australia            | 22           | 20          | 2             | 91%               |
| Austria              | 1            | 1           | -             | 100%              |
| Belgium              | 1            | 1           | -             | 100%              |
| Brazil               | 2            | 2           | -             | 100%              |
| Canada               | 28           | 28          | -             | 100%              |
| China Mainland       | 45           | 44          | 1             | 98%               |
| France               | 20           | 20          | -             | 100%              |
| Germany              | 16           | 16          | -             | 100%              |
| Hong Kong            | 6            | 6           | -             | 100%              |
| India                | 2            | 2           | -             | 100%              |
| Italy                | 17           | 17          | -             | 100%              |
| Japan                | 10           | 10          | -             | 100%              |
| Macau                | 2            | 2           | -             | 100%              |
| Mexico               | 2            | 2           | -             | 100%              |
| Netherlands          | 3            | 3           | -             | 100%              |
| Singapore            | 3            | 3           | -             | 100%              |
| South Korea          | 5            | 5           | -             | 100%              |
| Spain                | 11           | 11          | -             | 100%              |
| Sweden               | 3            | 2           | 1             | 67%               |
| Switzerland          | 4            | 4           | -             | 100%              |
| Taiwan               | 2            | 2           | -             | 100%              |
| Thailand             | 2            | 2           | -             | 100%              |
| Turkey               | 3            | 3           | -             | 100%              |
| United Arab Emirates | 4            | 4           | -             | 100%              |
| United Kingdom       | 40           | 38          | 2             | 95%               |
| <b>Total</b>         | <b>525</b>   | <b>516</b>  | <b>9</b>      | <b>98%</b>        |

Source: AlphaWise, Morgan Stanley Research, Apple.com

Exhibit 10: UPDATED: Looking month-over-month, the number of Apple stores that are open was flat M/M

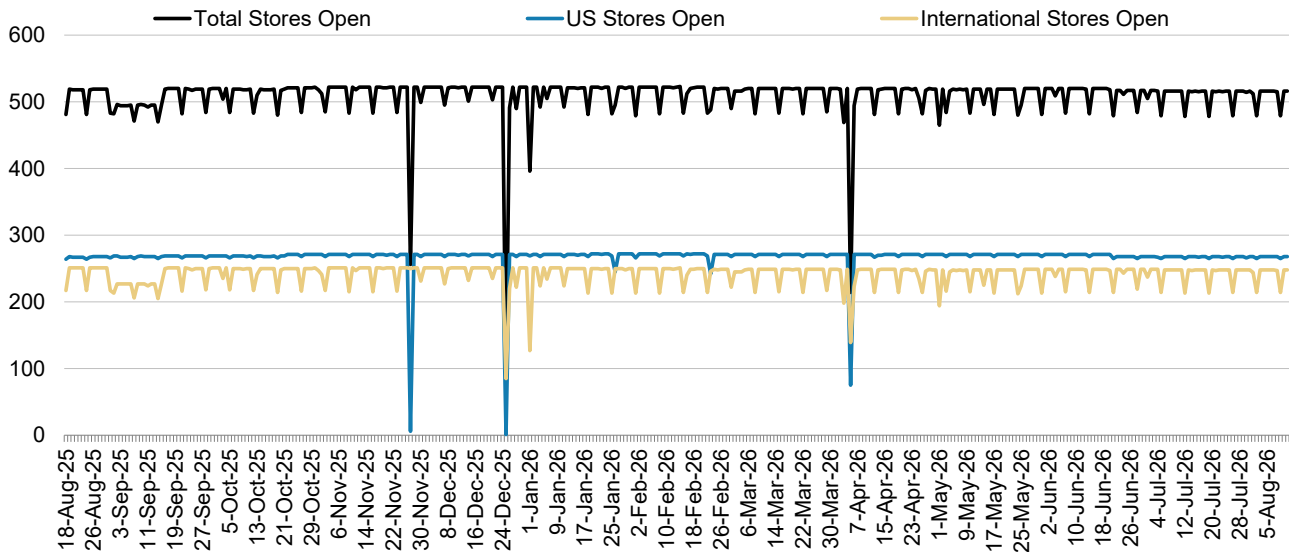
Apple Retail Store Openings/Closings By Week Ending:



Source: Apple.com, Morgan Stanley Research

Exhibit 11: UPDATED: Over the past 3 months, Apple's open store count has remained relatively steady, outside of holidays

Apple Retail Store Status Tracker



Source: Apple.com, Morgan Stanley Research

**Exhibit 12:** UPDATED: As of August 11th, Apple had 3 store closures in the US

Source: AlphaWise, Morgan Stanley Research, Apple.com

iPhone 17 Lead Time Tracker

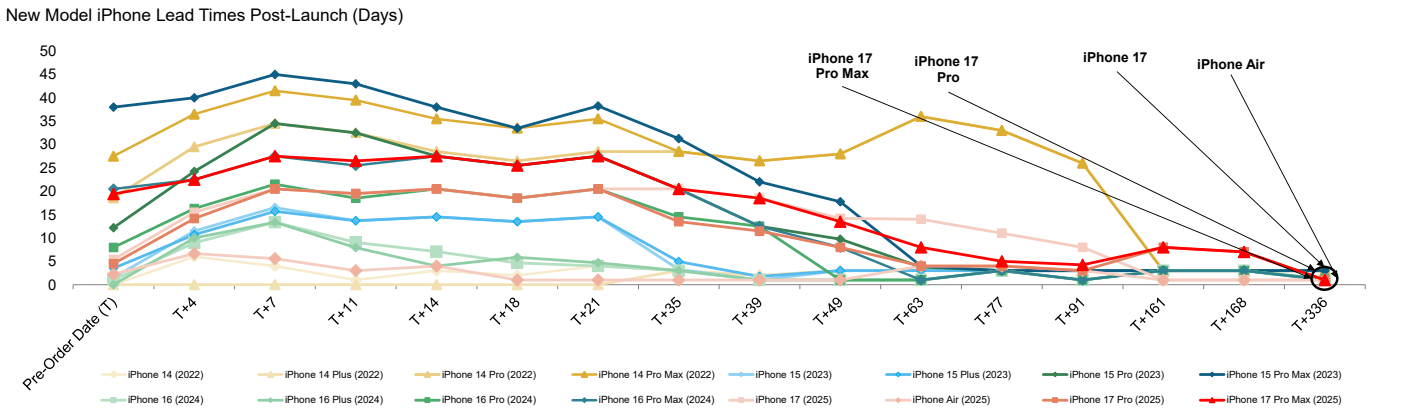
We have closely tracked iPhone lead times – the time between ordering and receiving a device – in the months after a new iPhone model launches back to the iPhone 7 and 7 Plus in 2016. In the exhibits below, we compare iPhone 17 lead times with past iPhone cycles.

Exhibit 13: UPDATED: As of August 14th, lead times for the iPhone 17e, iPhone 17 Pro Max , iPhone 17 Pro, iPhone 17 Air, and iPhone 17 were 11, 1, 1, 1, and 1 days, respectively.

| Summary of iPhone Lead Times Post Launch (Days)* |                    |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |
|--------------------------------------------------|--------------------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|
| Model (Launch Year)                              | Pre-Order Date (T) | T+4  | T+7  | T+11 | T+14 | T+18 | T+21 | T+35 | T+39 | T+49 | T+63 | T+77 | T+91 | T+161 | T+168 | T+336 |
| iPhone 7 (2016)                                  |                    |      | 19.0 | 19.0 |      |      |      |      |      |      | 7.8  |      |      |       |       |       |
| iPhone 7 Plus (2016)                             |                    |      | 21.6 | 21.0 |      |      |      |      |      |      | 24.5 |      |      |       |       |       |
| iPhone 8 (2017)                                  | 2.7                |      | 4.0  |      | 7.5  |      | 14.5 | 8.8  |      | 1.0  | 1.0  | 1.0  | 1.0  |       |       |       |
| iPhone 8 Plus (2017)                             | 2.8                |      | 4.6  |      | 6.4  |      | 8.0  | 6.0  |      | 1.0  | 1.0  | 1.0  | 1.0  |       |       |       |
| iPhone X (2017)                                  | 31.5               | 35.5 | 24.5 | 24.5 | 31.5 | 24.5 | 17.5 | 10.0 | 8.0  | 3.2  | 1.0  | 5.0  | 1.0  |       |       |       |
| iPhone XS (2018)                                 | 7.0                | 14.5 | 17.5 | 12.0 | 4.2  | 1.0  | 3.0  | 4.0  | 1.0  | 3.0  | 4.0  | 3.0  | 3.0  |       |       |       |
| iPhone XS Max (2018)                             | 10.5               | 14.5 | 17.5 | 17.5 | 16.0 | 15.0 | 14.0 | 4.0  | 1.0  | 3.0  | 4.0  | 3.0  | 3.0  |       |       |       |
| iPhone XR (2018)                                 | 2.1                | 14.5 | 13.0 | 12.0 | 5.8  | 2.7  | 3.1  | 4.0  | 2.0  | 3.0  | -    |      |      |       |       |       |
| iPhone 11 (2019)                                 | 4.7                | 11.0 | 12.0 | 6.0  | 12.2 | 9.4  | 10.0 | 11.5 | 7.1  | 3.1  | 3.2  | 4.0  | 1.0  |       |       |       |
| iPhone 11 Pro (2019)                             | 14.2               | 21.5 | 24.5 | 24.5 | 24.5 | 24.5 | 24.5 | 18.8 | 11.5 | 3.5  | 1.5  | 4.0  | 1.0  |       |       |       |
| iPhone 11 Pro Max (2019)                         | 10.7               | 21.5 | 24.5 | 24.5 | 24.5 | 24.5 | 24.5 | 13.5 | 3.5  | 1.5  | 1.5  | 4.0  | 1.0  |       |       |       |
| iPhone 12 (2020)                                 | 0.2                | 10.1 | 11.0 | 9.0  | 3.3  | 1.1  | 4.0  | 1.6  | 0.5  | 1.5  | 2.0  | 2.5  | 1.5  | 1.5   | 1.5   |       |
| iPhone 12 Pro (2020)                             | 11.2               | 18.5 | 24.5 | 24.5 | 26.0 | 25.0 | 27.5 | 28.5 | 22.3 | 28.3 | 27.3 | 26.0 | 12.3 | 1.5   | 1.5   |       |
| iPhone 12 mini (2020)                            | 0.8                | 15.4 | 12.0 | 10.0 | 2.5  | 0.5  | 4.0  | 1.5  | 0.5  | 3.0  | 1.5  | 1.5  | 1.5  | 1.5   | 2.0   |       |
| iPhone 12 Pro Max (2020)                         | 3.4                | 23.5 | 28.5 | 26.5 | 28.5 | 18.6 | 20.5 | 9.6  | 8.7  | 9.6  | 4.8  | 1.7  | 1.6  | 1.5   | 2.0   |       |
| iPhone 13 mini (2021)                            | 3.1                | 16.9 | 19.0 | 12.5 | 15.5 | 14.5 | 15.5 | 11.0 | 9.0  | 11.0 | 8.5  | 3.0  | 4.0  | 3.0   | 3.0   | 3.0   |
| iPhone 13 (2021)                                 | -                  | 16.5 | 21.5 | 19.5 | 23.5 | 22.5 | 23.5 | 11.0 | 9.0  | 11.0 | 8.5  | 3.0  | 4.0  | 3.0   | 3.0   | 3.0   |
| iPhone 13 Pro (2021)                             | 27.5               | 29.5 | 34.5 | 32.5 | 36.5 | 34.5 | 36.5 | 35.0 | 25.5 | 28.5 | 21.5 | 13.5 | 5.0  | 3.0   | 3.0   | 3.0   |
| iPhone 13 Pro Max (2021)                         | 27.5               | 29.5 | 34.5 | 32.5 | 36.5 | 34.5 | 36.5 | 35.0 | 25.5 | 28.5 | 21.5 | 13.5 | 5.0  | 3.0   | 3.0   | 3.0   |
| iPhone SE3 (2022)                                | -                  | 12.0 | 12.6 | 7.0  | 3.0  | 1.0  | 3.0  | 3.0  | 1.0  | 3.0  | 3.0  | 3.0  | 3.0  | 3.0   | 3.0   | 4.9   |
| iPhone 14 (2022)                                 | -                  | 6.0  | 4.0  | 1.0  | 3.0  | 2.0  | 4.0  | 3.0  | 2.0  | 3.0  | 3.0  | 4.0  | 3.0  | 3.0   | 3.0   | 3.0   |
| iPhone 14 Plus (2022)                            | -                  | -    | -    | -    | -    | -    | -    | 3.0  | 2.0  | 3.0  | 3.0  | 4.0  | 3.0  | 3.0   | 3.0   | 3.0   |
| iPhone 14 Pro (2022)                             | 18.6               | 29.5 | 34.5 | 32.5 | 28.5 | 26.5 | 28.5 | 28.5 | 26.5 | 28.0 | 36.0 | 33.0 | 26.0 | 3.0   | 3.0   | 3.0   |
| iPhone 14 Pro Max (2022)                         | 27.5               | 36.5 | 41.5 | 39.5 | 35.5 | 33.5 | 35.5 | 28.5 | 26.5 | 28.0 | 36.0 | 33.0 | 26.0 | 3.0   | 3.0   | 3.0   |
| iPhone 15 (2023)                                 | 1.5                | 11.5 | 16.5 | 13.7 | 14.5 | 13.5 | 14.5 | 3.3  | 1.0  | 3.0  | 3.0  | 3.0  | 3.0  | 3.0   | 3.0   | 3.0   |
| iPhone 15 Plus (2023)                            | 3.5                | 10.7 | 15.7 | 13.7 | 14.5 | 13.5 | 14.5 | 5.0  | 1.8  | 3.0  | 3.0  | 3.0  | 3.0  | 3.0   | 3.0   | 3.0   |
| iPhone 15 Pro (2023)                             | 12.2               | 24.3 | 34.5 | 32.5 | 27.5 | 25.5 | 27.5 | 20.5 | 12.5 | 9.8  | 3.8  | 3.0  | 3.0  | 3.0   | 3.0   | 3.0   |
| iPhone 15 Pro Max (2023)                         | 38.0               | 40.0 | 45.0 | 43.0 | 38.0 | 33.5 | 38.3 | 31.3 | 22.0 | 17.8 | 4.0  | 3.0  | 3.0  | 3.0   | 3.0   | 3.0   |
| iPhone 16 (2024)                                 | 1.0                | 9.0  | 13.4 | 9.0  | 7.1  | 4.7  | 4.0  | 3.0  | 1.0  | 1.0  | 1.0  | 3.0  | 1.0  | 3.0   | 3.0   | 1.0   |
| iPhone 16 Plus (2024)                            | -                  | 10.0 | 13.4 | 7.9  | 4.0  | 5.9  | 4.7  | 3.0  | 1.0  | 1.0  | 1.0  | 3.0  | 1.0  | 3.0   | 3.0   | 1.0   |
| iPhone 16 Pro (2024)                             | 8.0                | 16.3 | 21.5 | 18.5 | 20.5 | 18.5 | 20.5 | 14.5 | 12.5 | 1.0  | 1.0  | 3.0  | 1.0  | 3.0   | 3.0   | 1.2   |
| iPhone 16 Pro Max (2024)                         | 20.5               | 22.5 | 27.5 | 25.5 | 27.5 | 25.5 | 27.5 | 20.5 | 12.5 | 8.0  | 1.0  | 3.0  | 1.0  | 3.0   | 3.0   | 1.0   |
| iPhone 16e (2025)                                | -                  | -    | 4.0  | 3.9  | 4.0  | 1.0  | 3.0  | 3.0  | 1.0  | 3.0  | 1.0  | 1.0  | 1.0  | 1.0   | 1.0   | 5.0   |
| iPhone 17 (2025)                                 | 5.4                | 15.5 | 20.5 | 19.5 | 20.5 | 18.5 | 20.5 | 20.5 | 18.5 | 14.2 | 14.0 | 11.0 | 8.0  | 1.0   | 1.0   | 1.0   |
| iPhone Air (2025)                                | 2.3                | 6.7  | 5.6  | 3.0  | 4.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 4.0  | 4.0  | 3.0  | 1.0   | 1.0   | 1.0   |
| iPhone 17 Pro (2025)                             | 4.5                | 14.2 | 20.5 | 19.5 | 20.5 | 18.5 | 20.5 | 13.5 | 11.5 | 8.0  | 4.0  | 4.0  | 3.0  | 8.0   | 7.0   | 1.0   |
| iPhone 17 Pro Max (2025)                         | 19.4               | 22.5 | 27.5 | 26.5 | 27.5 | 25.5 | 27.5 | 20.5 | 18.5 | 13.5 | 8.0  | 5.0  | 4.3  | 8.0   | 7.0   | 1.0   |
| iPhone 17e (2026)                                | -                  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 3.0  | 1.0  | 1.0  | 11.0  |       |       |

Source: Apple.com, Morgan Stanley Research; \*For the lead time data points prior to iPhone availability (T and T+4), the lead time is calculated from the date of first availability (T+7), not from the pre-order date (T=0)

Exhibit 14: UPDATED: As of August 14th, lead times for the iPhone 17e, iPhone 17 Pro Max , iPhone 17 Pro, iPhone 17 Air, and iPhone 17 were 11, 1, 1, 1, and 1 days, respectively



Source: Apple.com, Morgan Stanley Research; \*For the lead time data points prior to iPhone availability (T and T+4), the lead time is calculated from the date of first availability (T+7), not from the pre-order date (T=0)

**Exhibit 15:** UPDATED: Internationally, the iPhone Air, iPhone 17 Pro, and iPhone 17 iPhone Pro Max and lead times are longer than in the US, while iPhone 17e lead times are the same.

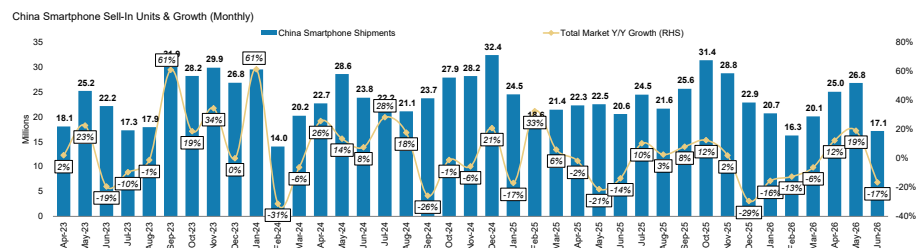
| iPhone 17 Lead Times by Region |                          |      |      |      |      |      |      |       |       |       |       |       |
|--------------------------------|--------------------------|------|------|------|------|------|------|-------|-------|-------|-------|-------|
| Region                         | Model                    | T+4  | T+7  | T+11 | T+14 | T+18 | T+28 | T+102 | T+154 | T+158 | T+273 | T+336 |
| US                             | iPhone 17 Pro Max (2025) | 22.5 | 27.5 | 26.5 | 27.5 | 25.5 | 3.0  | 2.0   | 8.0   | 7.0   | 1.2   | 1.0   |
|                                | iPhone 17 Pro (2025)     | 14.2 | 20.5 | 19.5 | 20.5 | 18.5 | 3.0  | 2.0   | 8.0   | 7.0   | 1.6   | 1.0   |
|                                | iPhone Air (2025)        | 6.7  | 5.6  | 3.0  | 4.0  | 1.0  | 3.0  | 2.0   | 1.0   | 1.0   | 1.0   | 1.0   |
|                                | iPhone 17 (2025)         | 15.5 | 20.5 | 19.5 | 20.5 | 18.5 | 3.0  | 2.0   | 1.0   | 1.0   | 1.0   | 1.0   |
|                                | iPhone 17e (2026)        | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0   | 1.0   | 9.0   |       |       |
| China                          | iPhone 17 Pro Max (2025) | 30.5 | 35.5 | 33.5 | 35.5 | 35.5 | 1.0  | 1.0   | 12.0  | 8.0   | 2.0   | 1.0   |
|                                | iPhone 17 Pro (2025)     | 30.5 | 28.5 | 26.5 | 21.5 | 21.5 | 1.0  | 1.0   | 12.0  | 8.0   | 2.0   | 1.0   |
|                                | iPhone Air (2025)        | -    | -    | -    | -    | -    | 1.0  | 1.0   | 7.0   | 3.0   | 2.0   | 1.0   |
|                                | iPhone 17 (2025)         | 30.5 | 35.5 | 33.5 | 35.5 | 35.5 | 1.0  | 1.0   | 7.0   | 3.0   | 2.0   | 1.0   |
|                                | iPhone 17e (2026)        | 2.0  | 2.0  | 3.1  | 1.5  | 1.5  | 2.0  | 2.0   | 8.0   | 9.8   |       |       |
| Japan                          | iPhone 17 Pro Max (2025) | 24.0 | 29.5 | 27.5 | 28.5 | 26.5 | 6.0  | 4.0   | 11.5  | 8.5   | 2.8   | 7.1   |
|                                | iPhone 17 Pro (2025)     | 16.5 | 22.0 | 20.0 | 20.0 | 19.5 | 1.0  | 1.0   | 6.0   | 5.0   | 1.0   | 1.0   |
|                                | iPhone Air (2025)        | 9.5  | 2.0  | 8.7  | 5.2  | 4.3  | 1.0  | 1.0   | 1.0   | 1.0   | 1.0   | 1.0   |
|                                | iPhone 17 (2025)         | 24.0 | 22.0 | 20.0 | 20.0 | 19.5 | 7.0  | 5.0   | 1.0   | 1.0   | 1.0   | 1.0   |
|                                | iPhone 17e (2026)        | 2.0  | 1.0  | 1.0  | 2.0  | 2.0  | 2.0  | 2.0   | 9.0   | 8.0   |       |       |
| UK                             | iPhone 17 Pro Max (2025) | 22.5 | 27.5 | 25.5 | 27.5 | 25.5 | 5.8  | 7.0   | 8.0   | 7.0   | 1.0   | 1.0   |
|                                | iPhone 17 Pro (2025)     | 15.5 | 20.5 | 18.5 | 20.5 | 18.5 | 3.0  | 7.0   | 8.0   | 7.0   | 1.0   | 1.0   |
|                                | iPhone Air (2025)        | 9.5  | 14.5 | 1.0  | 1.0  | 1.0  | 2.0  | 7.0   | 1.0   | 1.0   | 1.0   | 1.0   |
|                                | iPhone 17 (2025)         | 15.5 | 20.5 | 18.5 | 18.5 | 18.5 | 6.7  | 7.0   | 1.0   | 1.0   | 1.0   | 1.0   |
|                                | iPhone 17e (2026)        | 1.0  | 3.0  | 1.0  | 1.0  | 1.0  | 1.0  | 1.0   | 11.0  | 9.0   |       |       |
| France                         | iPhone 17 Pro Max (2025) | 22.5 | 27.5 | 25.5 | 27.5 | 25.5 | 3.0  | 4.4   | 8.0   | 7.0   | 3.0   | 3.0   |
|                                | iPhone 17 Pro (2025)     | 15.5 | 20.5 | 18.5 | 20.5 | 18.5 | 3.0  | 3.5   | 8.0   | 7.0   | 3.0   | 3.0   |
|                                | iPhone Air (2025)        | 9.5  | 14.5 | 2.0  | 5.3  | 2.7  | 3.0  | 3.5   | 3.0   | 2.0   | 3.0   | 3.0   |
|                                | iPhone 17 (2025)         | 15.5 | 20.5 | 18.5 | 20.5 | 18.5 | 3.3  | 3.5   | 3.0   | 2.0   | 3.0   | 3.0   |
|                                | iPhone 17e (2026)        | 1.0  | 3.0  | 2.0  | 3.0  | 1.0  | 3.0  | 1.0   | 11.0  | 9.0   |       |       |
| Germany                        | iPhone 17 Pro Max (2025) | 23.5 | 28.5 | 26.5 | 28.5 | 28.5 | 3.0  | 7.0   | 8.0   | 7.0   | 3.0   | 3.0   |
|                                | iPhone 17 Pro (2025)     | 16.0 | 21.5 | 19.5 | 21.5 | 19.5 | 3.0  | 7.0   | 8.0   | 7.0   | 3.0   | 3.0   |
|                                | iPhone Air (2025)        | 3.0  | 4.0  | 2.0  | 3.0  | 2.0  | 3.0  | 5.8   | 3.0   | 2.7   | 3.0   | 3.0   |
|                                | iPhone 17 (2025)         | 16.0 | 21.5 | 19.5 | 21.5 | 19.5 | 3.0  | 7.0   | 3.0   | 3.0   | 3.0   | 3.0   |
|                                | iPhone 17e (2026)        | 1.0  | 3.0  | 2.0  | 1.0  | 1.0  | 3.0  | 1.0   | 11.0  | 9.0   |       |       |
| India                          | iPhone 17 Pro Max (2025) | 21.0 | 24.0 | 25.0 | 25.0 | 25.0 | 3.0  | 2.5   | 6.0   | 5.5   | 2.0   | 4.0   |
|                                | iPhone 17 Pro (2025)     | 14.5 | 18.0 | 18.0 | 18.0 | 17.0 | 3.0  | 1.0   | 6.0   | 5.5   | 2.0   | 4.0   |
|                                | iPhone Air (2025)        | 8.0  | 9.7  | 2.0  | 4.0  | 2.0  | 3.0  | 1.0   | 3.0   | 2.0   | 2.0   | 4.0   |
|                                | iPhone 17 (2025)         | 12.5 | 18.0 | 18.0 | 18.0 | 17.0 | 19.5 | 19.0  | 3.0   | 2.0   | 2.0   | 29.5  |
|                                | iPhone 17e (2026)        | 2.0  | 3.0  | 2.0  | 3.0  | 2.0  | 2.0  | 2.0   | 9.0   | 9.0   |       |       |
| Int'l Average                  | iPhone 17 Pro Max (2025) | 24.6 | 29.7 | 27.7 | 29.5 | 28.3 | 3.8  | 4.7   | 9.5   | 7.5   | 2.4   | 3.0   |
|                                | iPhone 17 Pro (2025)     | 18.8 | 22.6 | 20.6 | 20.8 | 19.5 | 2.2  | 3.9   | 8.4   | 6.8   | 2.0   | 1.8   |
|                                | iPhone Air (2025)        | 7.9  | 8.8  | 3.4  | 3.6  | 2.5  | 2.0  | 3.7   | 3.0   | 1.9   | 2.0   | 1.8   |
|                                | iPhone 17 (2025)         | 20.3 | 24.0 | 22.0 | 23.2 | 22.3 | 4.2  | 4.7   | 3.0   | 2.0   | 2.0   | 1.8   |
|                                | iPhone 17e (2026)        | 1.4  | 2.4  | 1.8  | 1.7  | 1.3  | 2.2  | 1.4   | 10.0  | 9.0   |       |       |

Source: Apple.com, Morgan Stanley Research; \*For the lead time data points prior to iPhone availability (T and T+4), the lead time is calculated from the date of first availability (T+7)

## Monthly China Smartphone Shipments

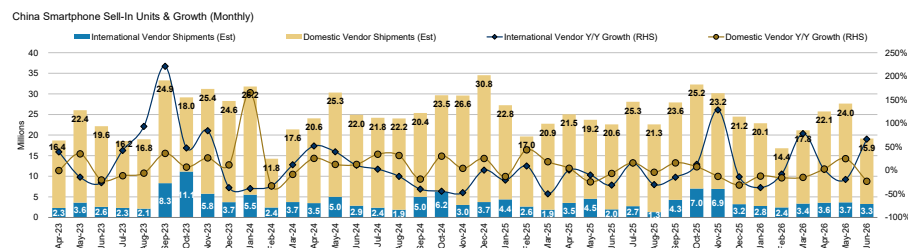
- The Chinese government, through the China Academy of Information and Communications Technology (CAICT), releases smartphone sell-in data in China each month. We also received monthly China smartphone shipment data from IDC on a lag, which is included at the end of this section.

**Exhibit 16:** UPDATED: Total smartphone shipments in China reached 19.1M units in June 2026, down 15% Y/Y



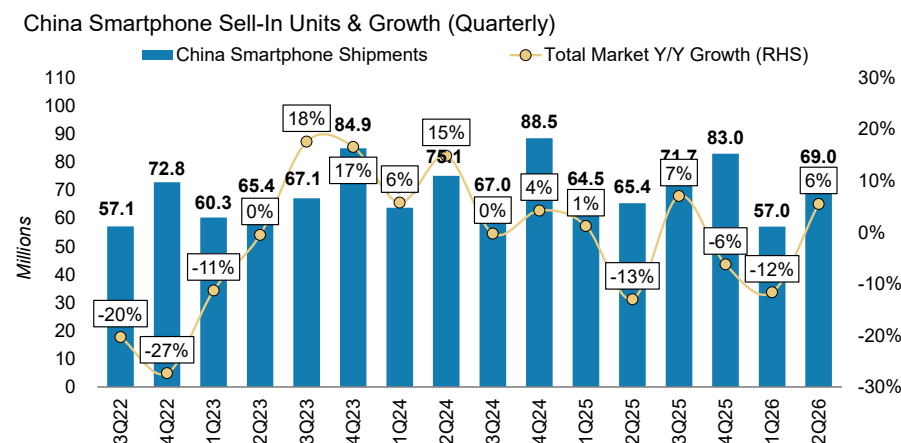
Source: CAICT, Morgan Stanley Research

**Exhibit 17:** UPDATED: Based on international vs. domestic vendor disclosures from the CAICT, we estimate iPhone shipments in China increased 66% Y/Y in June, while domestic Chinese brands decreased 23% Y/Y



Source: CAICT, Morgan Stanley Research

**Exhibit 18:** UPDATED: According to the CAICT, total smartphone shipments in China reached 69.0M units in the June quarter, up 6% Y/Y

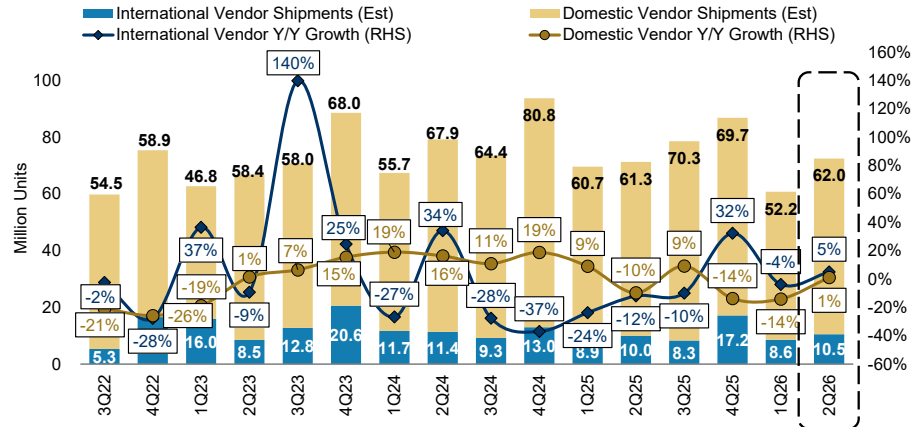


Source: CAICT, Morgan Stanley Research



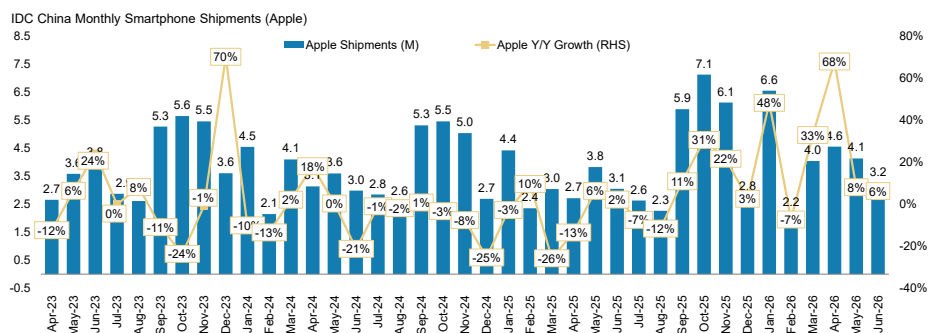
**Exhibit 19:** UPDATED: Monthly CAICT disclosures suggest iPhone shipments grew 5% Y/Y in the June quarter, outperforming domestic Chinese smartphone vendors that saw shipments of 1% Y/Y growth in the quarter

China Smartphone Sell-In Units & Growth (Quarterly)



Source: CAICT, Morgan Stanley Research

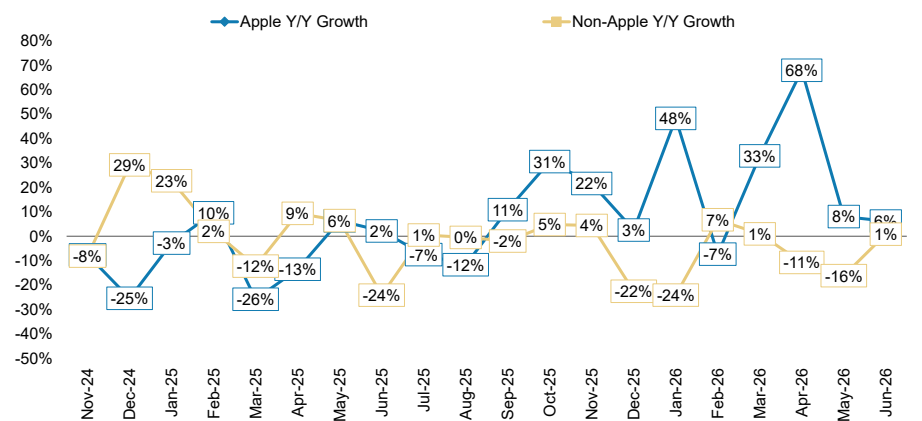
**Exhibit 20:** UPDATED: Monthly IDC data suggest iPhone shipments grew 6% Y/Y in June



Source: IDC, Morgan Stanley Research

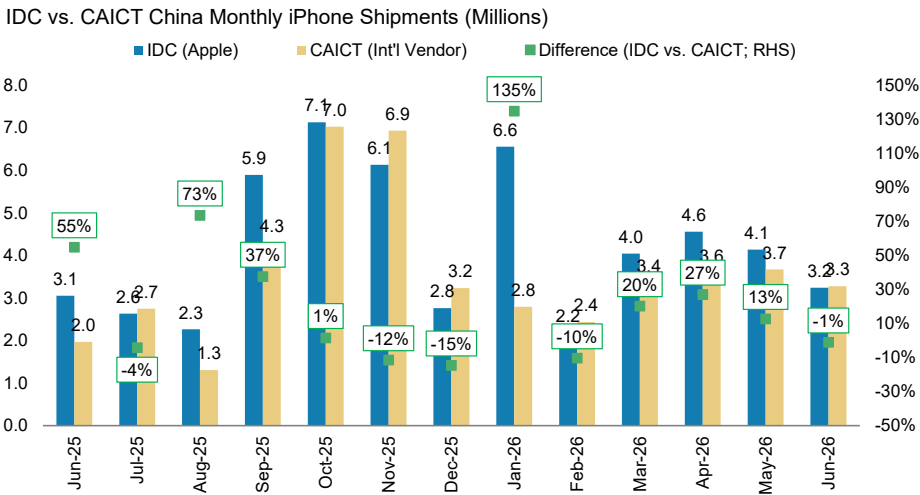
**Exhibit 21:** UPDATED: According to IDC's monthly data for China's smartphone, non-Apple (Android + Huawei) smartphone shipments grew 1% Y/Y in June, underperformed 5 points vs. Apple shipments of +6% Y/Y

China Monthly Smartphone Shipments (Apple vs. Others)



Source: IDC, Morgan Stanley Research

**Exhibit 22:** UPDATED: There is a consistent gap between the IDC and CAICT shipments for Apple/international vendor in China

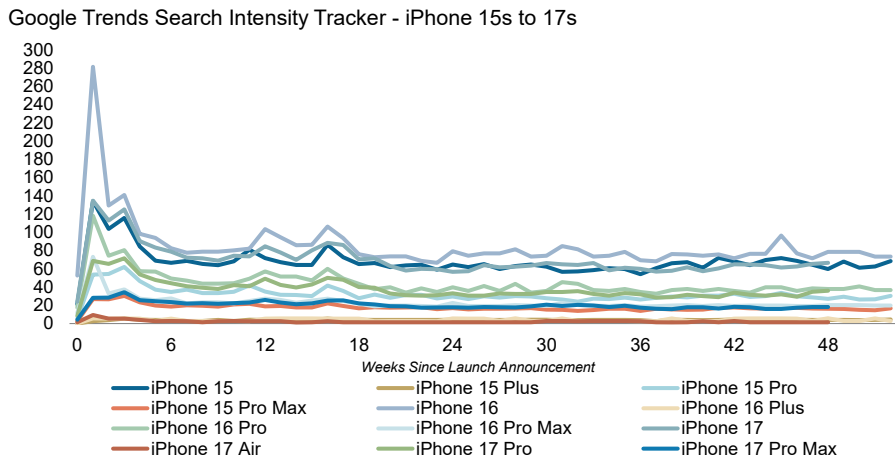


Source: IDC, CAICT, Morgan Stanley Research

## Google Search iPhone Tracker

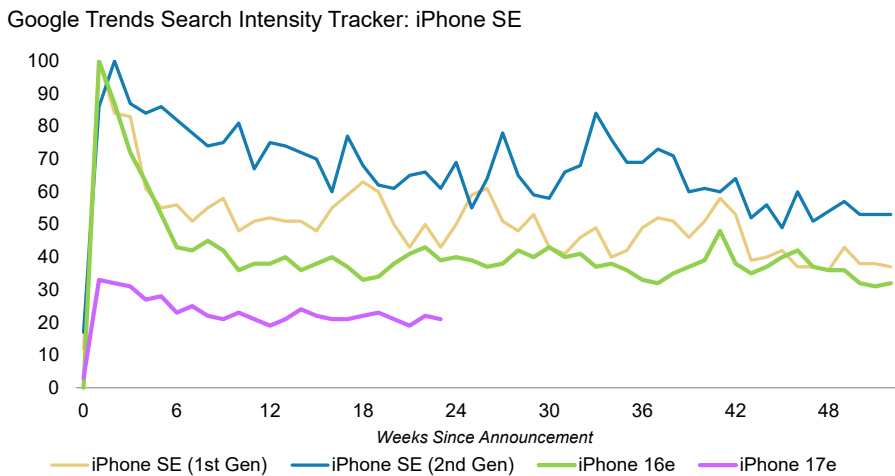
In partnership with our AlphaWise colleagues, we track the intensity of global Google searches for the iPhone on a weekly basis (with a one- or two-week lag).

**Exhibit 23:** UPDATED: Google search intensity for the iPhone 17 is roughly in line with the iPhone 16, 48 weeks after launch



Source: AlphaWise, Morgan Stanley Research; Google; Data for the week ending August 2nd, 2026

**Exhibit 24:** UPDATED: Google search intensity for the iPhone 17e is below the search intensity for the first-generation iPhone SE, the second-generation iPhone SE, and the iPhone 16e (23 weeks from announcement)



Source: AlphaWise, Morgan Stanley Research; Google; Data for the week ending August 2nd, 2026.

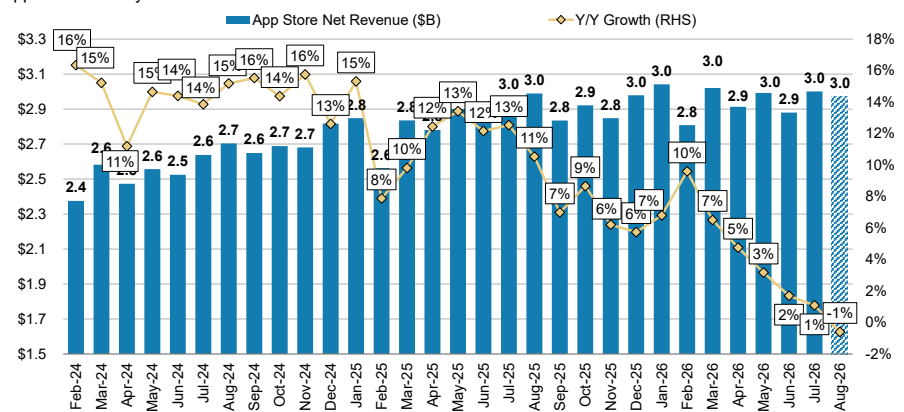
# Apple Services

## App Store Net Revenue and Downloads

- Using data from Sensor Tower, we track Apple's App Store monthly and quarterly net revenue and download data across 110+ countries and 25 app categories.

**Exhibit 25:** UPDATED: We estimate total App Store net revenue grew 1.1% Y/Y in July and month-to-date through August 17, App Store is tracking to 0.5% Y/Y growth

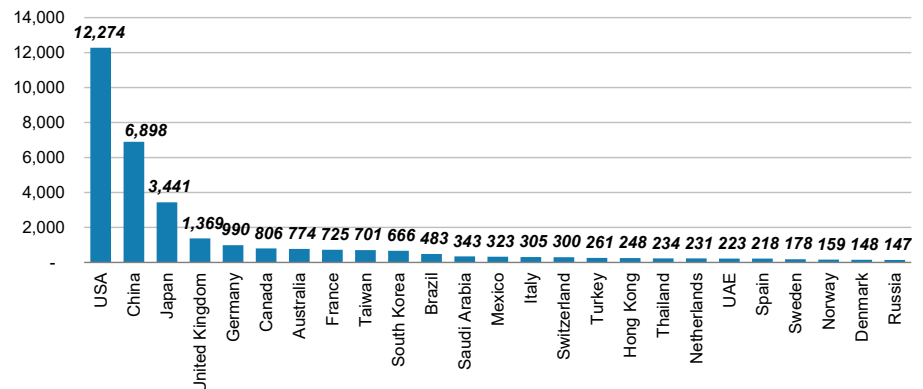
App Store Monthly Net Revenue & Growth



Source: Sensor Tower, Morgan Stanley Research

**Exhibit 26:** UPDATED: The US, China, and Japan collectively represent 64% of App Store trailing 12-month net revenue as of July 2026. The App Store's top 10 markets collectively represent 81% of T12M net revenue

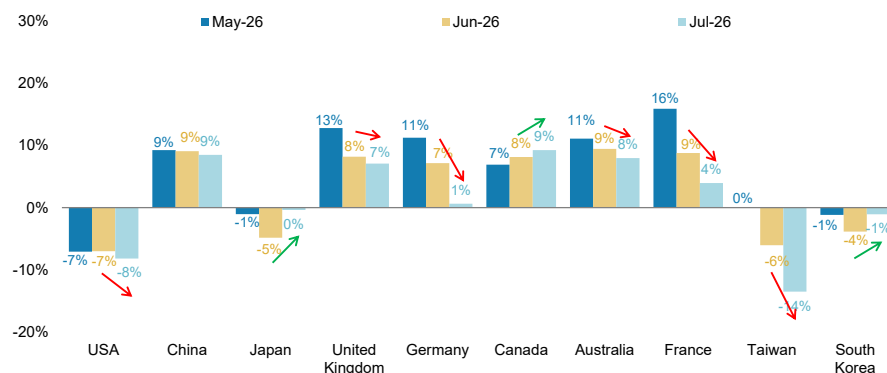
Largest App Store Markets (By T12M Net Revenue; \$M)



Source: Sensor Tower, Morgan Stanley Research

**Exhibit 27:** UPDATED: 7 of the 10 largest App Store markets saw growth deceleration in the month of July

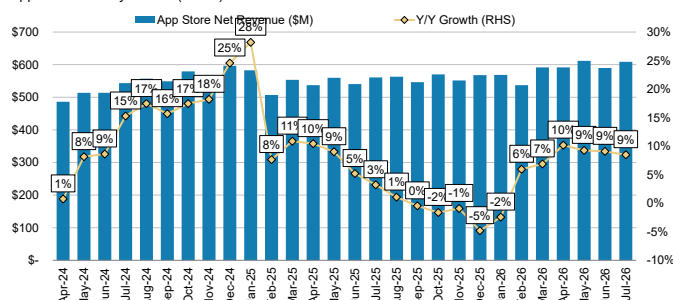
Y/Y Net Revenue Growth For 10 Largest App Store Markets



Source: Sensor Tower, Morgan Stanley Research

**Exhibit 28:** UPDATED: App Store net revenue growth in China stabilized at +9% Y/Y in July

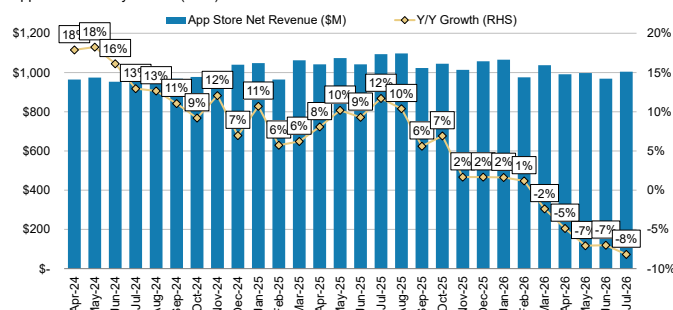
App Store Monthly Trends (China)



Source: Sensor Tower, Morgan Stanley Research

**Exhibit 29:** UPDATED: App Store net revenue growth in the US decelerated 1 point to -8% Y/Y in July

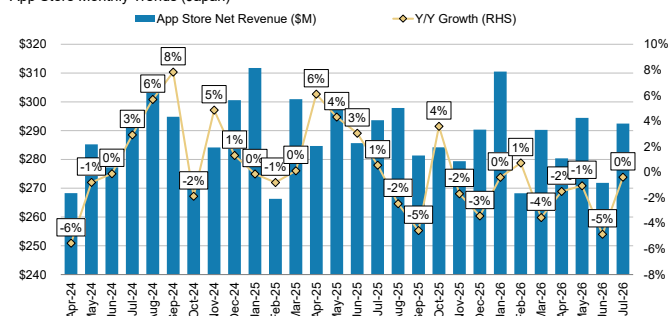
App Store Monthly Trends (USA)



Source: Sensor Tower, Morgan Stanley Research

**Exhibit 30:** UPDATED: App Store net revenue growth in Japan accelerated 5 points from June to flat Y/Y in July

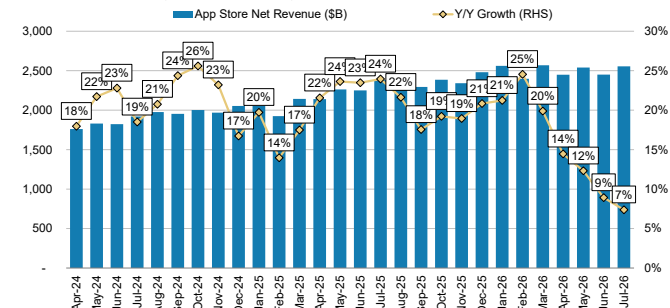
App Store Monthly Trends (Japan)



Source: Sensor Tower, Morgan Stanley Research

**Exhibit 31:** UPDATED: Outside of the 3 largest countries (USA, China and Japan), Y/Y App Store net revenue growth decelerated 2 points to +7% Y/Y in July

App Store Net Revenue (ROW Outside Top 3)

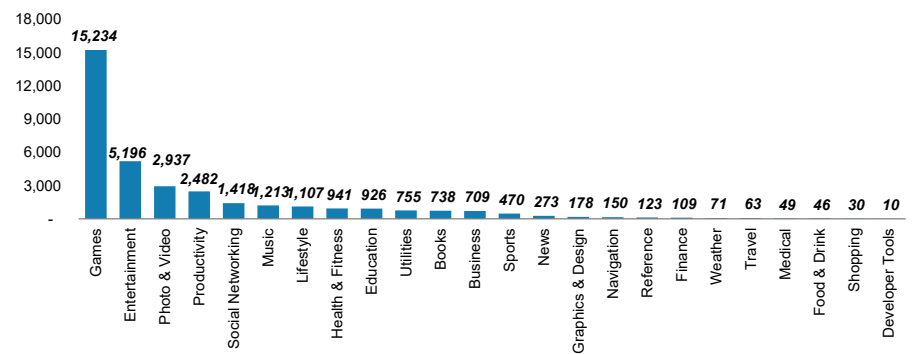


Source: Sensor Tower, Morgan Stanley Research

Exhibit 32:

UPDATED: Gaming represents the largest revenue-generating app category in the App Store by far, accounting for 43% of T12M App Store net revenue as of July 2026

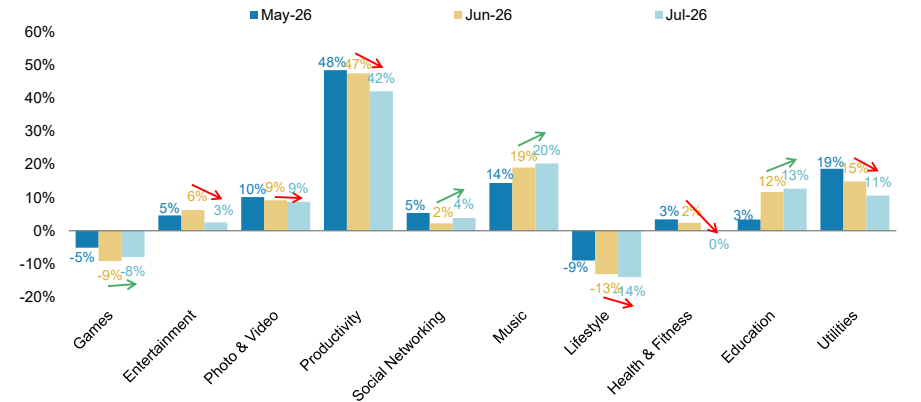
Largest App Store App Categories (By T12M Net Revenue; \$M)



Source: Sensor Tower, Morgan Stanley Research

Exhibit 33: UPDATED: 6 of the 10 largest App Store app categories saw a deceleration in net revenue growth in July

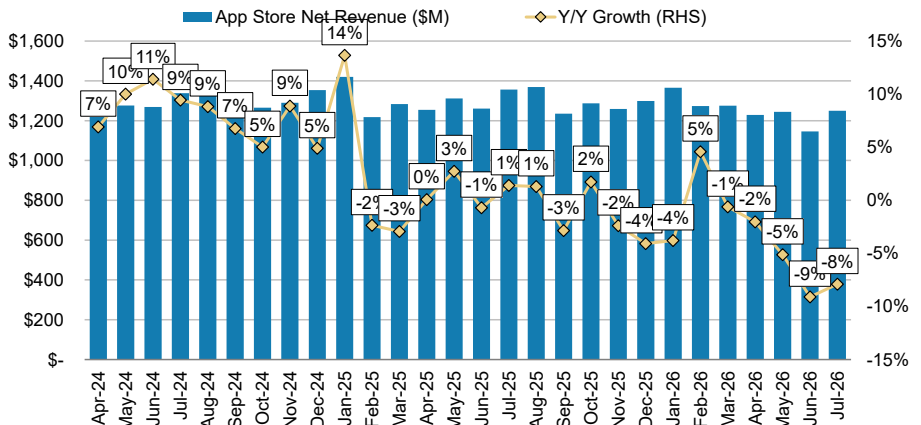
Y/Y Net Revenue Growth For 10 Largest App Store Categories



Source: Sensor Tower, Morgan Stanley Research

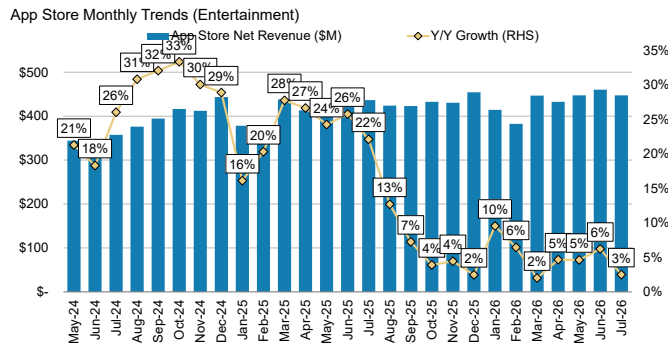
Exhibit 34: UPDATED: Gaming net revenue growth accelerated 1 point M/M to -8% Y/Y in July

App Store Monthly Trends (Gaming)



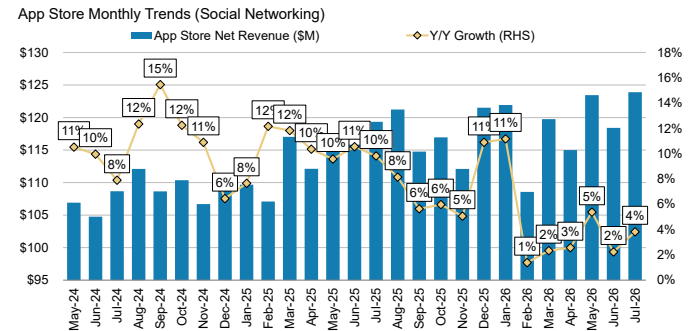
Source: Sensor Tower, Morgan Stanley Research

**Exhibit 35:** UPDATED: Entertainment-related App Store net revenue growth decelerated 2 points M/M to +3% Y/Y in July



Source: Sensor Tower, Morgan Stanley Research

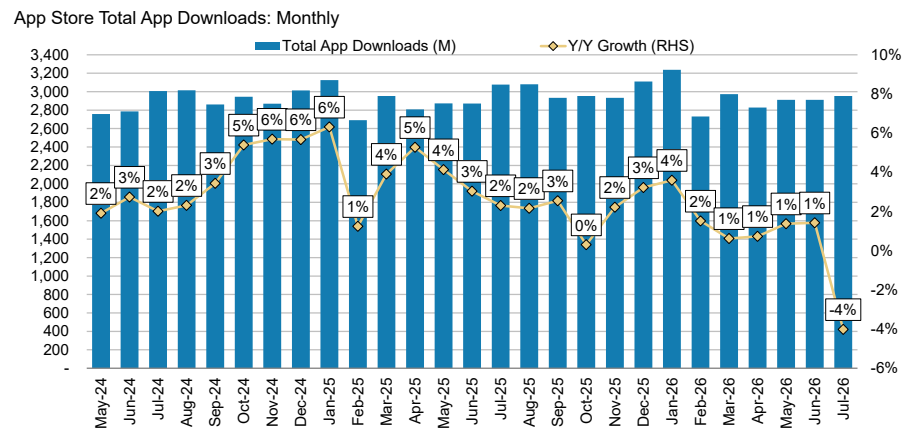
**Exhibit 36:** UPDATED: Social Networking-related App Store net revenue growth accelerated 2 points M/M to +4% Y/Y in July



Source: Sensor Tower, Morgan Stanley Research

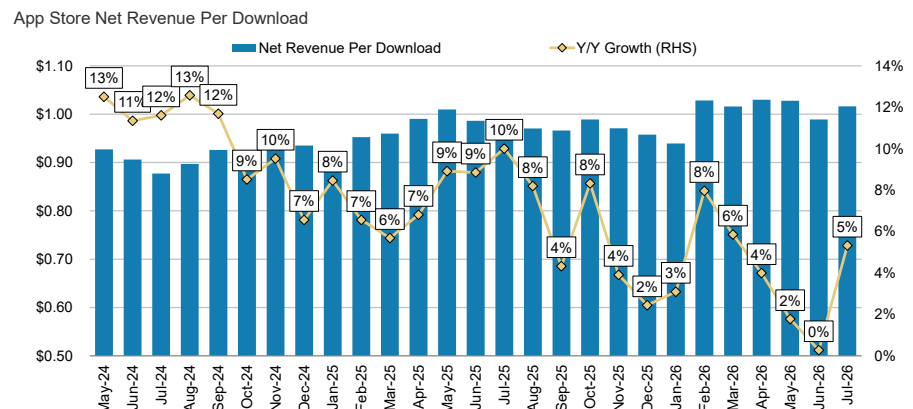
**Exhibit 37:**

UPDATED: Total App Store app downloads were -4% Y/Y in July, decelerating 5 points from June



Source: Sensor Tower, Morgan Stanley Research

**Exhibit 38:** UPDATED: App Store net revenue per download grew 5% Y/Y in July, accelerating 5 points from June

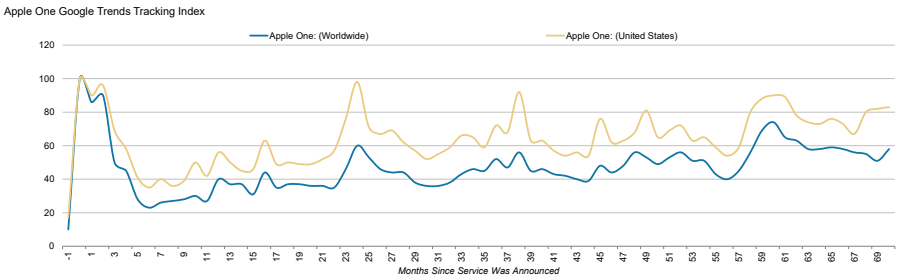


Source: Sensor Tower, Morgan Stanley Research

## Apple Services Google Search Trackers

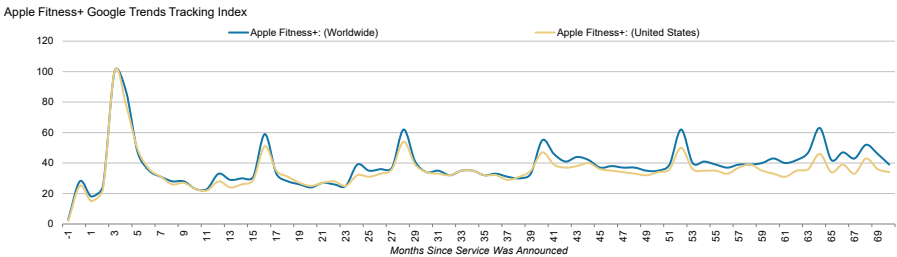
In partnership with our AlphaWise research colleagues, we track the intensity of both US and global Google searches for all of Apple's most recent Services launches, including Apple One, Apple Fitness+, Apple TV, Apple News+, Apple Card, Apple Arcade and Apple TV Channels, on a monthly basis.

**Exhibit 39:** Google Search intensity for the Apple One bundle in the United States upticked in July



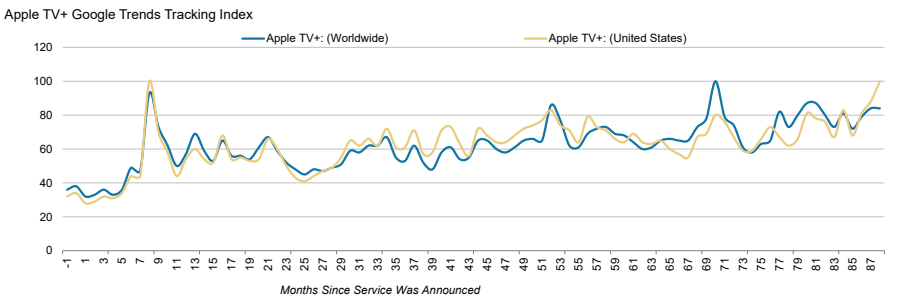
Source: AlphaWise, Morgan Stanley Research; Google; Latest data for the month of July.

**Exhibit 40:** Google Search intensity for the Apple Fitness+ bundle spiked in the week of December 13th, 2020 when the service was made available and downticked in July



Source: AlphaWise, Morgan Stanley Research; Google; Latest data for the month of July.

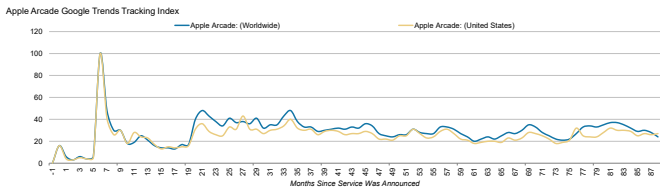
**Exhibit 41:** Apple TV+ Google search intensity downticked M/M in July



Source: AlphaWise, Morgan Stanley Research; Google; Latest data for the month of July.

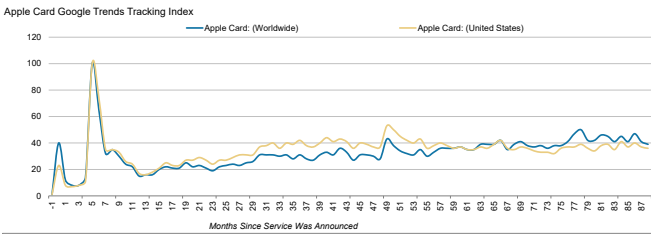


**Exhibit 42:** Search intensity for Apple Arcade remains significantly below its September 2019 peak



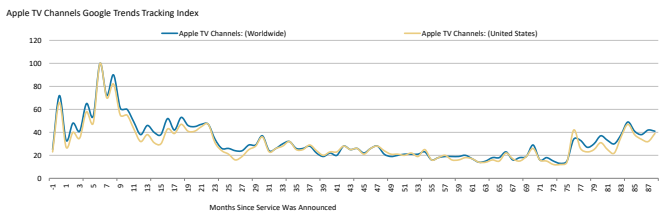
Source: AlphaWise, Morgan Stanley Research; Google; Latest data as of the month of July.

**Exhibit 43:** Search intensity for the Apple Card remains well below its August 2019 peak



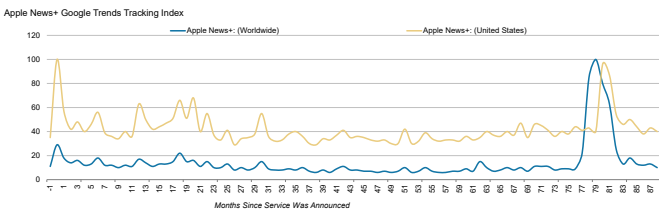
Source: AlphaWise, Morgan Stanley Research; Google; Latest data as of the month of July.

**Exhibit 44:** Search intensity for Apple TV Channels downticked in the month of July



Source: AlphaWise, Morgan Stanley Research; Google; Latest data as of the month of July.

**Exhibit 45:** Search intensity for Apple News+ downticked in July



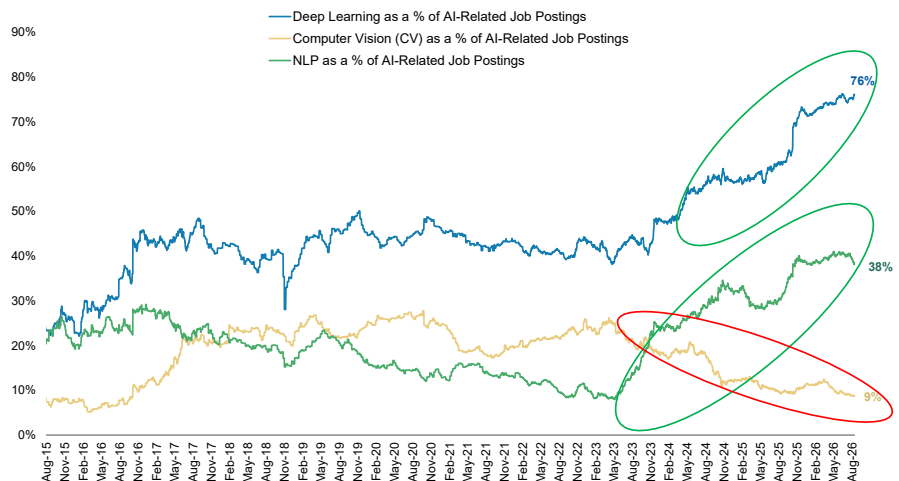
Source: AlphaWise, Morgan Stanley Research; Google; Latest data as of the month of July.

# Apple AI Job Postings

- We view employee count as another indicator of Apple's ramp on its AI efforts and will now add Apple AI Job postings to our monthly data tracker moving forward.

**Exhibit 46:** UPDATED: Over the past several months, deep learning and NLP have risen as a % of AI jobs postings, with Deep Learning appearing in 76% of Apple's AI job postings (up since June) and NLP appearing in 38% of postings (down since June)

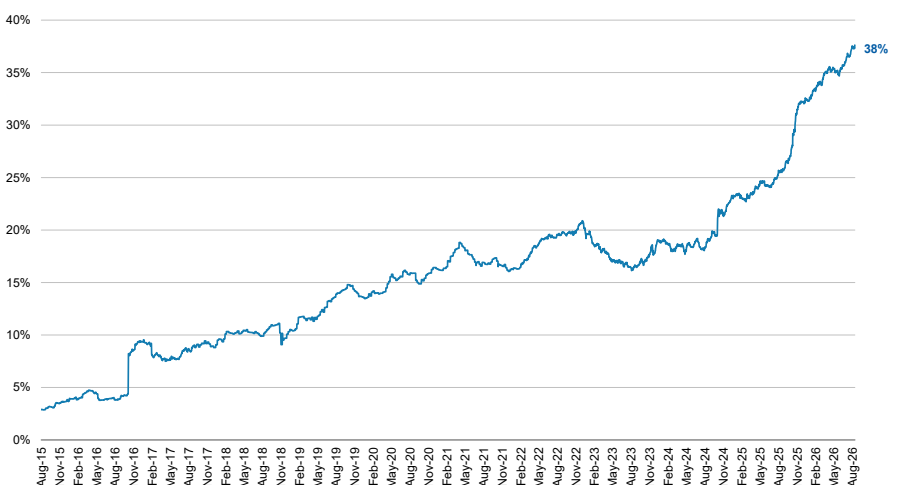
Apple AI-Related Job Postings by Topic



Source: LinkUp, Morgan Stanley Research

**Exhibit 47:** UPDATED: We estimate that 38% of Apple's current job postings are AI-related, up from just 17% nearly 6 years ago, showing the steady ramp to Apple's AI-technologies with recent moderation over the past month

Apple AI-Related Job Postings as % Total Job Postings



Source: LinkUp, Morgan Stanley Research

# Apple Vision Pro Demand

## Apple Vision Pro Lead Time Tracker

- We track Vision Pro lead times – the time between ordering and receiving a device – in the weeks after the pre-order date of the Vision Pro. In the exhibits below, we compare lead times for the Vision Pro across models. In the exhibits below, we compare the 2025 Vision Pro lead times with past Vision Pro cycles.

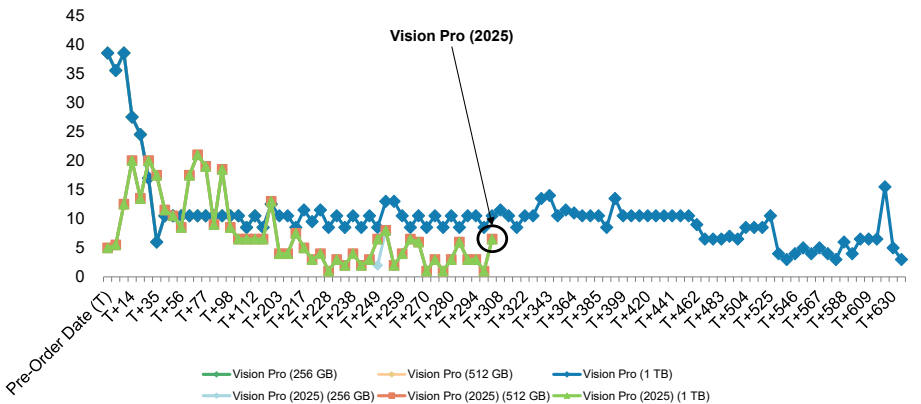
**Exhibit 48:** UPDATED: As of Friday, August 14th, lead times were 6.5 days for the 2025 Vision Pro (256 GB), 2025 Vision Pro (512 GB) and 2025 Vision Pro (1TB)

| Summary of Vision Pro Lead Times Post Launch (Days)* |                    |      |      |         |          |          |          |          |          |          |          |           |           |
|------------------------------------------------------|--------------------|------|------|---------|----------|----------|----------|----------|----------|----------|----------|-----------|-----------|
| Model (Launch Year)                                  | Pre-Order Date     |      |      | +1 Week | +2 Weeks | +3 Weeks | +4 Weeks | +5 Weeks | +6 Weeks | +7 Weeks | +8 Weeks | +16 Weeks | +43 Weeks |
| Model (Launch Year)                                  | Pre-Order Date (T) | T+4  | T+7  | T+14    | T+21     | T+28     | T+35     | T+42     | T+49     | T+56     | T+112    | T+301     |           |
| Vision Pro (256 GB)                                  | 38.5               | 35.5 | 38.5 | 27.5    | 24.5     | 17.0     | 6.0      | 10.5     | 10.5     | 10.5     | 10.5     | 10.5      |           |
| Vision Pro (512 GB)                                  | 38.5               | 35.5 | 38.5 | 27.5    | 24.5     | 17.0     | 6.0      | 10.5     | 10.5     | 10.5     | 10.5     | 10.5      |           |
| Vision Pro (1 TB)                                    | 38.5               | 35.5 | 38.5 | 27.5    | 24.5     | 17.0     | 6.0      | 10.5     | 10.5     | 10.5     | 10.5     | 10.5      |           |
| Vision Pro (2025) (256 GB)                           | 5.0                | 5.5  | 12.5 | 20.0    | 13.5     | 20.0     | 17.5     | 11.5     | 10.5     | 8.5      | 6.5      | 6.5       |           |
| Vision Pro (2025) (512 GB)                           | 5.0                | 5.5  | 12.5 | 20.0    | 13.5     | 20.0     | 17.5     | 11.5     | 10.5     | 8.5      | 6.5      | 6.5       |           |
| Vision Pro (2025) (1 TB)                             | 5.0                | 5.5  | 12.5 | 20.0    | 13.5     | 20.0     | 17.5     | 11.5     | 10.5     | 8.5      | 6.5      | 6.5       |           |

Source: Apple.com, Morgan Stanley Research

**Exhibit 49:** UPDATED: As of Friday, August 14th, lead times were 6.5 days for the 2025 Vision Pro (256 GB), 2025 Vision Pro (512 GB) and 2025 Vision Pro (1TB)

### Vision Pro Lead Times Post-Launch (Days)



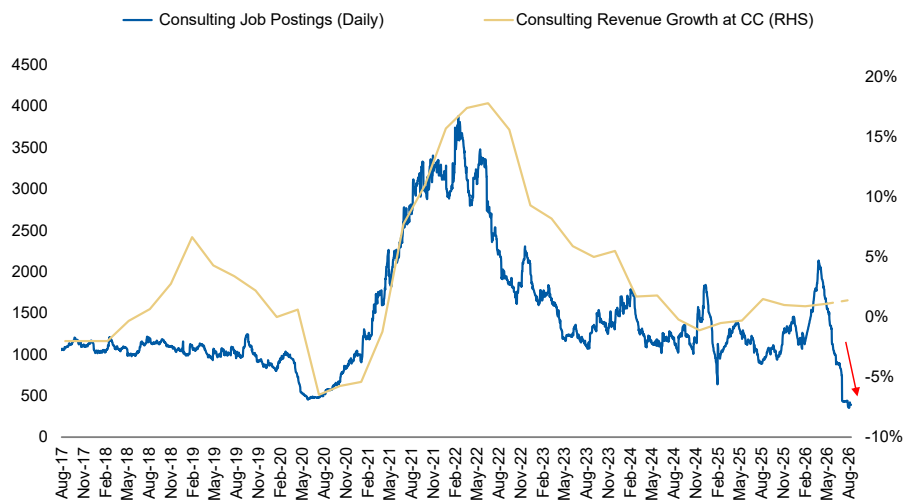
Source: Apple.com, Morgan Stanley Research

# IBM Consulting Job Postings and WatsonX Trends

- We view employee count as an important leading indicator for IBM's Consulting business and Google Trends on WatsonX as an important insight to IBM's Software business. In addition to our quarterly IBM previews, we will now add IBM active Consulting job postings and Google trends on IBM WatsonX to our monthly data tracker moving forward.

**Exhibit 50:** UPDATED: On a daily basis, IBM active Consulting job postings are down 53% from the end of 2Q

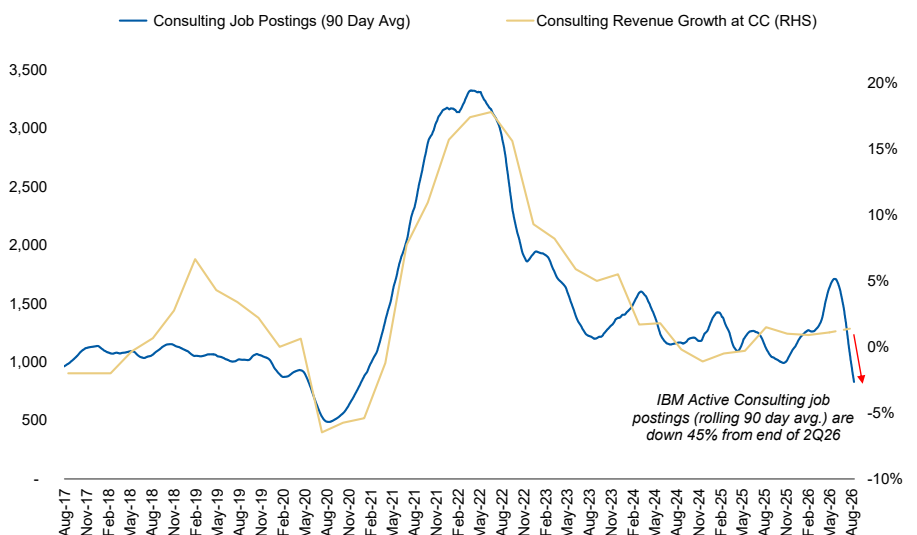
IBM Active Consulting Job Postings



Source: LinkUp, Morgan Stanley Research

**Exhibit 51:** UPDATED: On a rolling 90-day average basis, IBM active Consulting job postings are down 45% from end of 2Q26

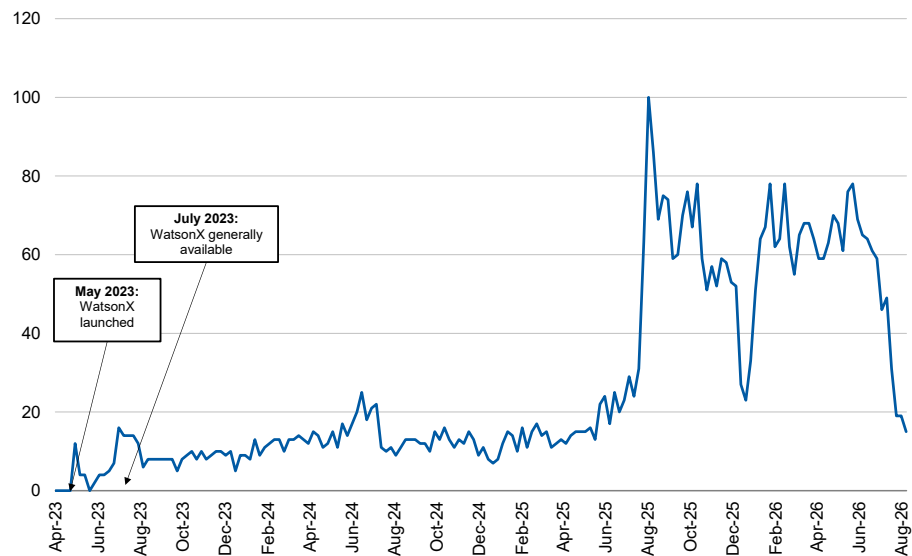
IBM Active Consulting Job Postings vs. Consulting Revenue Growth



Source: LinkUp, Morgan Stanley Research

**Exhibit 52:** UPDATED: IBM WatsonX Google Search decreased by 31 points to 15 on August 16th from 46 on July 12th. Google notes that the value of 100 is peak popularity for the term

**Google Trends: IBM WatsonX**



Source: Google, Morgan Stanley Research

# PC Supply and Demand

## Monthly Notebook ODM Builds

- Each month, our Greater China Technology Hardware team (led by Sharon Shih) publishes total monthly/quarterly Notebook build estimates from the leading Notebook original design manufacturers (ODMs).

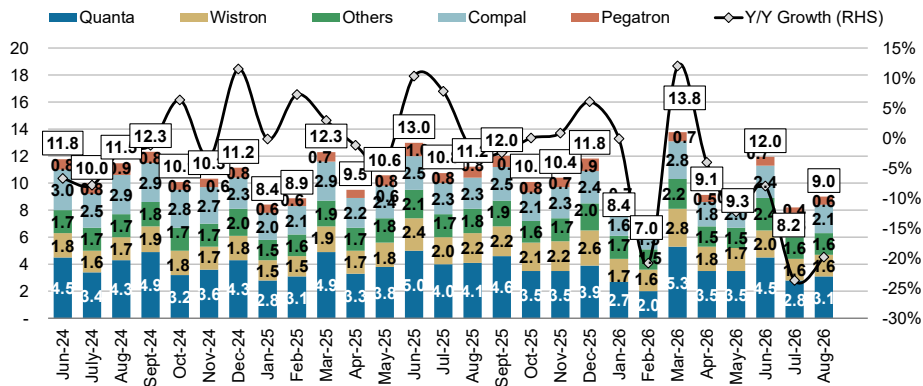
**Exhibit 53:** UPDATED: Our Greater China Tech Hardware research colleagues forecast top 5 Notebook ODM builds will reach 9M units in August 2026 (-20% Y/Y, +10% M/M)

|            | Jun-24 | July-24 | Aug-24 | Sept-24 | Oct-24 | Nov-24 | Dec-24 | Jan-25 | Feb-25 | Mar-25 | Apr-25 | May-25 | Jun-25 | Jul-25 | Aug-25 | Sept-25 | Oct-25 | Nov-25 | Dec-25 | Jan-26 | Feb-26 | Mar-26 | Apr-26 | May-26 | Jun-26 | Jul-26 | Aug-26 |
|------------|--------|---------|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Quanta     | 4,500  | 3,400   | 4,300  | 4,900   | 3,200  | 3,800  | 4,300  | 2,800  | 3,100  | 4,900  | 3,300  | 3,800  | 5,000  | 4,000  | 4,100  | 4,600   | 3,500  | 3,500  | 3,500  | 2,700  | 2,000  | 5,300  | 3,500  | 4,500  | 2,800  | 3,100  |        |
| Wistron    | 1,800  | 1,600   | 1,700  | 1,900   | 1,800  | 1,700  | 1,800  | 1,500  | 1,500  | 1,900  | 1,700  | 1,800  | 2,400  | 2,000  | 2,200  | 2,200   | 2,100  | 2,200  | 2,600  | 1,700  | 1,600  | 2,800  | 1,800  | 1,700  | 2,000  | 1,600  |        |
| Others     | 1,700  | 1,700   | 1,700  | 1,800   | 1,700  | 1,700  | 2,000  | 1,500  | 1,600  | 1,900  | 1,700  | 1,800  | 2,100  | 1,700  | 1,800  | 1,900   | 1,600  | 1,700  | 2,000  | 1,700  | 1,500  | 2,200  | 1,500  | 1,500  | 2,400  | 1,600  |        |
| Compal     | 3,000  | 2,500   | 2,900  | 2,900   | 2,800  | 2,700  | 2,300  | 2,000  | 2,100  | 2,900  | 2,200  | 2,400  | 2,500  | 2,300  | 2,300  | 2,500   | 2,100  | 2,300  | 2,400  | 1,600  | 1,500  | 2,800  | 1,800  | 2,000  | 2,400  | 1,800  |        |
| Pegatron   | 775    | 775     | 875    | 825     | 575    | 625    | 750    | 610    | 575    | 700    | 610    | 775    | 990    | 750    | 825    | 825     | 775    | 710    | 925    | 700    | 425    | 675    | 525    | 575    | 650    | 400    |        |
| Total      | 11,775 | 9,975   | 11,475 | 12,325  | 10,075 | 10,325 | 11,150 | 8,410  | 8,875  | 12,300 | 9,510  | 10,575 | 12,990 | 10,750 | 11,225 | 12,025  | 10,075 | 10,410 | 11,825 | 8,400  | 7,025  | 13,775 | 9,125  | 9,275  | 11,950 | 8,200  |        |
| M/M Growth |        |         |        |         |        |        |        |        |        |        |        |        |        |        |        |         |        |        |        |        |        |        |        |        |        |        |        |
| Quanta     | 13%    | -24%    | 26%    | 14%     | -35%   | 13%    | 19%    | -35%   | 11%    | 58%    | -33%   | 15%    | 32%    | -20%   | 2%     | 12%     | -24%   | 0%     | 11%    | -31%   | -26%   | 165%   | -34%   | 0%     | 29%    | -38%   | 11%    |
| Wistron    | 6%     | -11%    | 6%     | 12%     | -5%    | 6%     | 6%     | -17%   | 0%     | 27%    | -11%   | 6%     | 33%    | -17%   | 10%    | 0%      | -9%    | 5%     | 18%    | -35%   | -8%    | 75%    | -36%   | -6%    | 18%    | -20%   | 0%     |
| Others     | 0%     | 0%      | 0%     | 6%      | -6%    | 0%     | 18%    | -25%   | 7%     | 19%    | -11%   | 6%     | 17%    | -19%   | 6%     | 6%      | -16%   | 6%     | 18%    | -15%   | -12%   | 47%    | -32%   | 0%     | 60%    | -33%   | 0%     |
| Compal     | 3%     | -17%    | 16%    | 0%      | -3%    | -4%    | -15%   | -13%   | 5%     | 38%    | -24%   | 9%     | 4%     | -8%    | 0%     | 9%      | -16%   | 10%    | 4%     | -33%   | -4%    | 87%    | -36%   | 11%    | 20%    | -25%   | 17%    |
| Pegatron   | 6%     | 0%      | 13%    | -6%     | -30%   | 9%     | 20%    | -19%   | -4%    | 22%    | -13%   | 27%    | 28%    | -24%   | 10%    | 0%      | -6%    | -8%    | 30%    | -24%   | -39%   | 59%    | -22%   | 10%    | 13%    | -38%   | 50%    |
| Total      | 7%     | -15%    | 15%    | 7%      | -18%   | 2%     | 8%     | -25%   | 6%     | 39%    | -23%   | 11%    | 23%    | -17%   | 4%     | 7%      | -16%   | 3%     | 14%    | -29%   | -16%   | 96%    | -34%   | 2%     | 29%    | -31%   | 10%    |
| Y/Y Growth |        |         |        |         |        |        |        |        |        |        |        |        |        |        |        |         |        |        |        |        |        |        |        |        |        |        |        |
| Quanta     | -12%   | -11%    | -7%    | 4%      | 10%    | -5%    | 16%    | -3%    | 7%     | 4%     | 3%     | -5%    | 11%    | 18%    | -5%    | -6%     | 9%     | -3%    | -9%    | -4%    | -35%   | 8%     | 6%     | -8%    | -10%   | -30%   | -24%   |
| Wistron    | 6%     | 7%      | 0%     | -5%     | 6%     | -6%    | 0%     | 15%    | 7%     | 0%     | 6%     | 33%    | 25%    | 29%    | 16%    | 17%     | 29%    | 44%    | 13%    | 7%     | 47%    | 6%     | -6%    | -17%   | -20%   | -27%   |        |
| Others     | 0%     | 13%     | 6%     | 0%      | 21%    | 6%     | 33%    | 1%     | 23%    | 12%    | 13%    | 6%     | 24%    | 0%     | 6%     | 6%      | -6%    | 0%     | 0%     | 13%    | -6%    | 16%    | -12%   | -17%   | 14%    | -6%    | -11%   |
| Compal     | -9%    | -24%    | 0%     | -6%     | -3%    | -7%    | -8%    | -9%    | -9%    | -3%    | -21%   | -17%   | -17%   | -8%    | -21%   | -14%    | -25%   | -15%   | 4%     | -20%   | -29%   | -3%    | -18%   | -17%   | -4%    | -22%   | -9%    |
| Pegatron   | -6%    | 7%      | -3%    | -6%     | 0%     | 9%     | 50%    | 14%    | 53%    | 8%     | 16%    | 6%     | 28%    | -3%    | -6%    | 0%      | 35%    | 14%    | 23%    | 15%    | -6%    | -4%    | -14%   | -26%   | -34%   | -47%   | -27%   |
| Total      | -7%    | -8%     | -2%    | -1%     | 6%     | -3%    | 12%    | 0%     | 7%     | 3%     | -1%    | -4%    | 10%    | 8%     | -2%    | -2%     | 0%     | 1%     | 6%     | 0%     | -21%   | 12%    | -4%    | -12%   | -8%    | -24%   | -20%   |

Source: Company data, Morgan Stanley Research

**Exhibit 54:** UPDATED: Our Greater China Tech Hardware research colleagues expect NB builds at all of the ODMs to decline Y/Y in August.

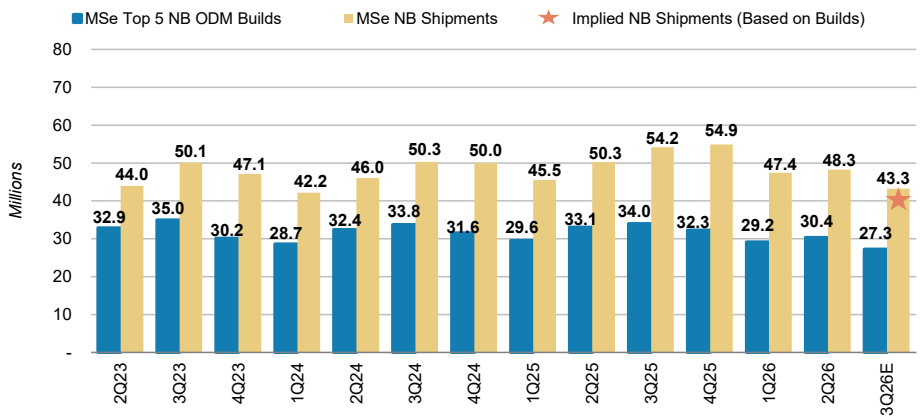
Monthly Top 5 Notebook ODM Builds (M)



Source: Company data, Morgan Stanley Research

**Exhibit 55:** UPDATED: Our Greater China Technology Hardware team now forecasts 27.3M NB ODM builds in C3Q26.

Quarterly Top 5 Notebook (NB) ODM Builds vs. MS NB Shipment Forecast



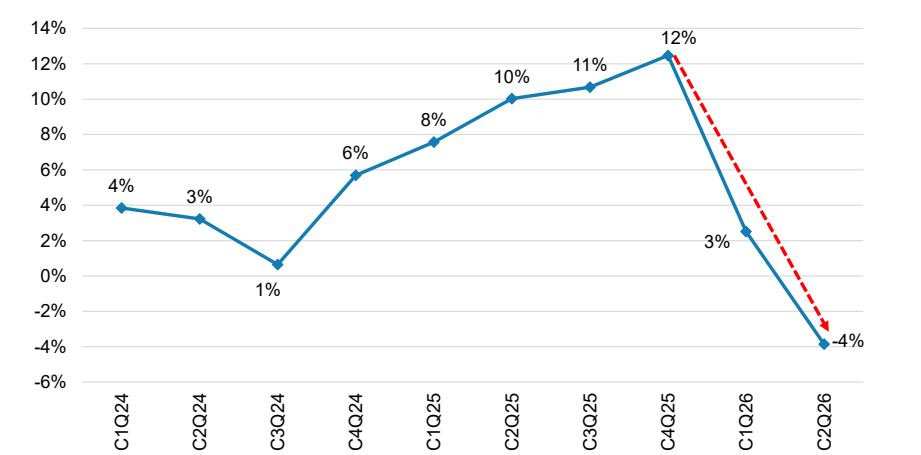
Source: Company data, Morgan Stanley Research estimates

## Quarterly PC Shipments & Pricing

- Using data from IDC, we track the global PC shipments, revenue, ASPs and vendor market share gains/losses on a quarterly basis.

**Exhibit 56:** PC shipments declined 4% Y/Y in C2Q, the first quarter of Y/Y declines in 2+ years.

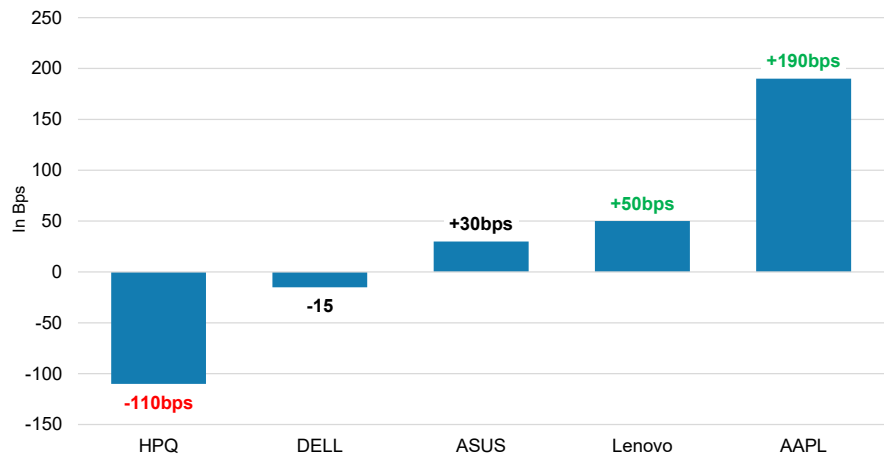
Y/Y PC Market Shipments



Source: IDC, Morgan Stanley Research

**Exhibit 57:** HPQ was the greatest share loser in the June quarter while AAPL was the greatest share gainer Y/Y.

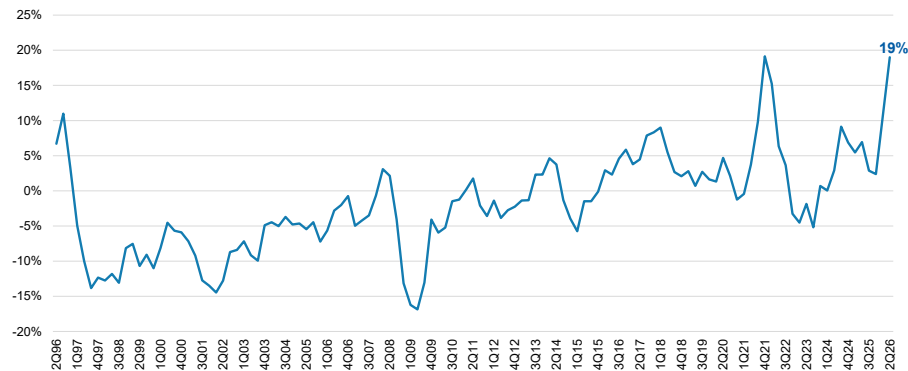
C2Q26 Y/Y PC Unit Market Share Gain/Loss



Source: IDC, Morgan Stanley Research

**Exhibit 58:** PC market ASPs increased 19% Y/Y in the June quarter, an acceleration from +11% Y/Y in the March quarter.

PC Market ASP Y/Y Growth



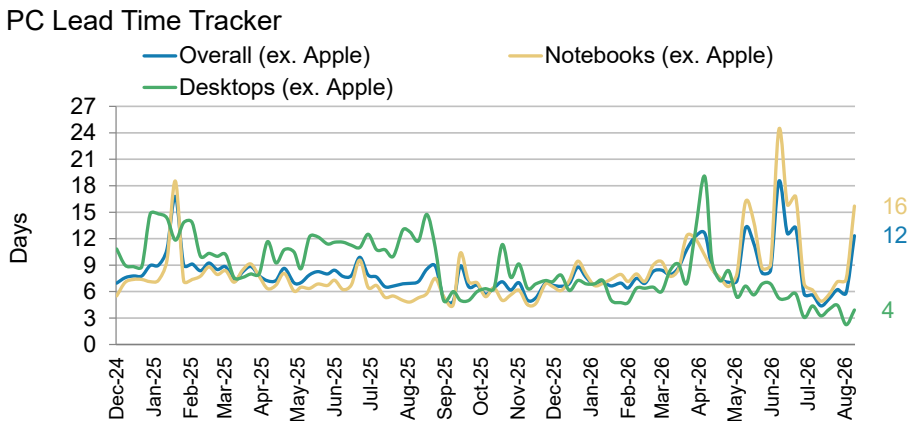
Source: IDC, Morgan Stanley Research



## Weekly PC Lead Time Tracker

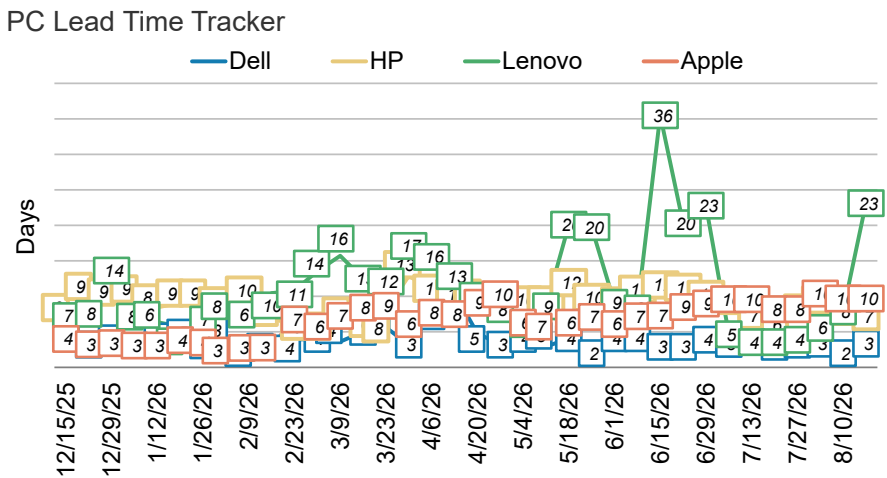
- Our proprietary PC Lead Time tracker collects data from leading PC vendors such as Dell, HP, Lenovo, and Apple. The tracker reflects the number of days between the order date and estimated delivery date for 40 different PC SKUs.

**Exhibit 59:** UPDATED: Overall lead times ex. Apple increased 8 days over the past month (12 days on average as of August 17th), with desktop lead times at 4 days and notebook lead times at 16 days



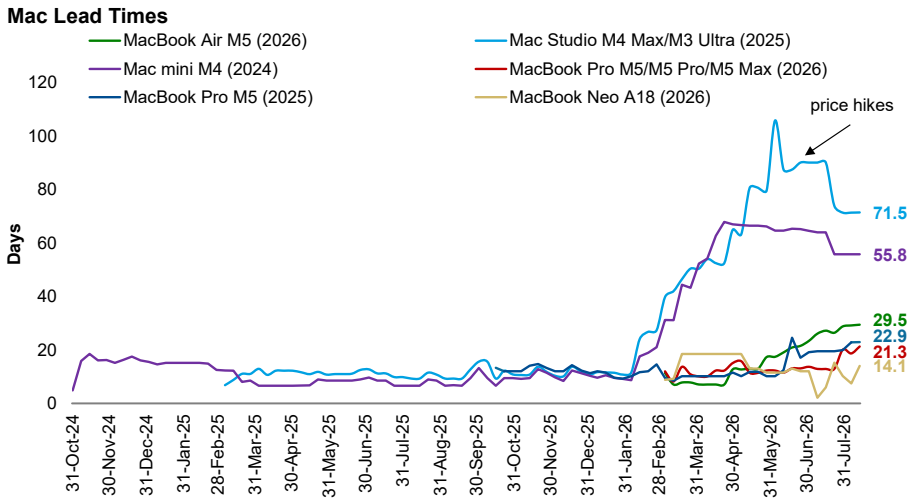
Source: Company websites, Morgan Stanley Research

**Exhibit 60:** UPDATED: Lead times for Lenovo, Apple and HP increased 20, 2 and 1 days compared to last month, respectively. While lead times for Dell remained unchanged during the same period (as of August 17th)



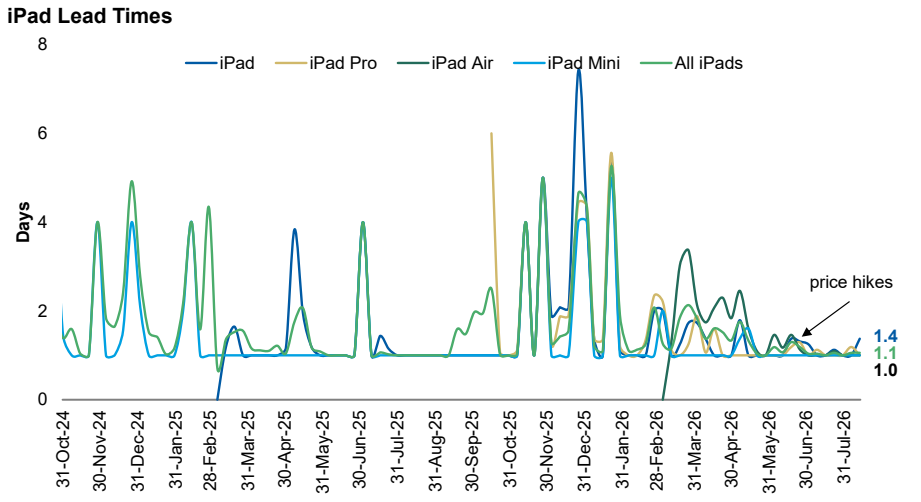
Source: Company websites, Morgan Stanley Research

**Exhibit 61:** UPDATED: Lead times for Apple Mac products had been falling from near-record highs, but the Mac Studio M4 Max/M3 Ultra (2025) is currently at ~71 days, followed by the Mac Mini M4 (2024) at ~56 days



Source: Company websites, Morgan Stanley Research

**Exhibit 62:** UPDATED: Lead times average for all Apple iPad products is currently at 1.1 days, with the average iPad lead times being the highest at 1.4 days, while the rest of the models have a current lead time average of 1 day



Source: Company websites, Morgan Stanley Research

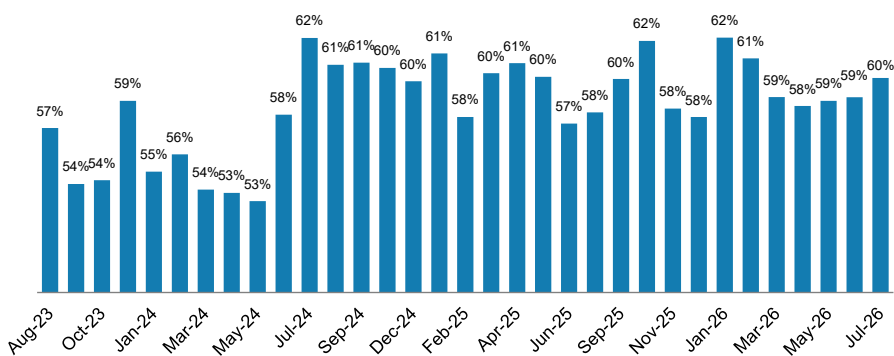
## AlphaWise Pulse Survey – Computers & Related Accessories

- AlphaWise, Morgan Stanley's longstanding evidence-based research capability, completed its 78th survey of ~2,000 US consumers on July 23 - July 27 ([AlphaWise US Consumer Pulse Survey Wave 78](#)).
- We focus on current spending (now vs. last month) and future spending intentions (next month vs. now) for computers and related accessories.

*Overall, are you spending more money, less or the same now compared to a month ago on computers & related accessories (monitor, laptop, keyboard, mouse, etc.)?*

**Exhibit 63:** UPDATED: Our AlphaWise survey data show that current consumer computer and related accessories purchasing intentions upticked 1 point vs. the last survey at 60%

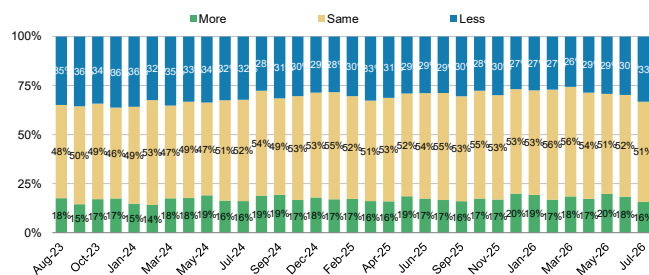
% of Respondents That Purchased Computers & Related Access. (Now vs. Month Ago)



Source: AlphaWise, Morgan Stanley Research; Note: x-axis corresponds to the days each survey was conducted

**Exhibit 64:** UPDATED: The latest survey shows a 2-points decrease in the % of respondents who spent more on computers & related accessories compared to last month, while the % who spent less increased 3 points

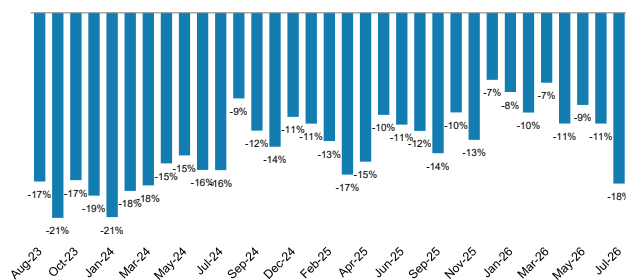
Computer & Related Access. Spending (Now vs. Month Ago)



Source: AlphaWise, Morgan Stanley Research. Reflects % on respondents that are purchasing; Note: x-axis corresponds to the days each survey was conducted

**Exhibit 65:** UPDATED: As a result, net spending intentions (today vs. last month) decreased to -18%

Computer & Related Access. Net Spending, More Minus Less (Now vs. Month Ago)

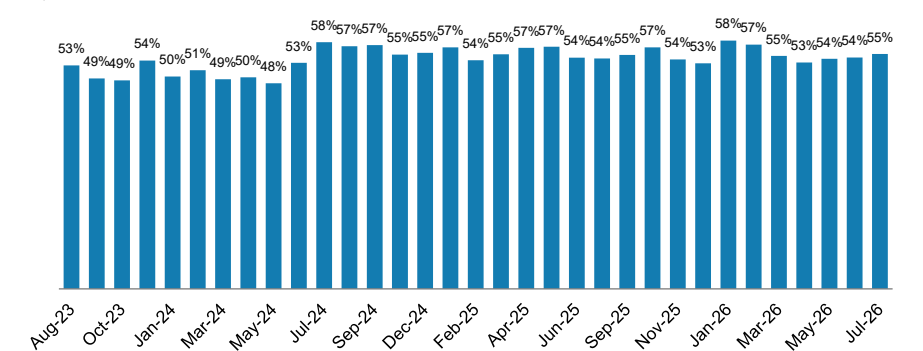


Source: AlphaWise, Morgan Stanley Research. NET: Percentage of consumers who said "More" – "Less"; Note: x-axis corresponds to the days each survey was conducted

*Over the next month, do you expect to spend more money, less or the same on computers & related accessories (monitor, laptop, keyboard, mouse, etc.)?*

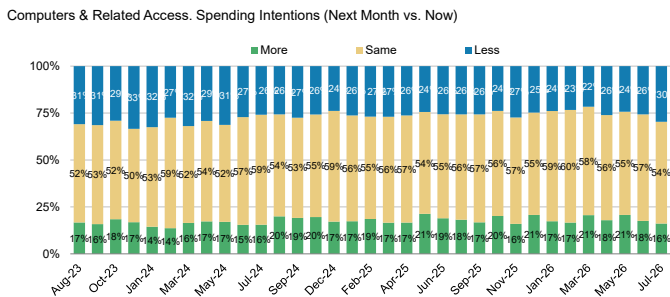
**Exhibit 66:** UPDATED: Forward-looking computer and related accessories spending intentions upticked 1 point M/M to 55%

% of Respondents That Will Purchase Computers & Related Access. (Next Month vs. Now)



Source: AlphaWise, Morgan Stanley Research; Note: x-axis corresponds to the days each survey was conducted

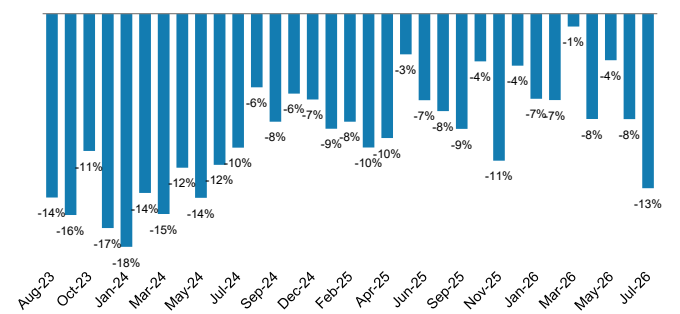
**Exhibit 67:** UPDATED: The latest survey shows that the % of respondents expecting to spend less on computers and related accessories next month relative to this month increased 4 points M/M, and the % expecting to spend more downticked 2 points...



Source: AlphaWise, Morgan Stanley Research. Reflects % on respondents that are purchasing; Note: x-axis corresponds to the days each survey was conducted

**Exhibit 68:** UPDATED: ... as a result, net future computer and related accessories spending intentions downticked at -13%

Computer & Related Access. Net Spending Intentions, More Minus Less (Next Month vs. Now)



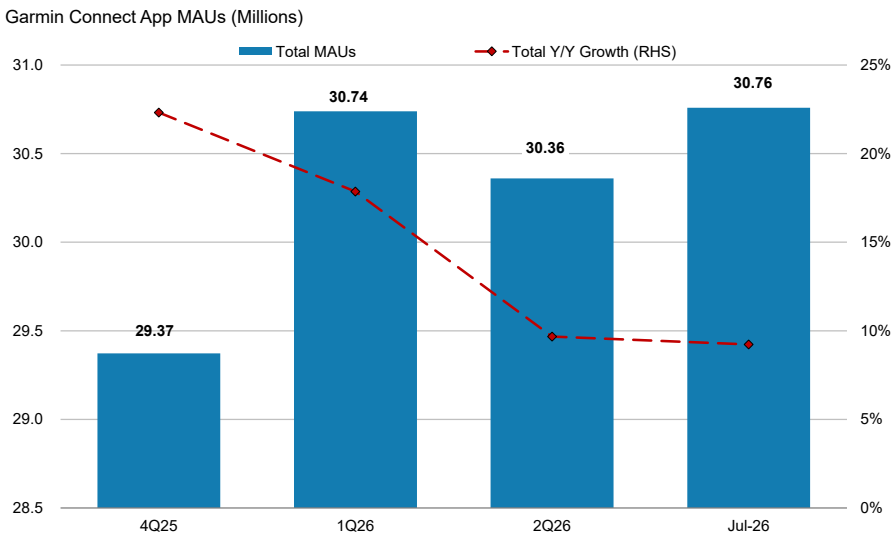
Source: AlphaWise, Morgan Stanley Research. NET: Percentage of consumers who said "More" – "Less"; Note: x-axis corresponds to the days each survey was conducted

# Consumer Electronics

## Monthly Active User App Data

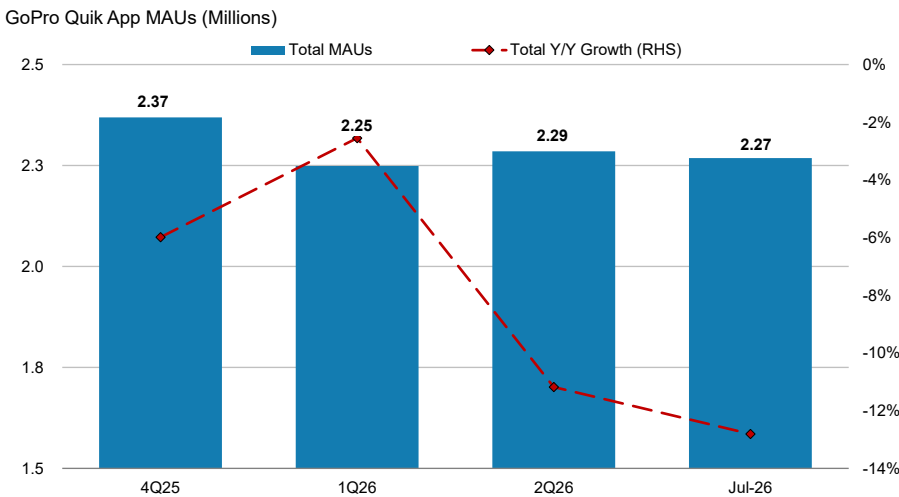
Using data from Sensor Tower, we track the total number of Monthly Active Users (MAUs) for each of 5 apps: 1) Garmin Connect, 2) GoPro Quik, 3) Sonos Controller, and 4) Cricut Design Space.

**Exhibit 69:** UPDATED: Sensor Tower estimates that the Garmin Connect app had 30.8M MAUs at the end of July, up 9% Y/Y



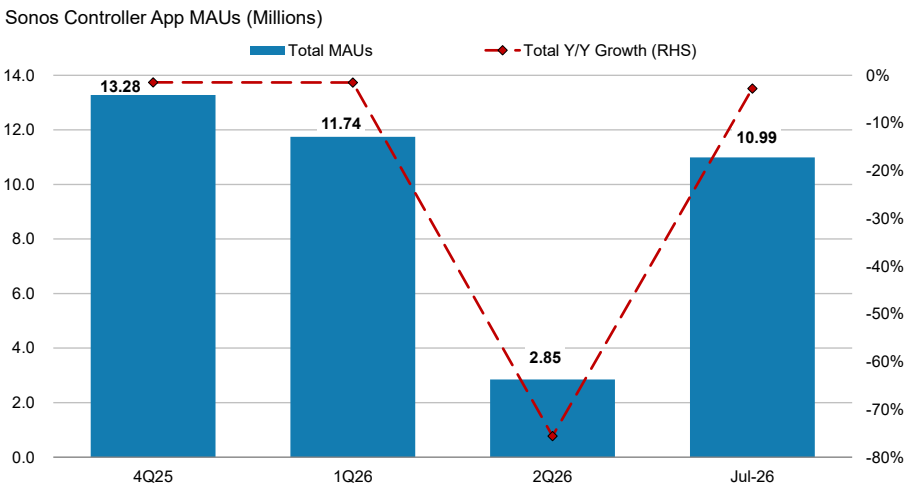
Source: Sensor Tower, Morgan Stanley Research

**Exhibit 70:** UPDATED: Sensor Tower estimates that the GoPro Quik app had 2.3M MAUs at the end of July, down 13% Y/Y



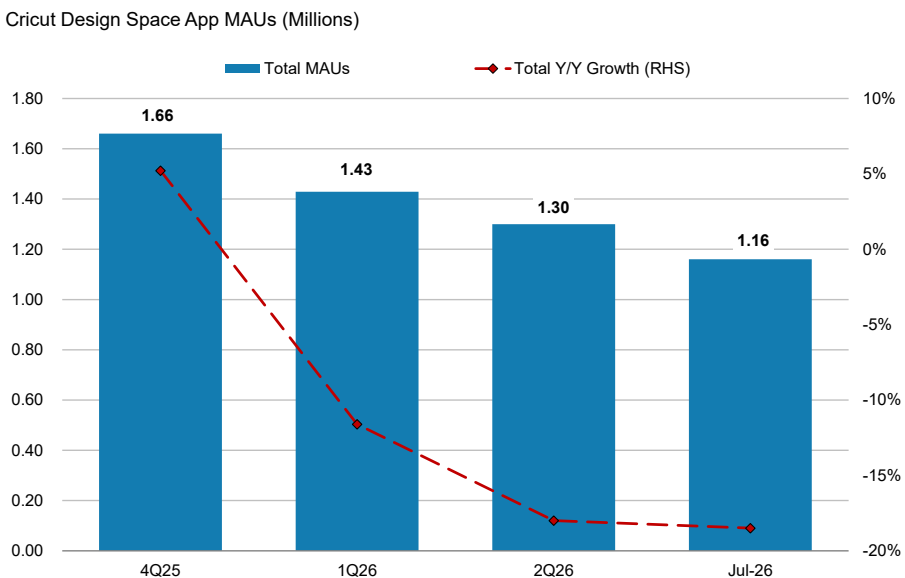
Source: Sensor Tower, Morgan Stanley Research

**Exhibit 71:** UPDATED: Sensor Tower estimates that the Sonos Controller apps had 11M MAUs at the end of July, down 3% Y/Y



Source: Sensor Tower, Morgan Stanley Research

**Exhibit 72:** UPDATED: Sensor Tower estimates that the Cricut Design Space app had 1.2M MAUs at the end of July, down 19% Y/Y

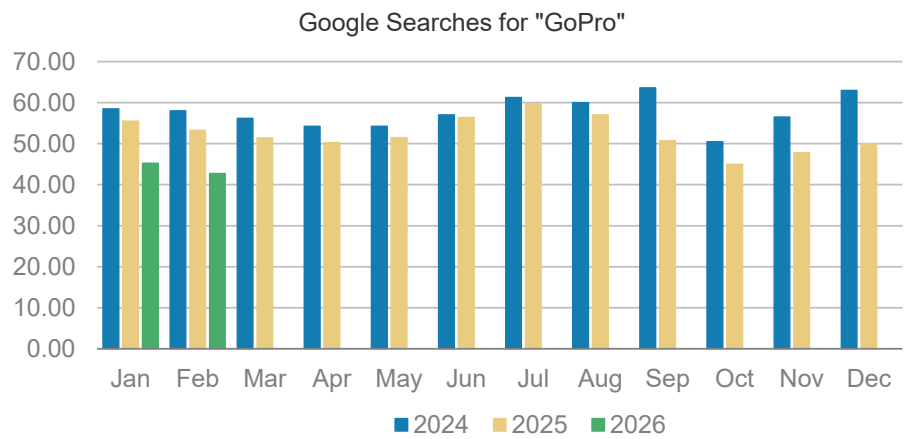


Source: Sensor Tower, Morgan Stanley Research

## AlphaWise Consumer Electronics Demand Trackers

- l In partnership with our AlphaWise colleagues, we track the intensity of global Google searches for a number of consumer electronics products, and analyze how those data impact our view on unit shipments, on a quarterly basis.

**Exhibit 73:** According to Google Trends data, search intensity for "GoPro" was down 20% Y/Y in February 2026

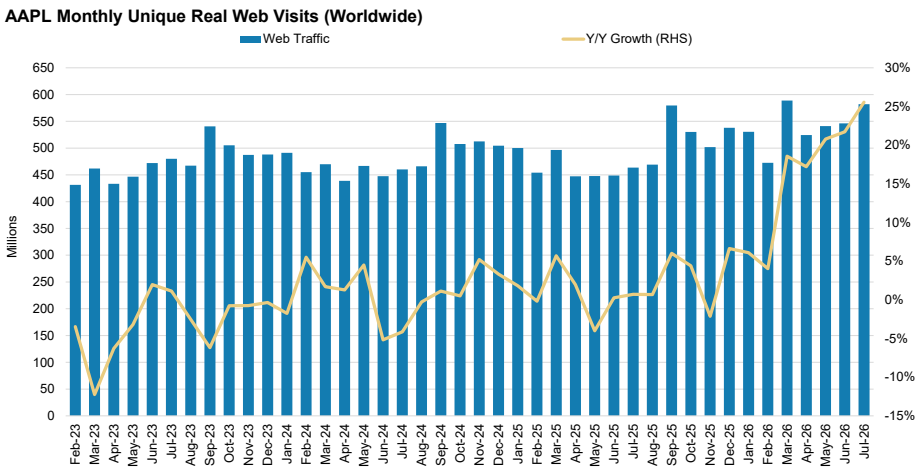


Source: AlphaWise, Morgan Stanley Research; Google

Consumer Electronics Web Traffic Data

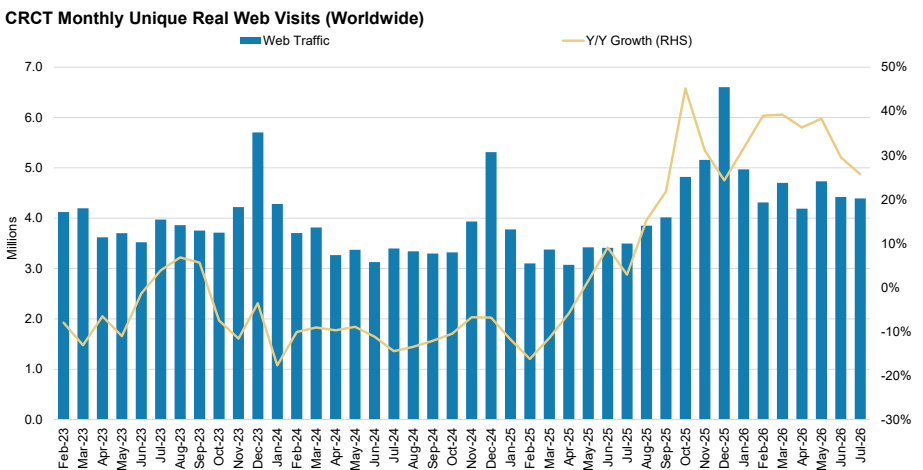
Using data from Similarweb, we track the total number of monthly unique website visits for 1) AAPL, 2) CRCT, 3) GPRO, 4) GRMN, 5) LOGI, and 6) SONO.

**Exhibit 74:** UPDATED: According to Similarweb web traffic data, unique website visits for Apple.com were up 26% Y/Y in the month of July.



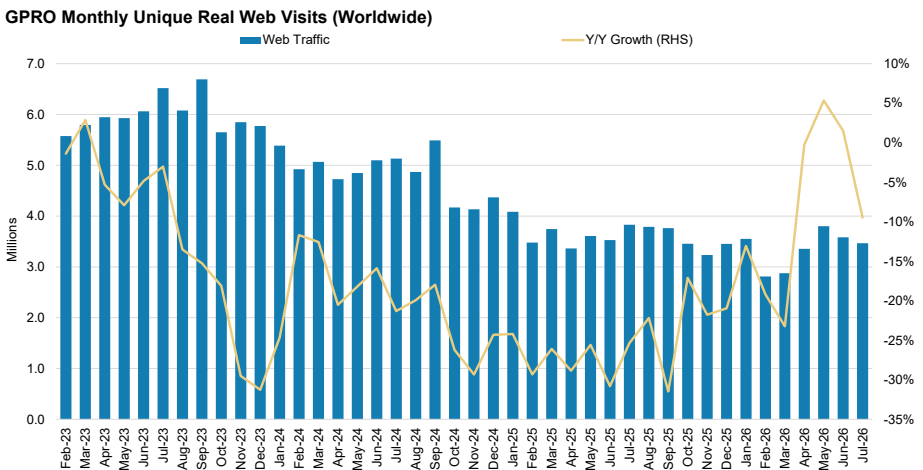
Source: Similarweb, Morgan Stanley Research

**Exhibit 75:** UPDATED: According to Similarweb web traffic data, unique website visits for Cricut.com grew 26% Y/Y in the month of July.



Source: Similarweb, Morgan Stanley Research

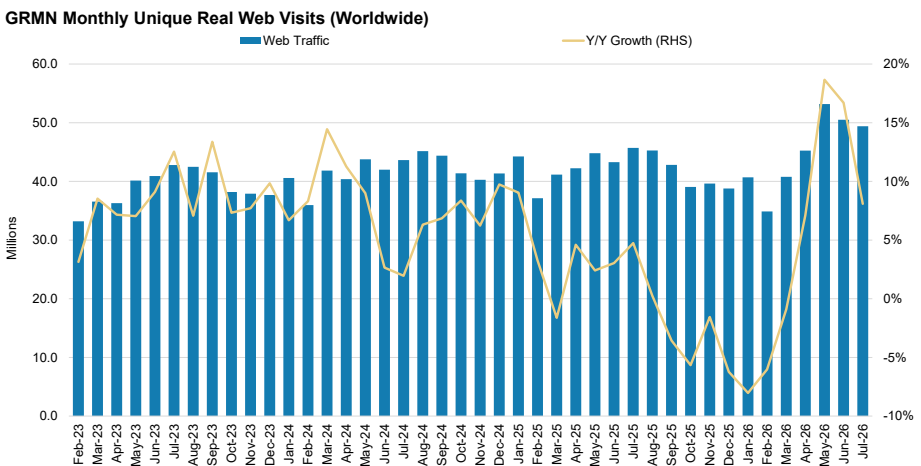
**Exhibit 76:** UPDATED: According to Similarweb web traffic data, unique website visits for Gopro.com were down 10% Y/Y in the month of July.



Source: Similarweb, Morgan Stanley Research

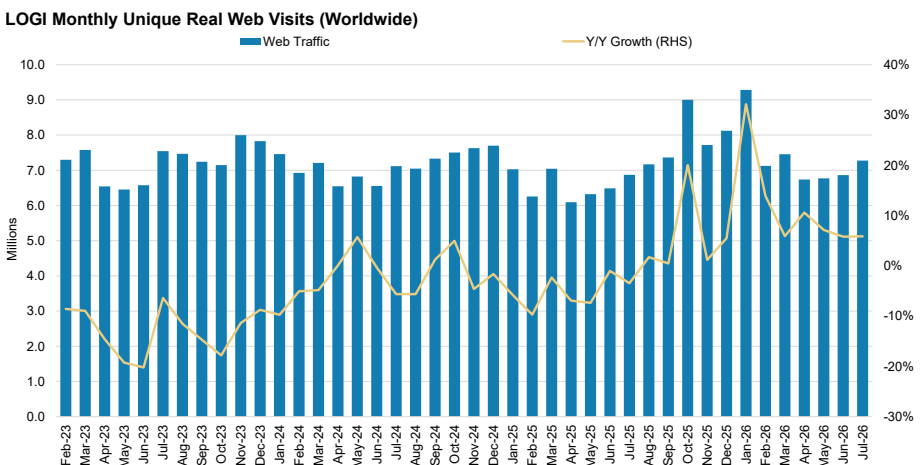


**Exhibit 77:** UPDATED: According to Similarweb web traffic data, unique website visits for Garmin.com were up 8% Y/Y in the month of July.



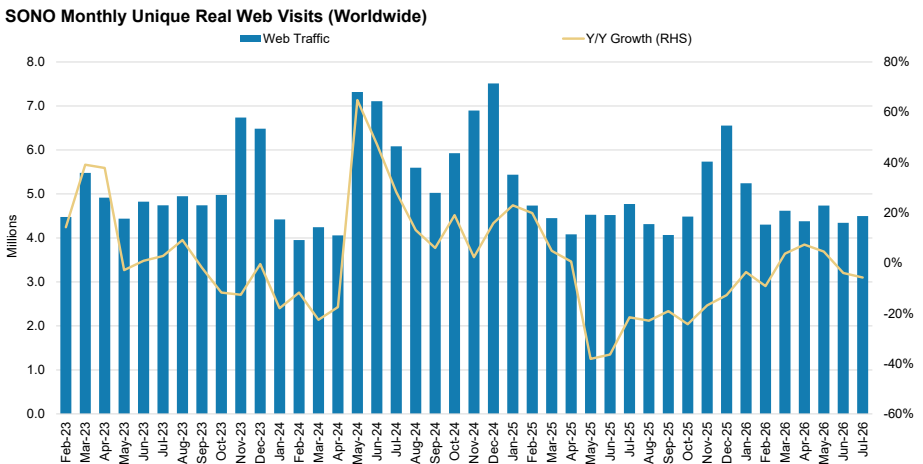
Source: Similarweb, Morgan Stanley Research

**Exhibit 78:** UPDATED: According to Similarweb web traffic data, unique website visits for Logitech.com were up 6% Y/Y in the month of July.



Source: Similarweb, Morgan Stanley Research

**Exhibit 79:** UPDATED: According to Similarweb web traffic data, unique website visits for Sonos.com were down 6% Y/Y in the month of July.



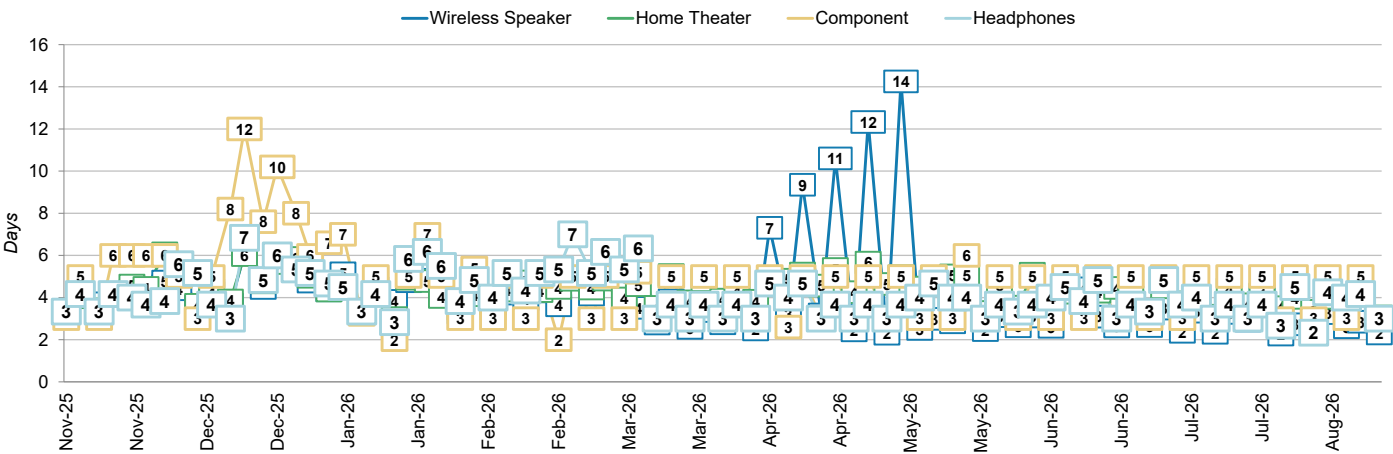
Source: Similarweb, Morgan Stanley Research

### Lead Time Trackers

- Our proprietary Sonos, GoPro and Logitech lead time trackers collect data points on lead times for various products from direct and indirect channels.

**Exhibit 80:** UPDATED: Sonos Wireless Speakers and Home Theater lead times are down 1 day M/M. While Component and Headphones lead times remained unchanged during the same time period

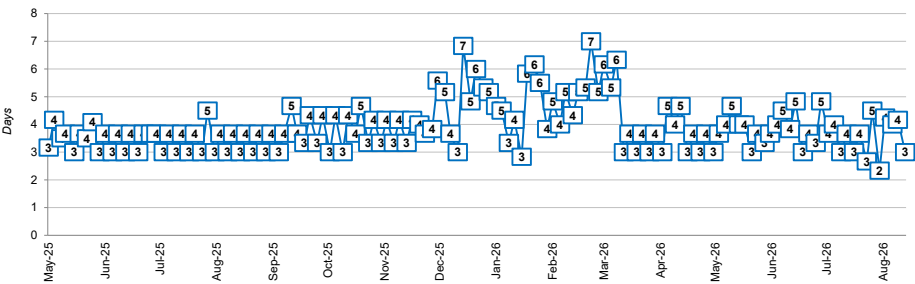
Sonos Lead Time Tracker by Product Category



Source: Sonos.com, Amazon.com, BestBuy.com, Morgan Stanley Research

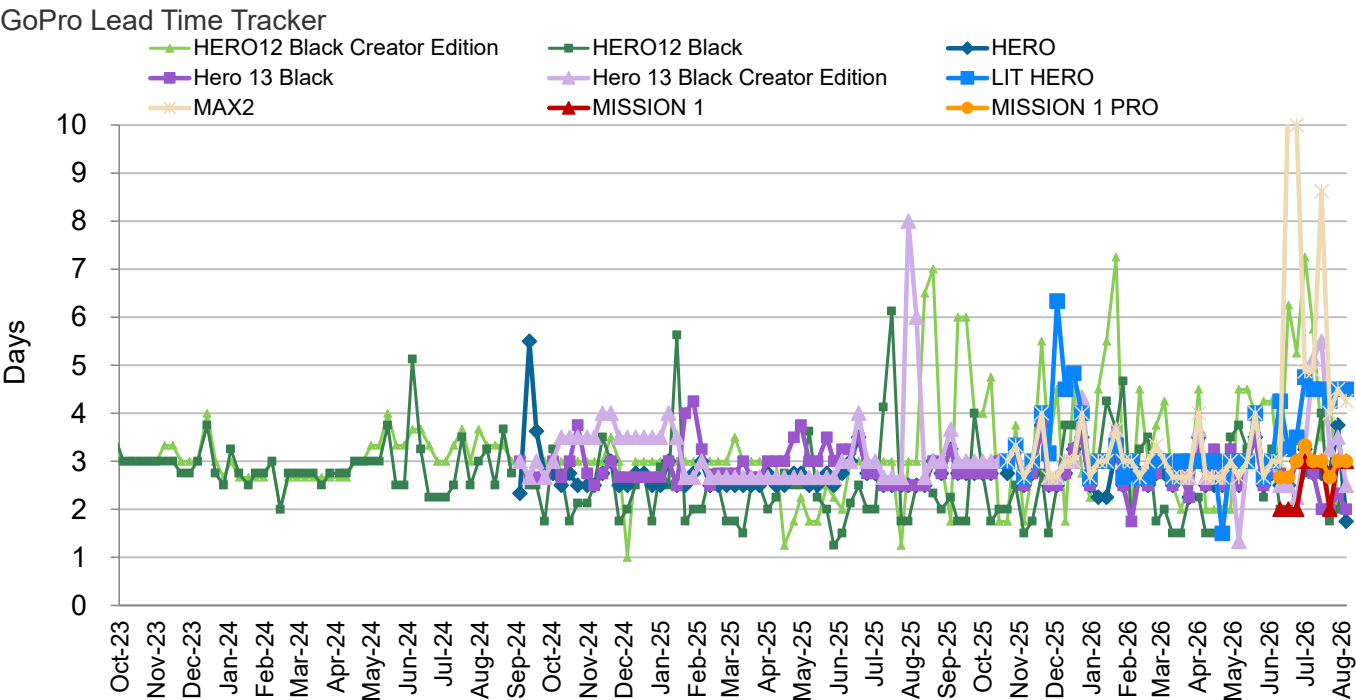
**Exhibit 81:** UPDATED: Sonos Headphones 'Ace' lead times were 3 days on August 18th, following the same 4-3 days average lead times cycle it has maintained since its release

Sonos Headphones 'Ace' Lead Time Tracker



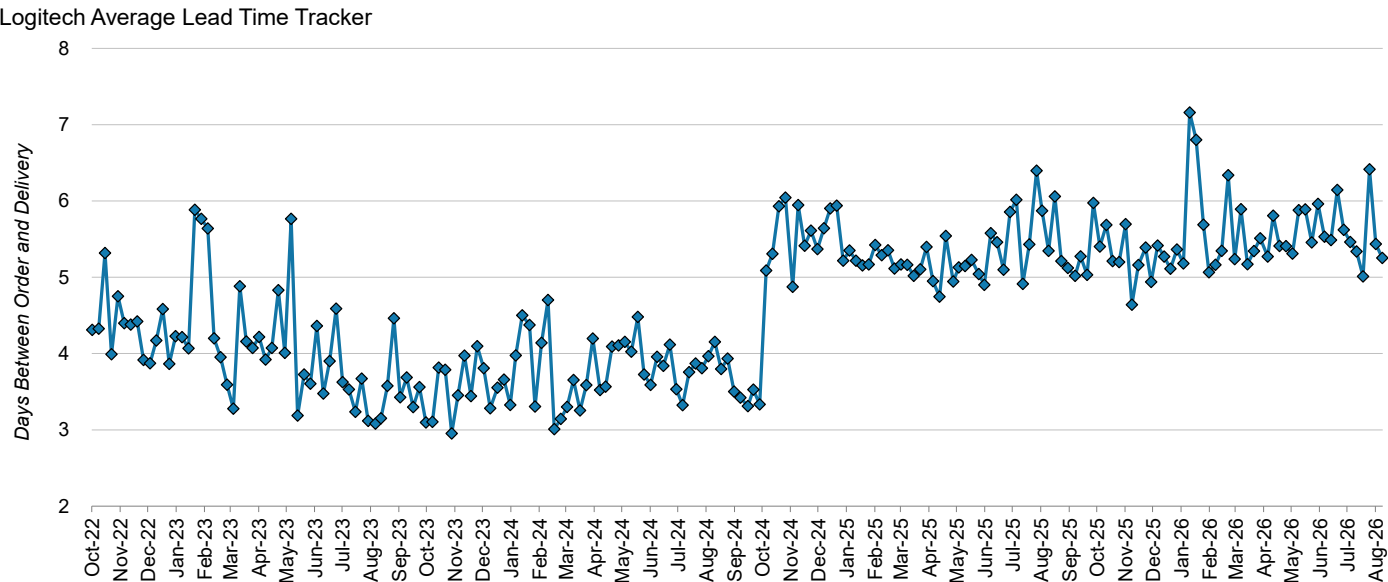
Source: Sonos.com, Amazon.com, BestBuy.com, Morgan Stanley Research

**Exhibit 82:** UPDATED: GoPro average lead times are ~3.3 days as of August 17th, down from ~4.4 days a month ago



Source: GoPro.com, BestBuy.com, Amazon.com, Target.com, Morgan Stanley Research

**Exhibit 83:** UPDATED: The average lead time for Logitech's PC, tablet, and video conferencing peripherals is ~5.3 days as of August 17th, down from ~6.4 three weeks ago



Source: Logitech.com, Amazon, Best Buy, Morgan Stanley Research

## Morning Consult Data on Sonos, Garmin and Apple

- l We now track high-frequency data sets on user sentiment and purchase consideration metrics in our monthly data tracker, specifically Net Promoter Score and Net Favorability from Morning Consult.

### Morning Consult Definitions

- **Net Promoter Score:** The likelihood of recommending a brand to a friend or a colleague based on survey responses from Morning Consult. The question is asked based on a standard 10 point scale. NPS is calculated by subtracting % of detractors (those who select 0-6) from the % of promoters (those who select 9-10). NPS is asked among users of the brand.
- **Net Favorability:** A net score on favorable/unfavorable impressions among survey respondents. The score is calculated by subtracting the % of respondents who select "somewhat unfavorable" and "very unfavorable" from those who select the top 2 positive choices of "very favorable" and "somewhat favorable".

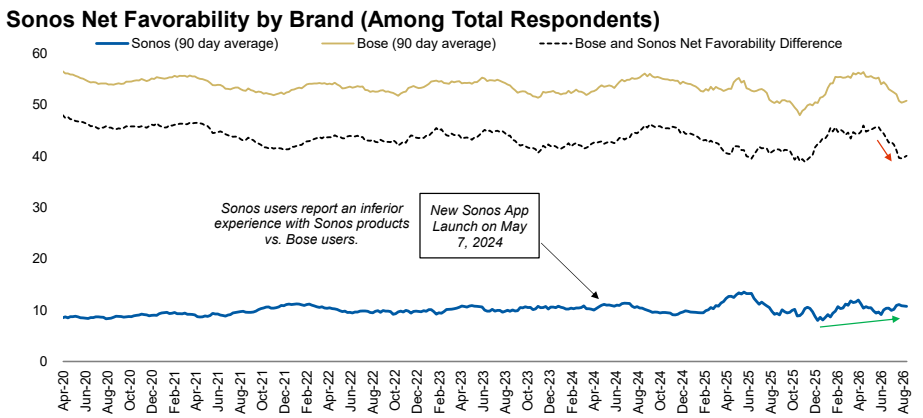
**Exhibit 84:** Sonos' average net promoter score has upticked in the past month, now at 11.6 from 4.9 last month

#### Sonos Net Promoter Score by Brand



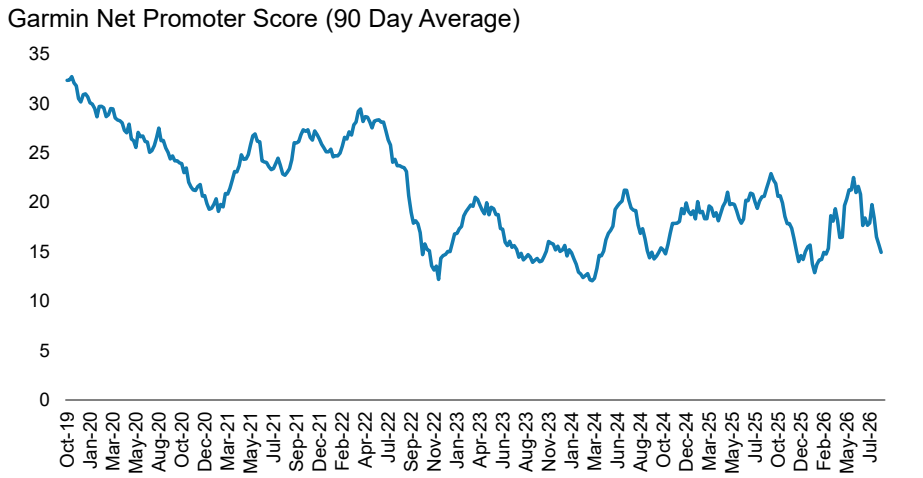
Source: Morning Consult, Morgan Stanley Research, Data as of August 10th.

**Exhibit 85:** Sonos' net favorability moved modestly lower, ticking down to 10.7 from 10.9 last month



Source: Morning Consult, Morgan Stanley Research, Data as of August 10th.

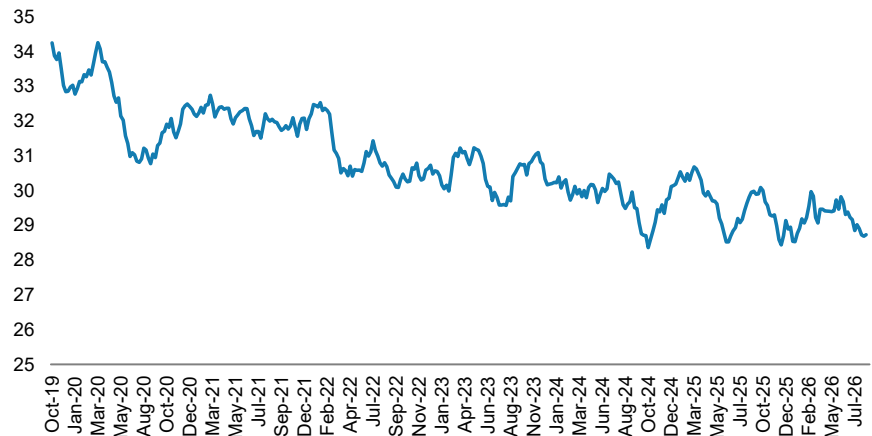
**Exhibit 86:** Garmin NPS has been on the decline over the past 5 years and recently downticked to 14.9 (from 19.8 a month ago)



Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 87:** Garmin Favorability has been on a slight decline over the past 5 years, and downticked in recent weeks to 28.7 (from 29.0 a month ago)

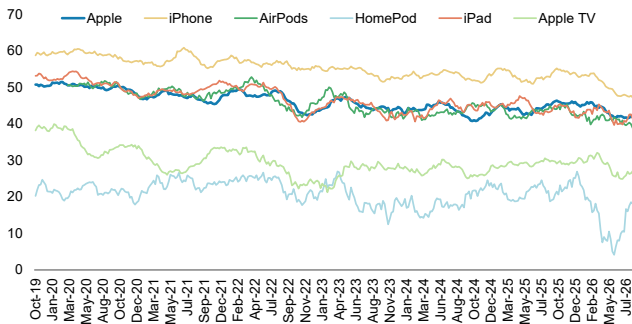
Garmin Net Favorability (90 Day Average)



Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 88:** On both the brand and product levels, Apple's overall NPS has been declining over the past 5 years...

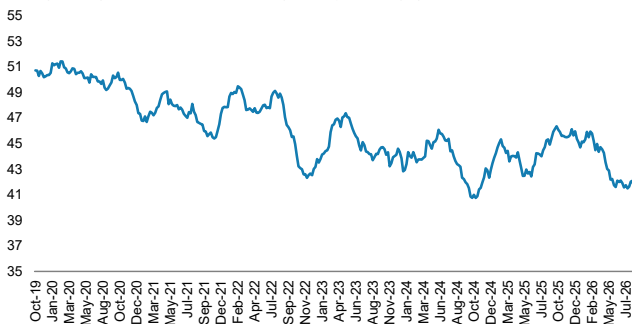
Apple Net Promoter Score By Product (90 day average)



Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 90:** Apple net promoter score has upticked by 1 point to 42.5 from 41.5 a month ago...

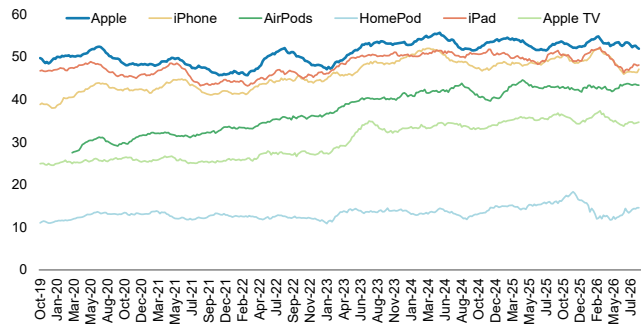
Apple (Brand) Net Promoter Score (90 Day Average)



Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 89:** ... however, Apple's Net favorability has been steadily increasing for all Apple products we track in the past 5 years

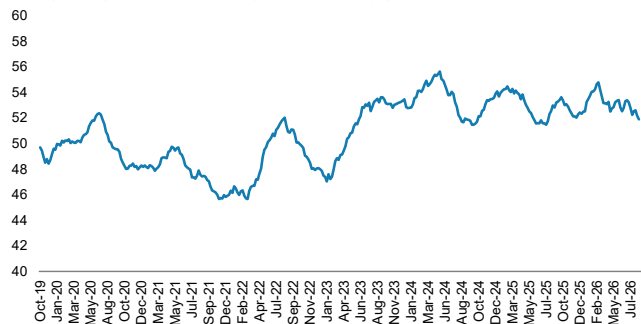
Apple Net Favorability By Product (90 day average)



Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

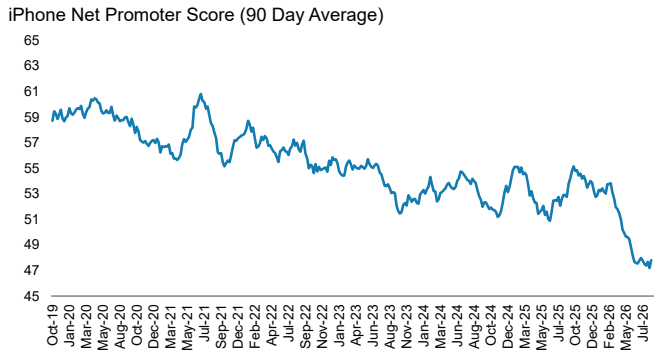
**Exhibit 91:** ... and net favorability slightly decreased to 51.9 from 52.2 a month ago

Apple (Brand) Net Favorability (90 Day Average)



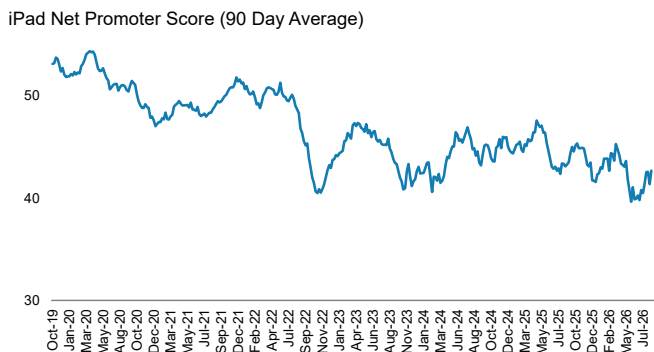
Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 92:** iPhone's NPS has been gradually declining over the past 5 years, but it showed some slight near-term uptick (now at 47.8, up from 47.5 a month ago)



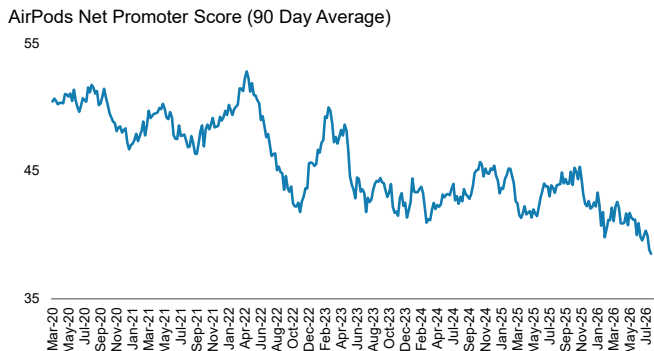
Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 94:** iPad's NPS has been declining over the past 5 years, now at 42.7...



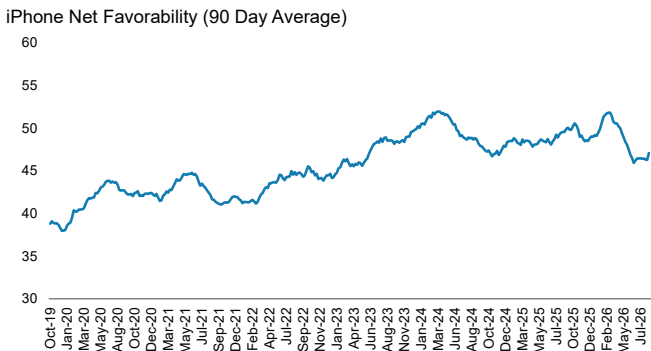
Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 96:** AirPods' NPS has been declining over the past 5 years, and it is now at 38.5...



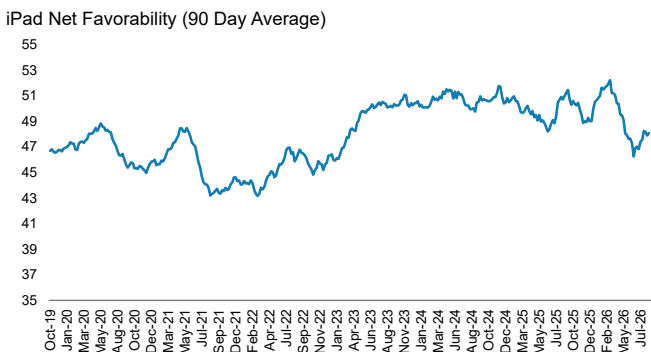
Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 93:** iPhone's Net Favorability has been on a general uptick, and recently showed a slight uptick to 47.1



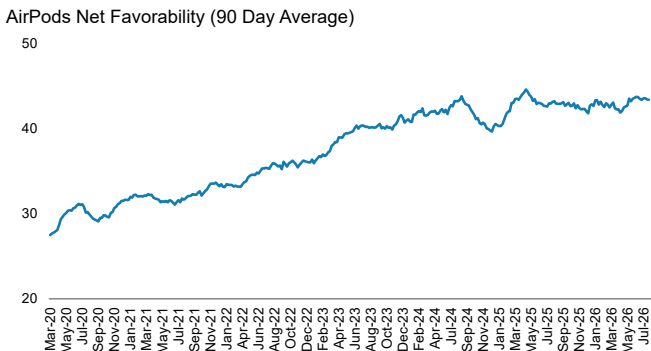
Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 95:** ... while Net Favorability has been on a general uptick, now up to 48.1



Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.

**Exhibit 97:** ... while Net Favorability has been on a general uptick, and it is now at 43.3



Source: Morning Consult, Morgan Stanley Research, Data as of August 17th.



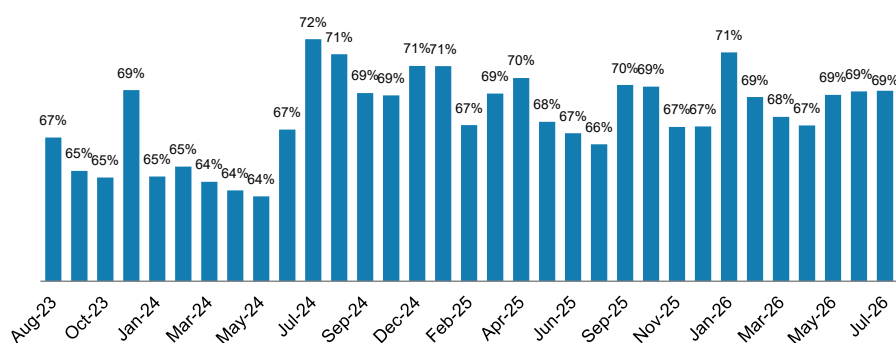
## AlphaWise Pulse Survey – Consumer Electronics

- AlphaWise, Morgan Stanley's longstanding evidence-based research capability, completed its 78th survey of ~2,000 US consumers on July 23 - July 27 ([AlphaWise US Consumer Pulse Survey Wave 78](#)).
- We focus on current spending (now vs. last month) and future spending intentions (next month vs. now) for consumer electronics.

Overall, are you spending more money, less or the same now compared to a month ago on consumer electronics (smartphone, tablet, TV, etc.)?

**Exhibit 98:** UPDATED: Our AlphaWise survey shows that current consumer electronics purchasing intentions remained unchanged from our prior survey at 69%

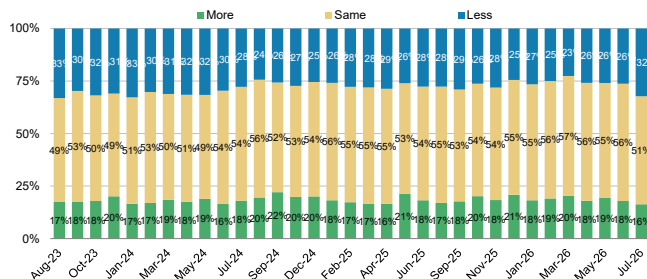
% of Respondents That Purchased Consumer Electronics (Now vs. Month Ago)



Source: AlphaWise, Morgan Stanley Research; Note: x-axis corresponds to the days each survey was conducted

**Exhibit 99:** UPDATED: The latest survey shows the percentage of respondents spending more on consumer electronics this month relative to last month decreased 2 points vs. the prior survey, while the percentage of respondents spending upticked by 6 points compared to last survey

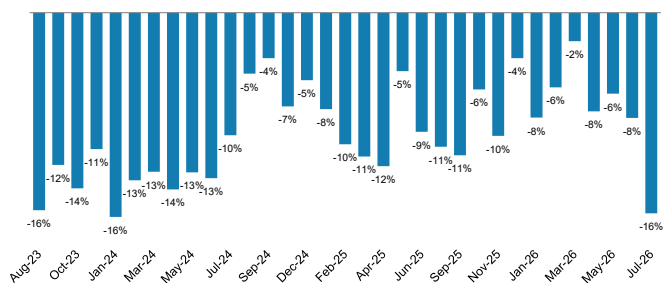
Consumer Electronics Spending (Now vs. Month Ago)



Source: AlphaWise, Morgan Stanley Research. Reflects % on respondents that are purchasing; Note: x-axis corresponds to the days each survey was conducted

**Exhibit 100:** UPDATED: As a result, net spending intentions on consumer electronics this month vs. last month downticked to -16%

Consumer Electronics Net Spending, More Minus Less (Now vs. Month Ago)

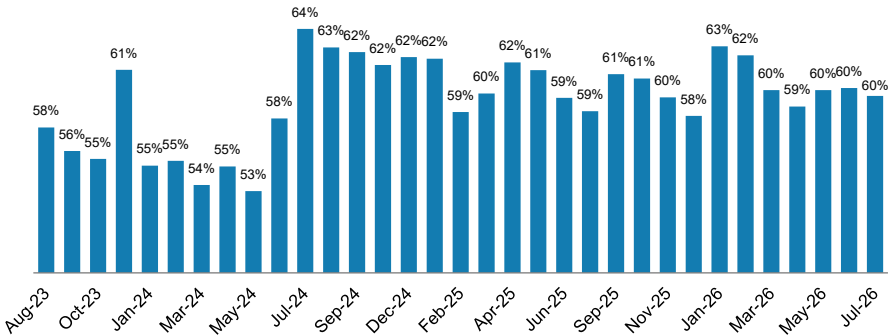


Source: AlphaWise, Morgan Stanley Research. NET: Percentage of consumers who said "More" – "Less"; Note: x-axis corresponds to the days each survey was conducted

Over the next month, do you expect to spend more money, less or the same on consumer electronics (smartphone, tablet, TV, etc.)?

**Exhibit 101:** UPDATED: Forward-looking consumer electronics spending intentions remained unchanged at 60% vs. our prior survey

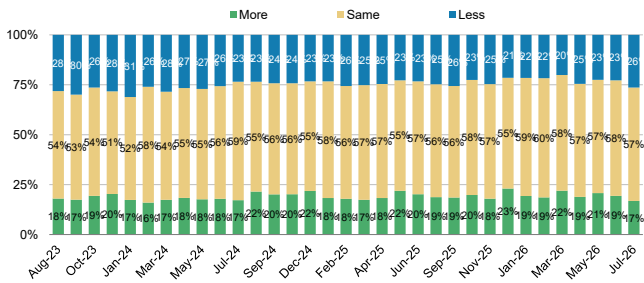
% of Respondents That Will Purchase Consumer Electronics (Next Month vs. Now)



Source: AlphaWise, Morgan Stanley Research; Note: x-axis corresponds to the days each survey was conducted

**Exhibit 102:** UPDATED: The latest survey data show the % of respondents expecting to spend more on consumer electronics next month vs. this month decreased 2 points to 17%, while the % expecting to spend less increased 3 points to 26%...

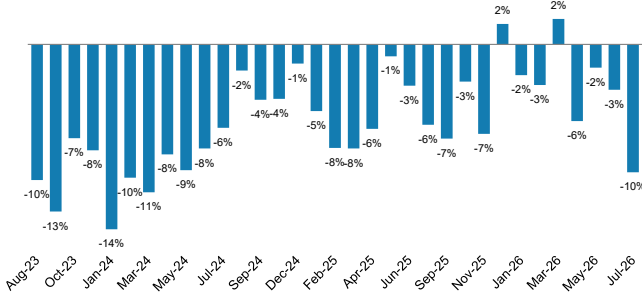
Consumer Electronics Spending Intentions (Next Month vs. Now)



Source: AlphaWise, Morgan Stanley Research. Reflects % on respondents that are purchasing; Note: x-axis corresponds to the days each survey was conducted

**Exhibit 103:** UPDATED: ... as a result, net spending intentions on consumer electronics next month vs. this month downticked to -10%

Consumer Electronics Net Spending Intentions, More Minus Less (Next Month vs. Now)

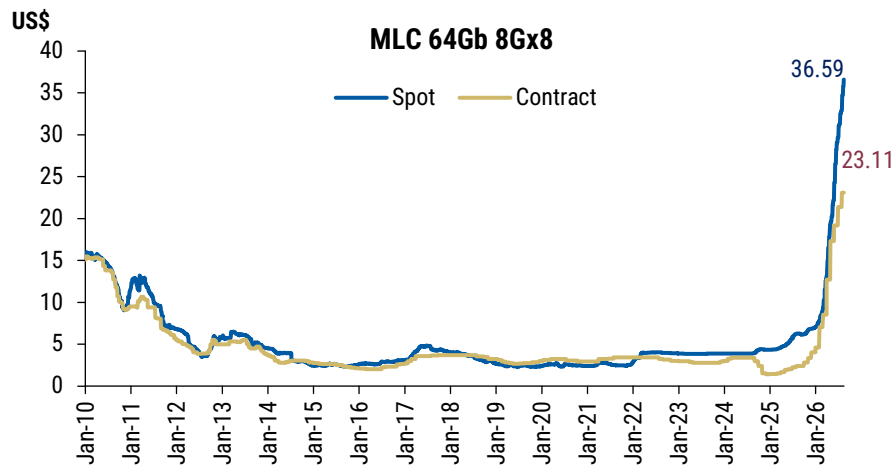


Source: AlphaWise, Morgan Stanley Research. NET: Percentage of consumers who said "More" - "Less"; Note: x-axis corresponds to the days each survey was conducted

## Memory Pricing Forecast

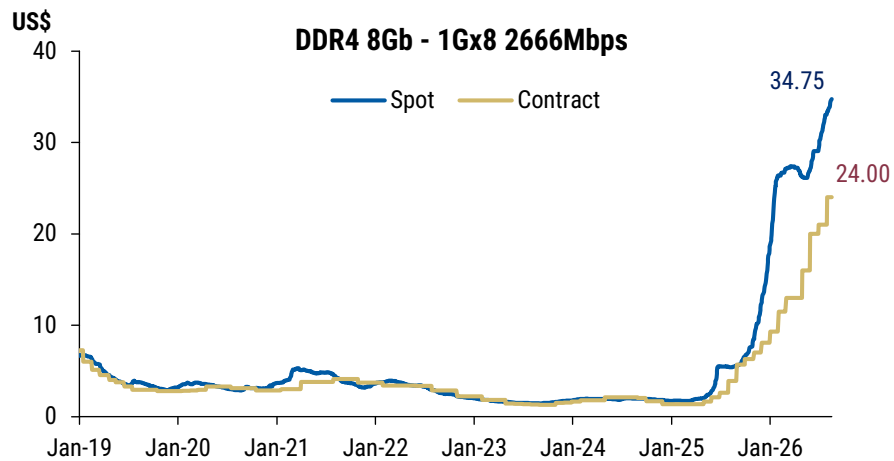
- l DRAMeXchange provides daily DRAM and NAND spot pricing data on a daily basis and TrendForce provides quarterly memory contract pricing forecasts.

**Exhibit 104:** NAND spot and contract prices are up 482% Y/Y and 860% Y/Y, respectively.



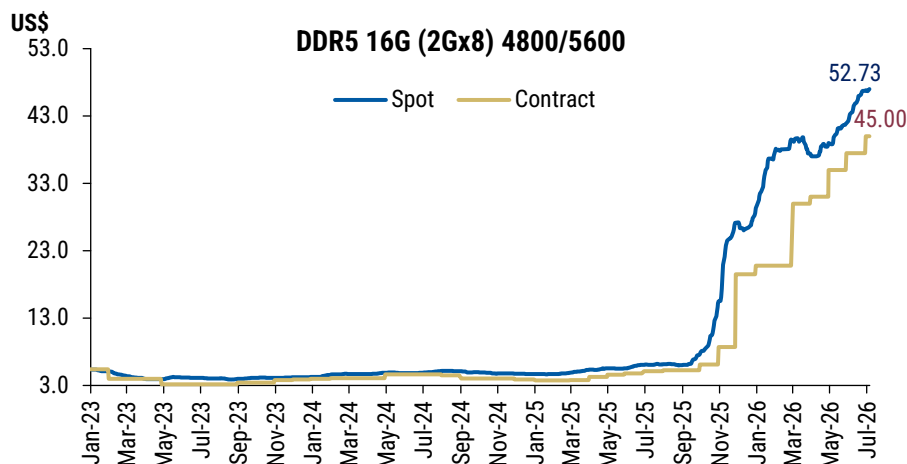
Source: DRAMeXchange, TrendForce, Morgan Stanley Research

**Exhibit 105:** DRAM (DDR4) spot and contract prices are up 536% Y/Y and 515% Y/Y, respectively.



Source: DRAMeXchange, TrendForce, Morgan Stanley Research

**Exhibit 106:** DRAM (DDR5) spot and contract prices are up 755% Y/Y and 757% Y/Y, respectively.



Source: DRAmExchange, TrendForce, Morgan Stanley Research

**Exhibit 107:** Based on the latest forecast from TrendForce, DRAM and NAND contract price estimates have been revised higher vs last month with DRAM and NAND expected to increase another 15-13% Q/Q in C3Q and by another 3-5% Q/Q in C4Q.

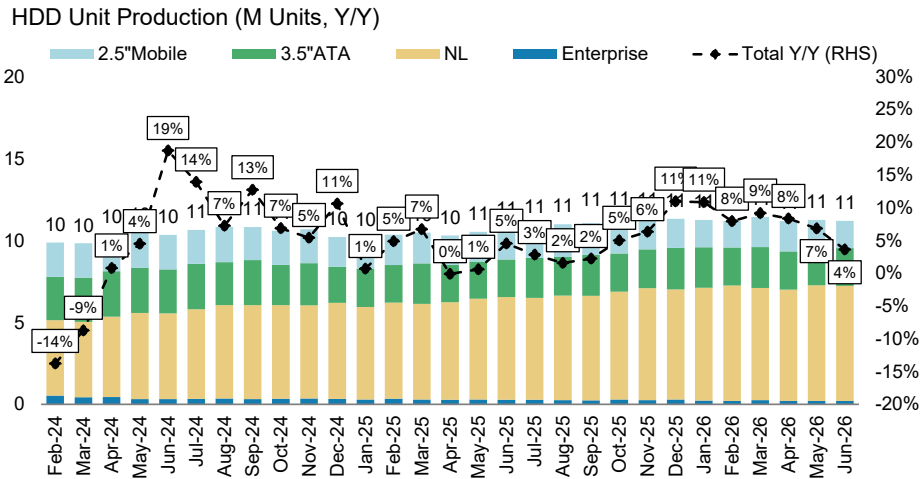
|                            | Mar-25<br>1Q25E - Old                                                        | Mar-25<br>1Q25E                                                              | May-26<br>2Q25E - Old                                                      | Jun-26<br>2Q25E - New                                                      | May-26<br>3Q25E - Old                           | Jun-26<br>3Q25E - New                                                      | May-26<br>4Q25E - Old                       | May-26<br>4Q25E - New                       |
|----------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------|
| PC DRAM                    | DDR4: up 120-125%<br>DDR5: up 110-115%<br>Blended ASP: up 110-115%           | DDR4: up 120-125%<br>DDR5: up 110-115%<br>Blended ASP: up 110-115%           | DDR4: up 35-40%<br>DDR5: up 43-48%<br>Blended ASP: up 43-48%               | DDR4: up 35-40%<br>DDR5: up 45-50%<br>Blended ASP: up 45-50%               | up 3-8%                                         | DDR4: up 15-20%<br>DDR5: up 15-20%<br>Blended ASP: up 15-20%               | up 0-5%                                     | up 3-8%                                     |
| Server DRAM                | DDR4: up 95-100%<br>DDR5: up 93-98%<br>Blended ASP: up 93-98%                | DDR4: up 95-100%<br>DDR5: up 93-98%<br>Blended ASP: up 93-98%                | DDR4: up 50-55%<br>DDR5: up 45-50%<br>Blended ASP: up 45-50%               | DDR4: up 50-55%<br>DDR5: up 53-58%<br>Blended ASP: up 53-58%               | up 10-15%                                       | DDR4: up 13-18%<br>DDR5: up 13-18%<br>Blended ASP: up 13-18%               | up 3-8%                                     | up 3-8%                                     |
| Mobile DRAM                | LPDDR4X: up 88-93%<br>LPDDR5(x): up 88-93%                                   | LPDDR4X: up 58-63%<br>LPDDR5(x): up 58-63%                                   | LPDDR4X: up 70-75%<br>LPDDR5(x): up 78-83%                                 | LPDDR4X: up 73-78%<br>LPDDR5(x): up 80-85%                                 | up 3-8%                                         | LPDDR4X: up 8-13%<br>LPDDR5(x): up 8-13%                                   | up 0-5%                                     | up 0-5%                                     |
| Graphics DRAM              | GDDR6: up 95-100%<br>GDDR7: up 95-100%<br>DDR5: up 75-80%<br>DDR4: up 75-80% | GDDR6: up 95-100%<br>GDDR7: up 95-100%<br>DDR5: up 75-80%<br>DDR4: up 75-80% | GDDR6: up 30-35%<br>GDDR7: up 40-45%<br>DDR5: up 45-50%<br>DDR4: up 45-50% | GDDR6: up 30-35%<br>GDDR7: up 40-45%<br>DDR5: up 55-60%<br>DDR4: up 55-60% | GDDR6: up 3-8%<br>GDDR7: up 10-15%<br>up 10-15% | GDDR6: up 15-20%<br>GDDR7: up 15-20%<br>DDR5: up 15-20%<br>DDR4: up 15-20% | GDDR6: up 0-5%<br>GDDR7: up 3-8%<br>up 3-8% | GDDR6: up 0-5%<br>GDDR7: up 3-8%<br>up 3-8% |
| Consumer DRAM              | Conventional DRAM: up 83-88%<br>HBM Blended: up 83-88%                       | Conventional DRAM: up 83-88%<br>HBM Blended: up 83-88%                       | Conventional DRAM: up 58-63%<br>HBM Blended: 53-58%                        | Conventional DRAM: up 58-63%<br>HBM Blended: 53-58%                        | Conventional DRAM: up 8-13%                     | Conventional DRAM: up 13-18%<br>HBM Blended: 8-13%                         | Conventional DRAM: up 3-8%                  | Conventional DRAM: up 3-8%                  |
| eMMC                       | up 95-100%                                                                   | up 95-100%                                                                   | up 75-80%                                                                  | up 75-80%                                                                  | up 5-10%                                        | up 5-10%                                                                   | mostly flat                                 | mostly flat                                 |
| UFS                        | up 75-80%                                                                    | up 75-80%                                                                    | up 48-53%                                                                  | up 48-53%                                                                  | up 13-18%                                       | up 18-23%                                                                  | up 3-8%                                     | up 3-8%                                     |
| Enterprise SSD             | up 105-110%                                                                  | up 105-110%                                                                  | up 70-75%                                                                  | up 70-75%                                                                  | up 5-10%                                        | up 13-18%                                                                  | mostly flat                                 | mostly flat                                 |
| Client SSD                 | up 75-80%                                                                    | up 75-80%                                                                    | up 28-33%                                                                  | up 28-33%                                                                  | up 5-10%                                        | up 5-10%                                                                   | mostly flat                                 | mostly flat                                 |
| 3D NAND Wafers (TLC & QLC) | up 85-90%                                                                    | up 85-90%                                                                    | up 55-60%                                                                  | up 55-60%                                                                  | up 8-13%                                        | up 10-15%                                                                  | up 0-5%                                     | up 0-5%                                     |
| Total NAND Flash           |                                                                              |                                                                              |                                                                            |                                                                            |                                                 |                                                                            |                                             |                                             |

Source: DRAmExchange, TrendForce, Morgan Stanley Research

### Monthly TSR HDD Production

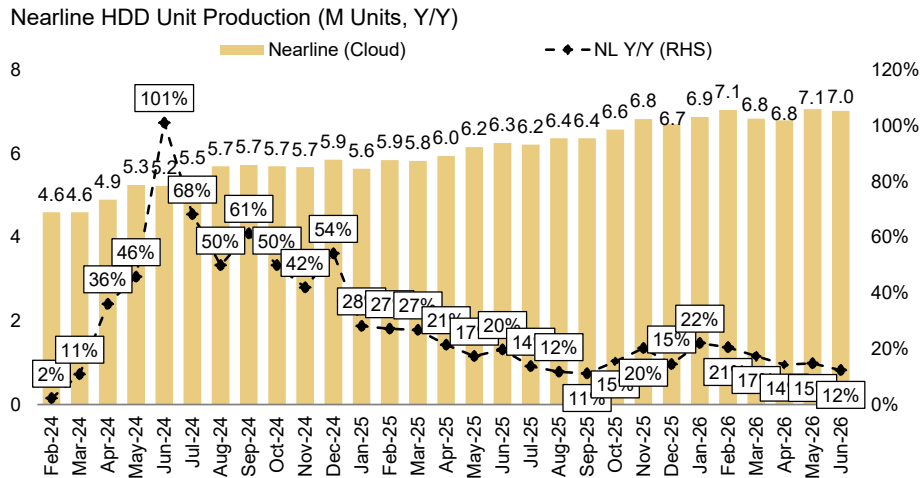
TSR, a data provider located in Japan, provides HDD unit production data on a monthly basis with a one-month lag.

**Exhibit 108:** UPDATED: In June, total HDD production was up 4% Y/Y at 11M units



Source: TSR, Morgan Stanley Research

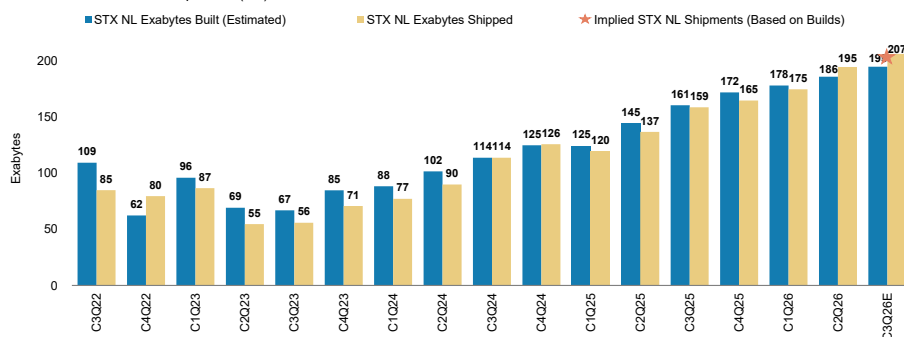
**Exhibit 109:** UPDATED: Nearline HDD unit production growth was +12% Y/Y in June, decelerating 3 points from May



Source: TSR, Morgan Stanley Research

**Exhibit 110:** UPDATED: Our STX nearline EB shipment tracker points to 205.0EB shipped in the September Q, 1% below our estimate of 206.7EB

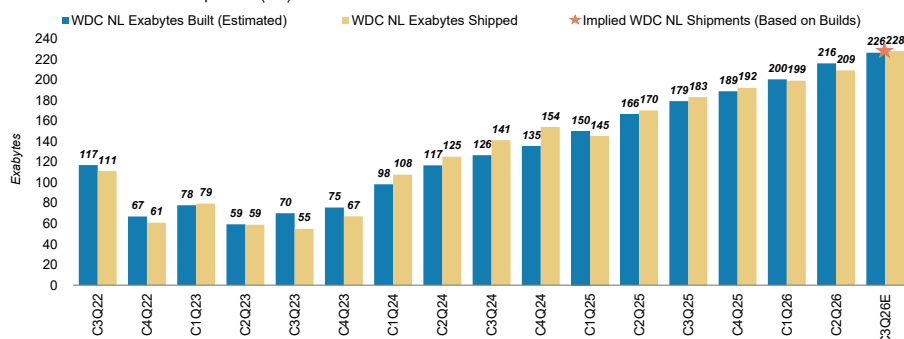
STX Nearline Builds vs. Shipments (EB)



Source: TSR, Morgan Stanley Research estimates

**Exhibit 111:** UPDATED: Our WDC nearline EB shipment tracker points to 228.5EB shipped in the September Q, largely in line with our estimate of 227.8EB

WDC Nearline Builds vs. Shipments (EB)



Source: TSR, Morgan Stanley Research estimates

## Quarterly TSR HDD Seminar

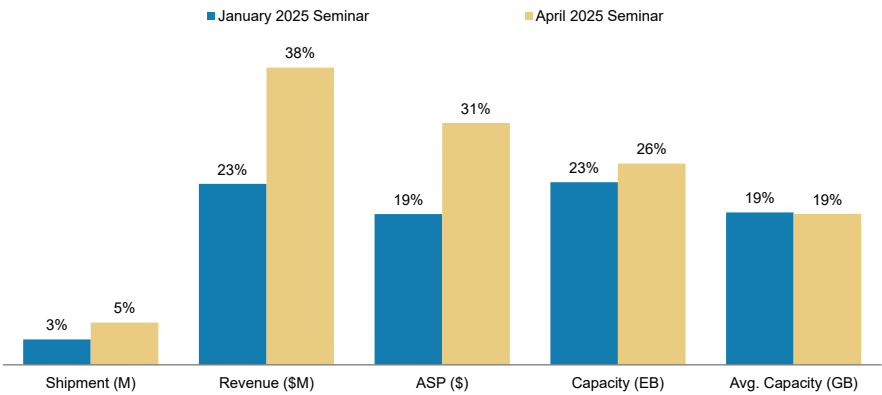
- Our team covering the HDD supply chain hosts a quarterly discussion with TSR on updated views of the HDD market. Below, we highlight key takeaways from the presentation.

### TSR Seminar Takeaways – July 22, 2026

At its **July seminar**, TSR raised its 2026 HDD revenue forecasts to **+38% Y/Y (from +23% prior)**. TSR forecasts 2026 HDD industry unit growth of 5% Y/Y (vs. 3% prior), revenue growth of 38% Y/Y (15 points higher than 3 months ago), ASP growth of 31% Y/Y (vs. 19% Y/Y prior), capacity growth of 26% Y/Y (3 points higher than 3 months ago), and average capacity per drive growth of 19% Y/Y (largely unchanged from 3 months ago). TSR forecasts that nearline (NL) HDD shipments will grow 13% Y/Y to 76.36M in 2026, which is now 2% above 2021 levels. While TSR expects a significant increase in capacity per drive and therefore substantial growth in both ASP and revenue, STX/WDC likely need increase external procurement of heads (TDK) and media (Resonac) – which have become bottlenecks – to expand nearline HDD production.

**Exhibit 112:** TSR now expects shipments to grow 5% Y/Y, and with ASP growing 31% Y/Y, HDD industry revenue will grow 38% Y/Y in CY26, following 26% Y/Y growth in CY25

CY26 HDD Industry Metrics Y/Y (TSR)



Source: TSR July 2026 Seminar, Morgan Stanley Research

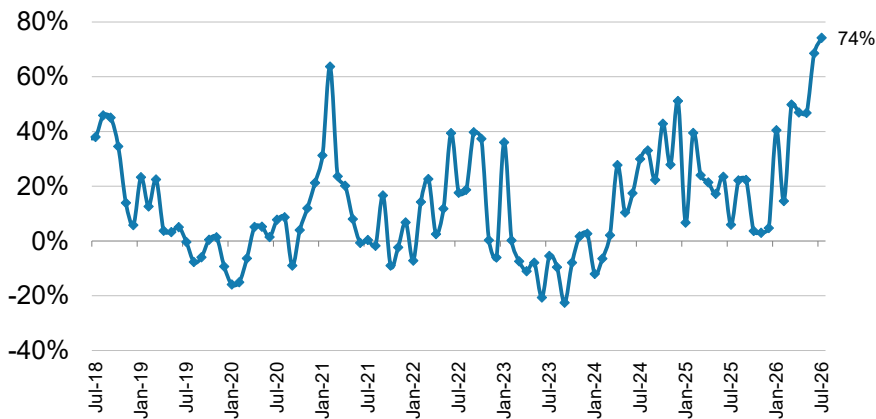
# Cloud and Enterprise Servers

## Monthly Taiwan ODM Revenue

- l We collect monthly sales data from Taiwan ODM vendors, including LandMark, Delta, Inventec, Nanya Tech, Aspeed, Hon Hai and Wiwynn as a proxy for cloud and enterprise server production. Monthly sales are typically released during the second week of the following month

**Exhibit 113:** UPDATED: As of July 31st, monthly sales from Taiwan ODM vendors grew 74% Y/Y

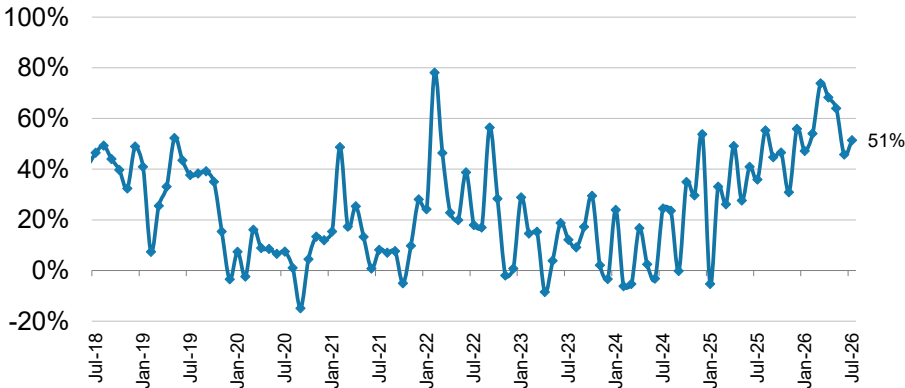
Taiwan ODM Monthly Revenue Y/Y



Source: Company data, Morgan Stanley Research

**Exhibit 114:** UPDATED: 2-year stacked revenue grew 51% Y/Y in July, a 5pt acceleration from June

2-Yr Stacked Taiwan ODM Monthly Revenue Y/Y



Source: Company data, Morgan Stanley Research

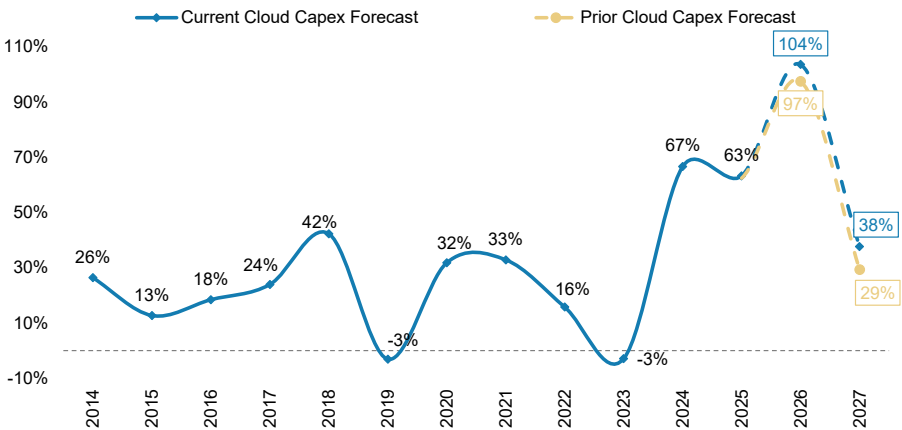


### Hyperscale Cloud Capex Tracker

- l We track capex spending from the top 14 hyperscalers by server installed base. These include Google, Amazon, Microsoft, Meta Platforms, CoreWeave, Nebius, IREN, Tesla, Oracle, Tencent, Baidu, Apple, and IBM. Dell Oro estimates that roughly ~75% of hyperscale capex goes towards data centers, which makes these companies a strong proxy for cloud data center equipment spending.

**Exhibit 115:** UPDATED: Our cloud capex tracker now points to 104% Y/Y growth in 2026 (up 7 points from a month ago), followed by 38% Y/Y growth in 2027 (up 9 points from a month ago)

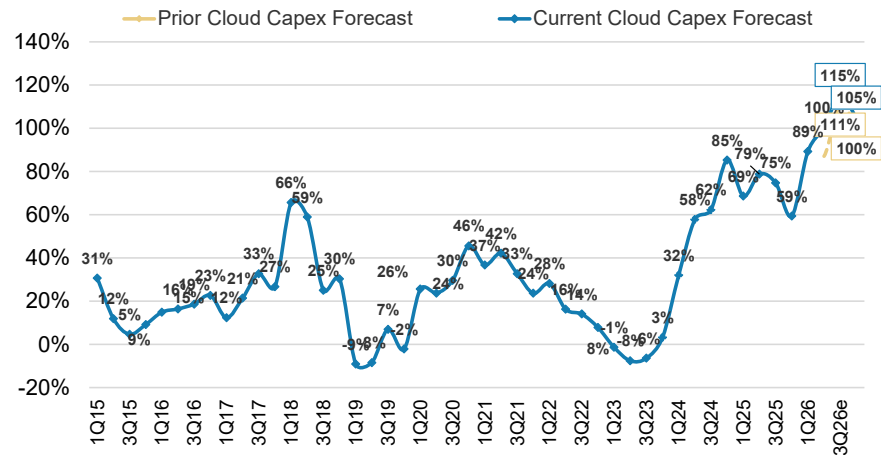
Top 15 Cloud Providers: Cloud Capex Y/Y Growth



Source: Company data, FactSet, Morgan Stanley Research, irs.gov. Note: Cloud capex includes capex from Amazon, Alphabet, Microsoft, Meta Platforms, SpaceX, CoreWeave, Nebius, IREN, Tesla, Tencent, Baidu, Alibaba, Apple, IBM, and Oracle.

**Exhibit 116:** UPDATED: Our cloud capex tracker now points to 115% Y/Y growth in C3Q26

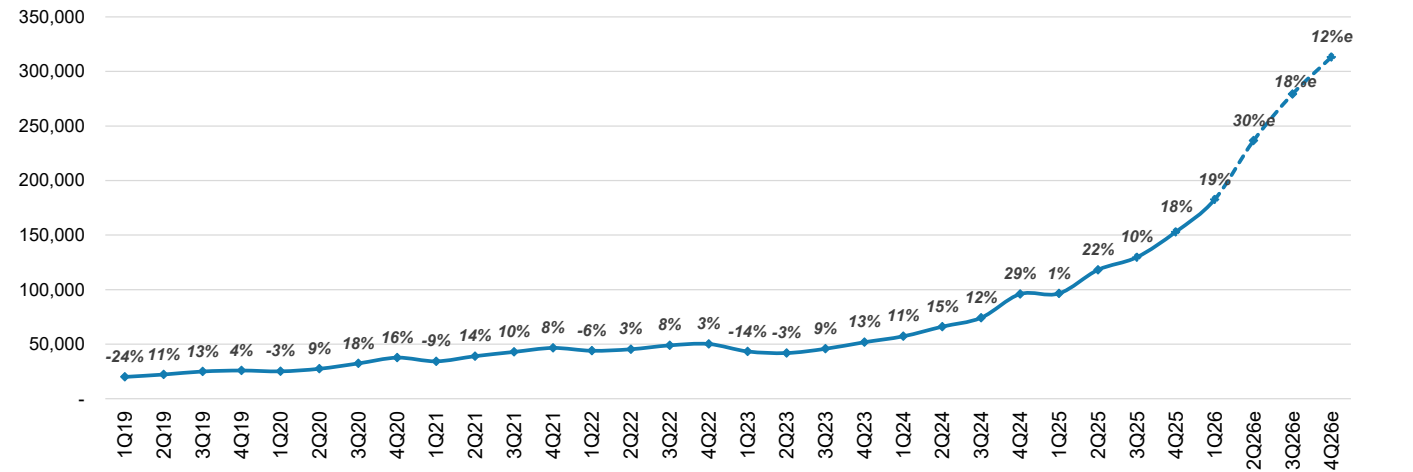
Quarterly Cloud Capex Growth Y/Y



Source: Company data, FactSet, Morgan Stanley Research. Note: Cloud capex includes capex from Amazon, Alphabet, Microsoft, Meta Platforms, SpaceX, CoreWeave, Nebius, IREN, Tesla, Tencent, Baidu, Alibaba, Apple, IBM, and Oracle.

**Exhibit 117:** UPDATED: While cloud capex spending patterns will ebb and flow, it's becoming clear that spending cycles are smoothing out from a sequential basis

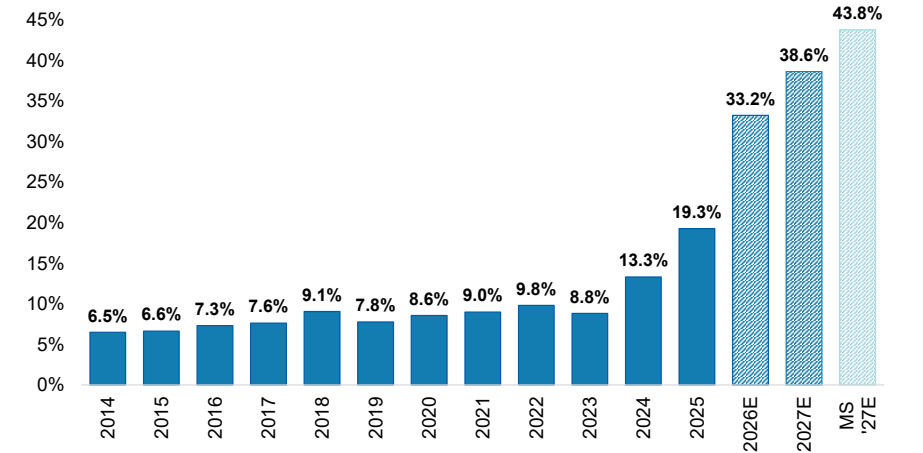
Top 15 Cloud Providers: Total Quarterly Capex (\$M) and Q/Q Growth



Source: Company data, FactSet, Morgan Stanley Research. Note: Cloud capex includes capex from Amazon, Alphabet, Microsoft, Meta Platforms, SpaceX, CoreWeave, Nebius, IREN, Tesla, Tencent, Baidu, Alibaba, Apple, IBM, and Oracle. Forward estimates include MSe for Tencent and Baidu, consensus est. for the others.

**Exhibit 118:** UPDATED: Our cloud capex tracker indicates that capital intensity will remain at a 10+ year high of 33.2% in 2026 (up 180bps from earlier this month), followed by 38.6% in 2027 (up 380bps from earlier this month)

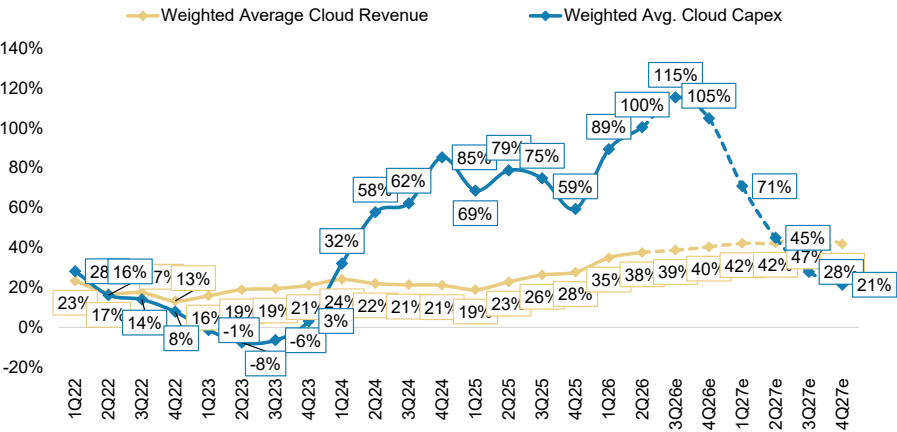
Top 15 Cloud Providers: Capital Intensity



Source: Company data, FactSet, Morgan Stanley Research. Note: Cloud capex includes capex from Alphabet, Amazon, Microsoft, Meta Platforms, SpaceX, CoreWeave, Nebius, IREN, Tesla, Tencent, Baidu, Alibaba, Apple, IBM, and Oracle.

**Exhibit 119:** UPDATED: Morgan Stanley estimates top 4 US hyperscaler cloud revenue growth to accelerate to 40-50% levels in 2027

Top 4 US Hyperscaler Cloud Revenue Y/Y vs. Cloud Capex Y/Y



Source: Company data, Morgan Stanley Research estimates. Note: Cloud capex includes capex from Alphabet, Amazon, Microsoft, Meta Platforms, SpaceX, CoreWeave, Nebius, IREN, Tesla, Tencent, Baidu, Alibaba, Apple, IBM, and Oracle; top 4 US hyperscalers include Alphabet, Meta Platforms, Microsoft, and Amazon.

## Public Cloud Use / Hyperscale Data Center Commentary

- l We collect anecdotes about public cloud use and commentary from cloud service providers and hyperscale data center vendors.

**Exhibit 120:** UPDATED: Commentary on public cloud use and hyperscaler data center capex commentary updated on a weekly basis

| Date      | Topic (Source)                | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8/11/2026 | NVDA (MSR)                    | More than just Hyperscale Capex: NVDA announcement indicates that AI infra requires 3P capital across the Capital structure, meaning ways for ails to raise, deploy and recycle capital.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8/5/2026  | DDOG, SNOW, MDB (MSR)         | DDOG SNOW MDB / Hyperscale Cloud revenue +48% y/y indicates key tailwinds for Infra SW: increasingly consumption-led growth, strengthening core business (buoyed by AI strength), and data & analytics demand.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7/30/2026 | ByteDance (FT)                | ByteDance is attempting a META-like transformation from a consumer advertising platform into a vertically integrated AI company spanning consumer applications, frontier models, video generation, enterprise cloud, data centres and custom chips.                                                                                                                                                                                                                                                                                                                                                                                                                |
| 7/28/2026 | META (PR)                     | META & BLK form a strategic venture to develop a 1GW data center campus in El Paso TX that will cost ~\$14B and bring capacity online starting in 2028. Including compute hardware, Bloomberg estimates a 1GW facility can cost ~\$35B-\$50B, highlighting the much larger all-in capital requirement.                                                                                                                                                                                                                                                                                                                                                             |
| 7/23/2026 | GOOGL (MSR)                   | GOOGL Cloud key positive as revenue grew 82% y/y with \$4.7B of sequential dollar growth, a 36% EBIT margin, and \$514B backlog.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7/21/2026 | ORCL (Bloomberg)              | AI and Credit / ORCL 5Y CDS hit highest level since late 2008: investor concerns on debt-funded AI and Cloud infra buildout are back in rotation as market fears that pressure on leverage and FCF will outweigh any future monetization.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7/17/2026 | Public Cloud (MSR)            | Share of workloads in colo or managed-hosting environments expected to remain stable through 2028 and public cloud gains are expected to displace on-prem infra.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7/16/2026 | Edge AI (MSR)                 | Latency costs matter the most, but partners consistently see edge AI as complementary of public cloud vs replacement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7/13/2026 | META, GOOGL, AMZN (MSR)       | META GOOGL AMZN / Brian Nowak raises Hyperscale Capex to ~\$1.2T in 2027 and ~\$1.4T in 2028 -- that's up ~10% vs prior in aggregate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7/8/2026  | META (Bloomberg)              | Meta is still hungry for even more computing power, the spokesperson said. It is still moving forward with plans for expensive new data centers and recently inked major computing deals with CoreWeave, Google and Oracle, among others.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7/7/2026  | NVDA (SA)                     | NVDA becoming the 'Central Bank of AI'? SemiAnalysis argues AI infrastructure financing is moving from hyperscaler cash-flow funding to a massive debt-financed market, with AI debt outstanding reaching ~\$7.1T by 2029 and cumulative AI capex reaching ~\$11.1T from 2024-2029                                                                                                                                                                                                                                                                                                                                                                                 |
| 7/6/2026  | META (MSR)                    | Brian Nowak remains bullish on META's neocloud optionality as a temporary way to monetize compute in a capacity-constrained AI world, with potential EPS upside of ~\$3+ / ~8% from leasing capacity.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6/22/2026 | GOOGLE (WSJ)                  | GOOGL also escalating from internal TPU user / cloud TPU provider into a more aggressive AI chip and AI infrastructure competitor to NVDA, using similar tactics such financial guarantees / financing, cloud partnerships, and direct chip sales.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 6/18/2026 | GOOGL, META, MSFT, AMZN (MSR) | AI Capex estimates also still too low. MS' bottom up build of cost per GW across 9 different chip generations and expected forward chip capacity per company helped inform MS' \$1 trillion of big 4 hyperscaler capex in '27 as the companies bring on ~20 total GW of capacity in '27. While '27 chip/rack numbers have less wiggle room, a substantial portion (in some cases over half) of capex is spent pre-building and starting data centers to come on in forward years. As such, Nowak is focused on '28/'29 chip capacity coming as this compute constrained environment may cause the hyperscalers to press and build/plan even more in forward years. |
| 6/9/2026  | GOOGL, META, MSFT, ORCL (MSR) | Incremental Capex revisions now accompanied by capital raises and/or innovative financing solutions as Hyperscale capital intensity >40% of sales. MS Analysts forecast capex-to-sales ratios of 36%, 44%, and 42% in 2026-28, well above the 32% dot-com fiber peak.                                                                                                                                                                                                                                                                                                                                                                                              |
| 6/3/2026  | CSCO (MSR)                    | Meta is OW on CSCO on opportunities for Cloud AI Capex cycle in conjunction with campus upgrade cycle. Unified silicon, networking, security, and software portfolio is a core vendor for customers looking to consolidate.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5/27/2026 | MSFT (MSR)                    | MSFT / MS' new Capex framework estimates Data Center footprint could scale from ~5GW in FY24 to ~20GW by FY28, and indicates materially higher monetization of incremental AI capacity vs market pricing GPU infra and Neoclouds. Significant upside opportunity in Azure growth (along with existing evidence of inflection in Copilot paid seats)                                                                                                                                                                                                                                                                                                                |

Source: Various news sources and Morgan Stanley Research

## Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at [www.morganstanley.com/eqr/disclosures/webapp/generalresearch](http://www.morganstanley.com/eqr/disclosures/webapp/generalresearch), or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

### Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Erik W Woodring.

### Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at [www.morganstanley.com/institutional/research/conflictolicies](http://www.morganstanley.com/institutional/research/conflictolicies). A Portuguese version of the policy can be found at [www.morganstanley.com.br](http://www.morganstanley.com.br)

### Important Regulatory Disclosures on Subject Companies

The analyst or strategist (or a household member) identified below owns the following securities (or related derivatives): Maya C Neuman - Apple, Inc.(common or preferred stock); Erik W Woodring - Alphabet Inc.(common or preferred stock), Amazon.com Inc(common or preferred stock), NVIDIA Corp.(common or preferred stock).

As of July 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: ACM Research Inc, Adobe Inc., Advanced Micro Devices, Advanced Micro-Fabrication Equipment Inc, Advanced Wireless Semiconductor Co, Aeva Technologies Inc, Airbnb Inc, Akamai Technologies, Inc., Alchip Technologies Ltd, AllRing Tech Co., Alphabet Inc., Amadeus IT Holdings S.A., Amazon.com Inc, Ambarella Inc, Amplitude Inc., Analog Devices Inc., AP Memory Technology Corp, Appian Corp, Apple, Inc., AppLovin Corp, Arm Holdings plc, Asana Inc, ASE Technology Holding Co. Ltd., ASMPT Ltd, Aspeed Technology, Astera Labs Inc, Atlassian Corporation PLC, Autodesk, BILL Holdings Inc, Blackline Inc, Booking Holdings Inc, Box Inc, Broadcom Inc., C3.ai, Cadence Design Systems Inc, Cambricon Technology Corporation, Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Cloudflare Inc, CoreWeave, Cricut Inc, Criteo SA, CrowdStrike Holdings Inc, Datadog, Inc., Dell Technologies Inc., Descartes Systems Group Inc, DocuSign Inc, DoorDash Inc, DoubleVerify Holdings Inc, Duolingo Inc, eBay Inc, Elastic NV, eMemory Technology Inc, Etsy Inc, Expedia Inc., Figma Inc, FIGS, Inc., Five9 Inc, Fortinet Inc., Freshworks Inc, Garmin Ltd, Gen Digital Inc, GigaDevice Semiconductor Beijing Inc, GitLab Inc, GlobalWafers Co Ltd, GoDaddy Inc, Grindr Inc., Gudeng Precision, Hangzhou Silan Microelectronics Co. Ltd., Hewlett Packard Enterprise, HP Inc., Hua-Hong Semiconductor Ltd, HubSpot, Inc., IBM, Instacart, Intel Corporation, Intuit, IonQ Inc, King Yuan Electronics Co Ltd, Klaviyo, Inc, Kornit Digital Ltd., Lightspeed Commerce Inc, Liveramp Holdings Inc, Logitech International SA, Lyft Inc, Macronix International Co Ltd, Manhattan Associates Inc., Marvell Technology Group Ltd, Match Group Inc, MediaTek, Meta Platforms Inc, Microchip Technology Inc., Micron Technology Inc., Microsoft, monday.com Ltd, MongoDB Inc, Montage Technology Co Ltd, NAURA Technology Group Co Ltd, Navitas Semiconductor Corp, Nebius Group NV, Nemetschek SE, NetApp Inc, NICE Ltd., Nuvoton Technology Corporation, NVIDIA Corp., NXP Semiconductor NV, Okta, Inc., ON Semiconductor Corp., Opendoor Technologies Inc, Oracle Corporation, PagerDuty, Inc., Palantir Technologies Inc., Palo Alto Networks Inc, Peloton Interactive, Inc., Phison Electronics Corp, Pinterest Inc, Powerchip Semiconductor Manufacturing Co, Qorvo Inc, Qualcomm Inc., Qualys Inc, Rapid7 Inc, Reddit Inc, Resideo Technologies Inc, RingCentral Inc, Roblox Corporation, Sage, Salesforce, Inc., Samsara Inc, SanDisk Corporation, SAP SE, Seagate Technology, Semtech Corp., SentinelOne, Inc., ServiceNow Inc, ServiceTitan Inc, SG Micro Corp., Shanghai Fudan Microelectronics, Shopify Inc, Silergy Corp., Silicon Laboratories Inc, Silicon Motion, Skyworks Solutions Inc, SmartRent, Inc., Snap Inc., Snowflake Inc., Sonos Inc., Sprinklr Inc, Sprout Social Inc, SPS Commerce Inc, StarPower Semiconductor Ltd, Synopsys Inc., Take-Two Interactive Software, TD Synnex Corporation, Temenos Group AG, Tenable Holdings Inc, Teradata, Texas Instruments, Toast, Inc., Trade Desk Inc, TSMC, Twilio Inc, Uber Technologies Inc, UiPath Inc, UMC, Unity Software Inc, Vanguard International Semiconductor, Varonis Systems, Inc., Vertex Inc., Western Digital, WIN Semiconductors Corp, Winbond Electronics Corp, WinWay Technology Co Ltd, Wix.Com Ltd, Wolfspeed, INC, Workday Inc, WPG Holdings, WT Microelectronics Co. Ltd., Yelp Inc, Zeta Global Holdings Corp, Zillow Group Inc, ZoomInfo Technologies Inc.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Advanced Micro Devices, Aeva Technologies Inc, Airbnb Inc, Akamai Technologies, Inc., Alchip Technologies Ltd, Alphabet Inc., Amazon.com Inc, Broadcom Inc., Capgemini, Cerebras Systems, Check Point Software Technologies Ltd., Cloudflare Inc, Compass, Inc., CoreWeave, Dell Technologies Inc., DigitalOcean Holdings Inc, eBay Inc, GlobalFoundries Inc, Iluvatar CoreX Semiconductor Co., Ltd., Ingram Micro, Intel Corporation, Intuit, Liff Off Mobile Inc., Meta Platforms Inc, Montage Technology Co Ltd, Navan Inc, Nebius Group NV, Netskope, Inc., NVIDIA Corp., ON Semiconductor Corp., Powerchip Semiconductor Manufacturing Co, Quantinuum, Salesforce, Inc., SAP SE, Semtech Corp., ServiceNow Inc, Uber Technologies Inc, Via Transportation Inc, Wix.Com Ltd.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Advanced Micro Devices, Aeva Technologies Inc, Airbnb Inc, Akamai Technologies, Inc., Allegro Microsystems Inc, Alphabet Inc., Amazon.com Inc, Amkor Technology Inc, Analog Devices Inc., ASMPT Ltd, Autodesk, Blackline Inc, Broadcom Inc., Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Cloudflare Inc, CoreWeave, Dassault Systemes SA, Dell Technologies Inc., DigitalOcean Holdings Inc, eBay Inc, Hexagon AB, HubSpot, Inc., IBM, Iluvatar CoreX Semiconductor Co., Ltd., Instacart, Intel Corporation, Intuit, IonQ Inc, Liff Off Mobile Inc., Meta Platforms Inc, Micron Technology Inc., Microsoft, Montage Technology Co Ltd, Navan Inc, Navitas Semiconductor Corp, Nebius Group NV, Netskope, Inc., NICE Ltd., NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Powerchip Semiconductor Manufacturing Co, Quantinuum, RingCentral Inc, Salesforce, Inc., SAP SE, Seagate Technology, Semtech Corp., ServiceNow Inc, Silicon Motion, Texas Instruments, Uber Technologies Inc, Via Transportation Inc, Wix.Com Ltd, Workday Inc, Zeta Global Holdings Corp.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Adobe Inc., Advanced Micro Devices, Advanced Micro-Fabrication Equipment Inc, Aeva Technologies Inc, Airbnb Inc, Akamai Technologies, Inc., Alchip Technologies Ltd, Allegro Microsystems Inc, Alphabet Inc., Amadeus IT Holdings S.A., Amazon.com Inc, Ambarella Inc, Amkor Technology Inc, Amplitude Inc., Analog Devices Inc., AP Memory Technology Corp, Appian Corp, Apple, Inc., AppLovin Corp, Arm Holdings plc, Asana Inc, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, Astera Labs Inc, Atlassian Corporation PLC, Autodesk, BILL Holdings Inc, Blackline Inc, Booking Holdings Inc, Box Inc, Broadcom Inc., Bumble Inc., C3.ai, Cadence Design Systems Inc, Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Cloudflare Inc, CoreWeave, Criteo SA, CrowdStrike Holdings Inc, Dassault Systemes SA, Datadog, Inc., Dell Technologies Inc., Descartes Systems Group Inc, DigitalOcean Holdings Inc, DocuSign Inc, DoorDash Inc, DoubleVerify Holdings Inc, Duolingo Inc, eBay Inc, Elastic NV, Espressif Systems, Etsy Inc, Everpure, Inc., Expedia Inc., Figma Inc, FIGS, Inc., Five9 Inc, Fortinet Inc,

Freshworks Inc, Garmin Ltd, Gen Digital Inc., GigaDevice Semiconductor Beijing Inc, GitLab Inc, GlobalFoundries Inc, GlobalWafers Co Ltd, GoDaddy Inc, GoPro Inc, Gudeng Precision, Hewlett Packard Enterprise, Hexagon AB, Himax Technologies Inc, HP Inc., Hua Hong Semiconductor Ltd, HubSpot, Inc., IBM, Iluvatar CoreX Semiconductor Co., Ltd., Ingram Micro, Innoscience, Instacart, Intel Corporation, Intuit, IONOS Group SE, IonQ Inc, JFrog Ltd, King Yuan Electronics Co Ltd, Klaviyo, Inc, Kornit Digital Ltd., Liftoff Mobile Inc., Lightspeed Commerce Inc., Liveramp Holdings Inc, Logitech International SA, Lyft Inc, Macronix International Co Ltd, Marvell Technology Group Ltd, Match Group Inc, MediaTek, Meta Platforms Inc, MetaX Integrated Circuits, Microchip Technology Inc., Micron Technology Inc., Microsoft, MNTN Inc, monday.com Ltd, MongoDB Inc, Montage Technology Co Ltd, Navan Inc, Navitas Semiconductor Corp, Nebius Group NV, Nemetschek SE, NetApp Inc, Netcompany Group A/S, Netskope, Inc., NICE Ltd., Novatek, Nutanix Inc, NVIDIA Corp., NXP Semiconductor NV, Okta, Inc., ON Semiconductor Corp., Opendoor Technologies Inc, Oracle Corporation, PagerDuty, Inc., Palantir Technologies Inc., Palo Alto Networks Inc, Peloton Interactive, Inc., Phison Electronics Corp, Pinterest Inc, Playtika Holding Corp, Powerchip Semiconductor Manufacturing Co, Qualcomm Inc., Qualys Inc, Quantinuum, Rapid7 Inc, Realtek Semiconductor, Reddit Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, Roblox Corporation, Sage, SailPoint Inc, Salesforce, Inc., Samsara Inc, SanDisk Corporation., SAP SE, Seagate Technology, Semtech Corp., SentinelOne, Inc., ServiceNow Inc, ServiceTitan Inc, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, Shopify Inc, SICC Co Ltd, Silergy Corp., Silicon Laboratories Inc., Silicon Motion, Skyworks Solutions Inc, SmartRent, Inc., Snap Inc., Snowflake Inc., Sonos Inc., Sprinklr Inc, Sprout Social Inc, SPS Commerce Inc, Synopsys Inc., Take-Two Interactive Software, Temenos Group AG, Tenable Holdings Inc, Teradata, Texas Instruments, Tieto Oyj, Toast, Inc., Trade Desk Inc, TSMC, Twilio Inc, Uber Technologies Inc, UiPath Inc, UMC, Unity Software Inc, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Varonis Systems, Inc., Vertex Inc., Via Transportation Inc, Webtoon Entertainment Inc, Western Digital, Winbond Electronics Corp, WinWay Technology Co Ltd, Wix.Com Ltd, Wolfspeed, INC, Workday Inc, WPG Holdings, WT Microelectronics Co. Ltd., Yelp Inc, Zeta Global Holdings Corp, Zillow Group Inc, Zoom Communications, Zscaler Inc.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Adobe Inc., Advanced Micro Devices, Airbnb Inc, Akamai Technologies, Inc., Allegro Microsystems Inc, Alphabet Inc., Amadeus IT Holdings S.A., Amazon.com Inc, Ambarella Inc, Amkor Technology Inc, Amplitude Inc., Analog Devices Inc., Apple, Inc., AppLovin Corp, Asana Inc, ASE Technology Holding Co. Ltd., Atlassian Corporation PLC, Autodesk, Blackline Inc, Booking Holdings Inc, Box Inc, Broadcom Inc., Cadence Design Systems Inc, Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Cloudflare Inc, CoreWeave, CrowdStrike Holdings Inc, Dassault Systemes SA, Dell Technologies Inc., DigitalOcean Holdings Inc, DocuSign Inc, DoorDash Inc, Dynatrace Inc, eBay Inc, Expedia Inc, Figma Inc, Five9 Inc, Freshworks Inc, Gen Digital Inc., GitLab Inc, GlobalFoundries Inc, GoPro Inc, HBX Group International PLC, Hewlett Packard Enterprise, Hexagon AB, HP Inc., HubSpot, Inc., IBM, Ingram Micro, Instacart, Intel Corporation, Intuit, JFrog Ltd., King Yuan Electronics Co Ltd, Klaviyo, Inc, Liftoff Mobile Inc., Logitech International SA, Marvell Technology Group Ltd, Match Group Inc, MediaTek, Meta Platforms Inc, Microchip Technology Inc, Micron Technology Inc., Microsoft, monday.com Ltd, MongoDB Inc, Nanya Technology Corp., NetApp Inc, NICE Ltd., Novatek, Nutanix Inc, Nuvoton Technology Corporation, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Oracle Corporation, Palantir Technologies Inc., Palo Alto Networks Inc, Peloton Interactive, Inc., Pinterest Inc, Playtika Holding Corp, Qorvo Inc, Qualcomm Inc., Realtek Semiconductor, Reddit Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, SailPoint Inc, Salesforce, Inc., SAP SE, Seagate Technology, Semtech Corp., SentinelOne, Inc., ServiceNow Inc, ServiceTitan Inc, Silicon Laboratories Inc., Silicon Motion, SMIC, Snap Inc., Snowflake Inc., Sonos Inc., Synopsys Inc., Take-Two Interactive Software, TD Synnex Corporation, Tenable Holdings Inc, Texas Instruments, Tieto Oyj, Toast, Inc., TSMC, Uber Technologies Inc, UiPath Inc, UMC, Unity Software Inc, Universal Scientific Ind. (Shanghai), Varonis Systems, Inc., Western Digital, Winbond Electronics Corp, Workday Inc, WT Microelectronics Co. Ltd., WW International Inc, Zeta Global Holdings Corp, Zillow Group Inc, Zoom Communications, Zscaler Inc.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Adobe Inc., Advanced Micro Devices, Advanced Micro-Fabrication Equipment Inc, Aeva Technologies Inc, Airbnb Inc, Akamai Technologies, Inc., Alchip Technologies Ltd, Allegro Microsystems Inc, Alphabet Inc., Amadeus IT Holdings S.A., Amazon.com Inc, Ambarella Inc, Amkor Technology Inc, Amplitude Inc., Analog Devices Inc., AP Memory Technology Corp, Appian Corp, Apple, Inc., AppLovin Corp, Arm Holdings plc, Asana Inc, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, Astera Labs Inc, Atlassian Corporation PLC, Autodesk, BILL Holdings Inc, Blackline Inc, Booking Holdings Inc, Box Inc, Broadcom Inc., Bumble Inc., C3.ai, Cadence Design Systems Inc, Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Cloudflare Inc, Compass, Inc., CoreWeave, Criteo SA, CrowdStrike Holdings Inc, Dassault Systemes SA, Datadog, Inc., Dell Technologies Inc., Descartes Systems Group Inc, DigitalOcean Holdings Inc, DocuSign Inc, DoorDash Inc, DoubleVerify Holdings Inc, Duolingo Inc, eBay Inc, Elastic NV, Espressif Systems, Etsy Inc, Everpure, Inc., Expedia Inc., Figma Inc, FIGS, Inc., Five9 Inc, Fortinet Inc., Freshworks Inc, Garmin Ltd, Gen Digital Inc., GigaDevice Semiconductor Beijing Inc, GitLab Inc, GlobalFoundries Inc, GlobalWafers Co Ltd, GoDaddy Inc, GoPro Inc, Gudeng Precision, Hewlett Packard Enterprise, Hexagon AB, Himax Technologies Inc, HP Inc., Hua Hong Semiconductor Ltd, HubSpot, Inc., IBM, Iluvatar CoreX Semiconductor Co., Ltd., Ingram Micro, Innoscience, Instacart, Intel Corporation, Intuit, IONOS Group SE, IonQ Inc, JFrog Ltd, King Yuan Electronics Co Ltd, Klaviyo, Inc, Kornit Digital Ltd., Liftoff Mobile Inc., Lightspeed Commerce Inc., Liveramp Holdings Inc, Logitech International SA, Lyft Inc, Macronix International Co Ltd, Marvell Technology Group Ltd, Match Group Inc, MediaTek, Meta Platforms Inc, MetaX Integrated Circuits, Microchip Technology Inc., Micron Technology Inc., Microsoft, MNTN Inc, monday.com Ltd, MongoDB Inc, Montage Technology Co Ltd, Navan Inc, Navitas Semiconductor Corp, Nebius Group NV, Nemetschek SE, NetApp Inc, Netcompany Group A/S, Netskope, Inc., NICE Ltd., Novatek, Nutanix Inc, NVIDIA Corp., NXP Semiconductor NV, Okta, Inc., ON Semiconductor Corp., Opendoor Technologies Inc, Oracle Corporation, PagerDuty, Inc., Palantir Technologies Inc., Palo Alto Networks Inc, Peloton Interactive, Inc., Phison Electronics Corp, Pinterest Inc, Playtika Holding Corp, Powerchip Semiconductor Manufacturing Co, Qualcomm Inc., Qualys Inc, Quantinuum, Rapid7 Inc, Realtek Semiconductor, Reddit Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, Roblox Corporation, Sage, SailPoint Inc, Salesforce, Inc., Samsara Inc, SanDisk Corporation., SAP SE, Seagate Technology, Semtech Corp., SentinelOne, Inc., ServiceNow Inc, ServiceTitan Inc, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, Shopify Inc, SICC Co Ltd, Silergy Corp., Silicon Laboratories Inc., Silicon Motion, Skyworks Solutions Inc, SmartRent, Inc., Snap Inc., Snowflake Inc., Sonos Inc., Sprinklr Inc, Sprout Social Inc, SPS Commerce Inc, Synopsys Inc., Take-Two Interactive Software, Temenos Group AG, Tenable Holdings Inc, Teradata, Texas Instruments, Tieto Oyj, Toast, Inc., Trade Desk Inc, TSMC, Twilio Inc, Uber Technologies Inc, UiPath Inc, UMC, Unity Software Inc, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Varonis Systems, Inc., Vertex Inc., Via Transportation Inc, Webtoon Entertainment Inc, Western Digital, Winbond Electronics Corp, WinWay Technology Co Ltd, Wix.Com Ltd, Wolfspeed, INC, Workday Inc, WPG Holdings, WT Microelectronics Co. Ltd., Yelp Inc, Zeta Global Holdings Corp, Zillow Group Inc, Zoom Communications, Zscaler Inc.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Adobe Inc., Advanced Micro Devices, Airbnb Inc, Akamai Technologies, Inc., Allegro Microsystems Inc, Alphabet Inc., Amadeus IT Holdings S.A., Amazon.com Inc, Ambarella Inc, Amkor Technology Inc, Amplitude Inc., Analog Devices Inc., Apple, Inc., AppLovin Corp, Asana Inc, ASE Technology Holding Co. Ltd., Atlassian Corporation PLC, Autodesk, Blackline Inc, Booking Holdings Inc, Box Inc, Broadcom Inc., Cadence Design Systems Inc, Capgemini, CCC Intelligent Solutions Holdings Inc, CDW Corporation, Cerebras Systems, Check Point Software Technologies Ltd., Chewy Inc, Cloudflare Inc, CoreWeave, CrowdStrike Holdings Inc, Dassault Systemes SA, Datadog, Inc., Dell Technologies Inc., DigitalOcean Holdings Inc, DocuSign Inc, DoorDash Inc, Dynatrace Inc, eBay Inc, Etsy Inc, Everpure, Inc., Expedia Inc., Figma Inc, Five9 Inc, Fortinet Inc., Freshworks Inc, Garmin Ltd, Gen Digital Inc., GitLab Inc, GlobalFoundries Inc, GoDaddy Inc, GoPro Inc, HBX Group International PLC, Hewlett Packard Enterprise, Hexagon AB, HP Inc., HubSpot, Inc., IBM, Iluvatar CoreX Semiconductor Co., Ltd., Ingram Micro, Instacart, Intel Corporation, Intuit, IonQ Inc, JFrog Ltd, King Yuan Electronics Co Ltd, Klaviyo, Inc, Liftoff Mobile Inc., Logitech International SA, Lyft Inc, Marvell Technology Group Ltd, Match Group Inc, MediaTek, Meta Platforms Inc, Microchip Technology Inc., Micron Technology Inc., Microsoft, monday.com Ltd, MongoDB Inc, Montage Technology Co Ltd, Nanya Technology Corp., Nebius Group NV, NetApp Inc, NICE Ltd., Novatek, Nutanix Inc, Nuvoton Technology Corporation, NVIDIA Corp., NXP Semiconductor NV, ON Semiconductor Corp., Opendoor Technologies Inc, Oracle Corporation, PagerDuty, Inc., Palantir Technologies Inc., Palo Alto Networks Inc, Peloton Interactive, Inc., Pinterest Inc, Playtika Holding Corp, Qorvo Inc, Qualcomm Inc.,

Qualys Inc, Realtek Semiconductor, Reddit Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, Sage, SailPoint Inc, Salesforce, Inc., SanDisk Corporation., SAP SE, Seagate Technology, Semtech Corp., SentinelOne, Inc., ServiceNow Inc, ServiceTitan Inc, Shopify Inc, Silicon Laboratories Inc., Silicon Motion, SMIC, Snap Inc., Snowflake Inc., Sonos Inc., Synopsys Inc., Take-Two Interactive Software, TD Synnex Corporation, Tenable Holdings Inc, Texas Instruments, Tieto Oyj, Toast, Inc., TSMC, Twilio Inc, Uber Technologies Inc, UiPath Inc, UMC, Unity Software Inc, Universal Scientific Ind. (Shanghai), Varonis Systems, Inc., Western Digital, Winbond Electronics Corp, Wix.Com Ltd, Wolfspeed, INC, Workday Inc, WT Microelectronics Co. Ltd., WW International Inc, Zeta Global Holdings Corp, Zillow Group Inc, Zoom Communications, Zscaler Inc.

An employee, director or consultant of Morgan Stanley is a director of eBay Inc, Elastic NV, HP Inc., Tenable Holdings Inc. This person is not a research analyst or a member of a research analyst's household.

Morgan Stanley & Co. LLC makes a market in the securities of Aeva Technologies Inc, Ambarella Inc, Amplitude Inc., Appian Corp, Asana Inc, Autodesk, Blackline Inc, Box Inc, C3.ai, CCC Intelligent Solutions Holdings Inc, Check Point Software Technologies Ltd., DoubleVerify Holdings Inc, Elastic NV, FIGS, Inc., Five9 Inc, Garmin Ltd, Grindr Inc., Liveramp Holdings Inc, Manhattan Associates Inc., PagerDuty, Inc., Playtika Holding Corp, Qualys Inc, Rapid7 Inc, Resideo Technologies Inc, Revolve Group Inc, RingCentral Inc, Sonos Inc., Sprinklr Inc, Sprout Social Inc, SPS Commerce Inc, TD Synnex Corporation, Twilio Inc, Varonis Systems, Inc., Vertex Inc., Webtoon Entertainment Inc, Yelp Inc, ZoomInfo Technologies Inc.

Morgan Stanley & Co. International plc is a corporate broker to Sage.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

## STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

## Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

| Stock Rating Category | Coverage Universe |            | Investment Banking Clients (IBC) |                |                      | Other Material Investment Services Clients (MISC) |                       |
|-----------------------|-------------------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
|                       | Count             | % of Total | Count                            | % of Total IBC | % of Rating Category | Count                                             | % of Total Other MISC |
| Overweight/Buy        | 1529              | 42%        | 442                              | 47%            | 29%                  | 750                                               | 43%                   |
| Equal-weight/Hold     | 1582              | 43%        | 396                              | 42%            | 25%                  | 773                                               | 44%                   |
| Not-Rated/Hold        | 3                 | 0%         | 1                                | 0%             | 33%                  | 1                                                 | 0%                    |
| Underweight/Sell      | 560               | 15%        | 97                               | 10%            | 17%                  | 217                                               | 12%                   |
| Total                 | 3,674             |            | 936                              |                |                      | 1741                                              |                       |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

## Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

## Analyst Industry Views



Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

## Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at [www.morganstanley.com/online/researchdisclosures](http://www.morganstanley.com/online/researchdisclosures).

For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

## Other Important Disclosures

A member of Research who had or could have had access to the research prior to completion owns securities (or related derivatives) in the Alphabet Inc., Amazon.com Inc, Analog Devices Inc., Apple, Inc., Atlassian Corporation PLC, Booking Holdings Inc, Broadcom Inc., Meta Platforms Inc, Microchip Technology Inc., Microsoft, NVIDIA Corp., NetApp Inc, Qualcomm Inc., Uber Technologies Inc, Workday Inc. This person is not a research analyst or a member of research analyst's household.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy ([http://www.morganstanley.com/privacy\\_pledge.html](http://www.morganstanley.com/privacy_pledge.html)), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy ([http://www.morganstanley.com/privacy\\_pledge.html](http://www.morganstanley.com/privacy_pledge.html)). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited ([https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms\\_and\\_conditions.pdf](https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf)) and the following link to review the audit report ([https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL\\_Morgan\\_Stanley\\_Research\\_Audit\\_Report.pdf](https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf)).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.



Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommended Regulation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of ASMP Ltd, GigaDevice Semiconductor Beijing Inc, Hua Hong Semiconductor Ltd, Montage Technology Co Ltd listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

### INDUSTRY COVERAGE: IT Hardware

| COMPANY (TICKER)                   | RATING (AS OF) | PRICE* (08/19/2026) |
|------------------------------------|----------------|---------------------|
| <b>Erik W Woodring</b>             |                |                     |
| Apple, Inc. (AAPL.O)               | O (05/26/2009) | \$316.83            |
| CDW Corporation (CDW.O)            | O (06/23/2026) | \$137.20            |
| Cricut Inc (CRCT.O)                | U (08/12/2021) | \$5.43              |
| Dell Technologies Inc. (DELL.N)    | E (06/01/2026) | \$437.55            |
| Everpure, Inc. (P.N)               | O (08/10/2026) | \$112.00            |
| Garmin Ltd (GRMN.N)                | E (02/18/2026) | \$295.67            |
| GoPro Inc (GPRO.O)                 | U (12/12/2023) | \$0.63              |
| Hewlett Packard Enterprise (HPE.N) | O (08/10/2026) | \$53.13             |
| HP Inc. (HPQ.N)                    | U (11/16/2025) | \$29.94             |
| IBM (IBM.N)                        | E (01/18/2023) | \$237.16            |
| Ingram Micro (INGM.N)              | E (06/11/2025) | \$27.06             |
| Kornit Digital Ltd. (KRNT.O)       | E (11/06/2025) | \$16.79             |
| Logitech International SA (LOGI.O) | U (01/20/2026) | \$100.47            |
| NetApp Inc (NTAP.O)                | E (08/10/2026) | \$194.48            |
| Resideo Technologies Inc (REZI.N)  | O (08/11/2025) | \$21.54             |
| Seagate Technology (STX.O)         | O (03/26/2024) | \$832.56            |
| SmartRent, Inc. (SMRT.N)           | ++             | \$1.38              |
| Sonos Inc. (SONO.O)                | E (11/06/2025) | \$16.28             |
| TD Synnex Corporation (SNX.N)      | O (06/11/2025) | \$251.78            |
| Teradata (TDC.N)                   | E (08/10/2026) | \$27.90             |
| Western Digital (WDC.O)            | O (04/16/2025) | \$462.09            |
| <b>Sanjit K Singh</b>              |                |                     |
| Nutanix Inc (NTNX.O)               | E (01/12/2026) | \$65.42             |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

### INDUSTRY COVERAGE: Internet

| COMPANY (TICKER)               | RATING (AS OF) | PRICE* (08/19/2026) |
|--------------------------------|----------------|---------------------|
| <b>Brian Nowak, CFA</b>        |                |                     |
| Airbnb Inc (ABNB.O)            | U (12/06/2022) | \$186.39            |
| Alphabet Inc. (GOOGL.O)        | O (08/11/2015) | \$344.72            |
| Amazon.com Inc (AMZN.O)        | O (04/24/2015) | \$265.84            |
| Booking Holdings Inc (BKNG.O)  | O (02/23/2026) | \$213.00            |
| DoorDash Inc (DASH.O)          | O (02/21/2024) | \$220.22            |
| Expedia Inc. (EXPE.O)          | E (01/09/2019) | \$326.27            |
| Instacart (CART.O)             | E (01/29/2024) | \$50.67             |
| Lyft Inc (LYFT.O)              | E (10/24/2019) | \$17.43             |
| Meta Platforms Inc (META.O)    | O (03/20/2023) | \$546.03            |
| Pinterest Inc (PINS.N)         | O (07/20/2025) | \$23.39             |
| Reddit Inc (RDDT.N)            | O (12/08/2024) | \$151.71            |
| Snap Inc. (SNAP.N)             | E (07/22/2024) | \$5.24              |
| Uber Technologies Inc (UBER.N) | ++             | \$78.04             |
| <b>Matthew Cost</b>            |                |                     |
| AppLovin Corp (APP.O)          | O (04/10/2025) | \$310.79            |
| Compass, Inc. (COMP.N)         | E (01/12/2026) | \$12.80             |

|                                        |                |          |
|----------------------------------------|----------------|----------|
| Criteo SA (CRO.O)                      | E (01/26/2016) | \$17.85  |
| DoubleVerify Holdings Inc (DV.N)       | E (06/25/2024) | \$13.29  |
| Liftoff Mobile Inc. (LFTO.O)           | E (06/29/2026) | \$19.51  |
| MNTN Inc (MNTN.N)                      | E (06/16/2025) | \$12.82  |
| Opendoor Technologies Inc (OPEN.O)     | E (07/24/2023) | \$3.58   |
| Playtika Holding Corp (PLTK.O)         | E (11/27/2022) | \$2.35   |
| Roblox Corporation (RBLX.N)            | O (11/04/2024) | \$38.69  |
| Take-Two Interactive Software (TTWO.O) | O (02/01/2018) | \$237.04 |
| Trade Desk Inc (TTD.O)                 | E (09/10/2025) | \$13.54  |
| Unity Software Inc (U.N)               | O (09/02/2024) | \$47.11  |
| Webtoon Entertainment Inc (WBTN.O)     | E (07/22/2024) | \$8.54   |
| Yelp Inc (YELP.N)                      | U (01/10/2019) | \$23.95  |
| Zillow Group Inc (Z.O)                 | E (04/18/2018) | \$36.58  |
| <b>Nathan Feather</b>                  |                |          |
| Bumble Inc. (BMBL.O)                   | E (03/08/2021) | \$2.79   |
| Chewy Inc (CHWY.N)                     | O (10/31/2023) | \$23.99  |
| Duolingo Inc (DUOL.O)                  | E (02/27/2026) | \$146.03 |
| eBay Inc (EBAY.O)                      | O (04/18/2024) | \$102.64 |
| Etsy Inc (ETSY.N)                      | E (07/20/2025) | \$80.46  |
| FIGS, Inc. (FIGS.N)                    | E (02/29/2024) | \$14.49  |
| Grindr Inc. (GRND.N)                   | O (07/01/2026) | \$15.59  |
| Match Group Inc (MTCH.O)               | E (04/18/2024) | \$39.70  |
| Peloton Interactive, Inc. (PTON.O)     | E (03/14/2022) | \$5.47   |
| Revolve Group Inc (RVLV.N)             | E (10/20/2024) | \$23.86  |
| WW International Inc (WW.O)            | E (08/01/2025) | \$14.36  |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Software

| COMPANY (TICKER)                               | RATING (AS OF) | PRICE* (08/19/2026) |
|------------------------------------------------|----------------|---------------------|
| <b>Adam Wood</b>                               |                |                     |
| Adobe Inc. (ADBE.O)                            | U (07/21/2026) | \$272.47            |
| Atlassian Corporation PLC (TEAM.O)             | O (01/13/2020) | \$174.23            |
| Intuit (INTU.O)                                | E (07/21/2026) | \$362.47            |
| Microsoft (MSFT.O)                             | O (07/21/2026) | \$484.31            |
| Salesforce, Inc. (CRM.N)                       | E (07/21/2026) | \$206.09            |
| Samsara Inc (IOT.N)                            | E (03/23/2023) | \$39.20             |
| ServiceNow Inc (NOW.N)                         | O (09/24/2025) | \$127.20            |
| Shopify Inc (SHOP.O)                           | O (04/19/2024) | \$146.58            |
| Workday Inc (WDAY.O)                           | U (07/21/2026) | \$198.42            |
| <b>Chris Quintero</b>                          |                |                     |
| BILL Holdings Inc (BILL.N)                     | E (06/10/2025) | \$47.71             |
| Blackline Inc (BL.O)                           | E (07/21/2026) | \$31.43             |
| Box Inc (BOX.N)                                | E (05/21/2024) | \$33.12             |
| CCC Intelligent Solutions Holdings Inc (CCC.O) | O (11/13/2024) | \$7.44              |
| Descartes Systems Group Inc (DSGX.O)           | O (01/15/2026) | \$77.58             |
| DocuSign Inc (DOCU.O)                          | E (01/16/2024) | \$62.11             |
| Manhattan Associates Inc. (MANH.O)             | E (10/21/2025) | \$205.99            |
| Navan Inc (NAVN.O)                             | O (11/24/2025) | \$29.18             |
| ServiceTitan Inc (TTAN.O)                      | O (01/20/2026) | \$95.02             |
| SPS Commerce Inc (SPSC.O)                      | U (07/21/2026) | \$80.76             |
| Vertex Inc. (VERX.O)                           | E (07/21/2026) | \$13.52             |
| Via Transportation Inc (VIA.N)                 | O (01/20/2026) | \$26.38             |

Elizabeth Porter, CFA

|                                                 |                |          |
|-------------------------------------------------|----------------|----------|
| Amplitude Inc. (AMPL.O)                         | O (01/15/2026) | \$12.83  |
| Asana Inc (ASAN.N)                              | U (05/20/2025) | \$9.57   |
| Autodesk (ADSK.O)                               | O (08/23/2024) | \$251.29 |
| Figma Inc (FIG.N)                               | E (08/25/2025) | \$26.79  |
| Five9 Inc (FIVN.O)                              | E (10/10/2022) | \$32.74  |
| Freshworks Inc (FRSH.O)                         | E (10/18/2021) | \$13.07  |
| GoDaddy Inc (GDDY.N)                            | E (07/19/2021) | \$96.84  |
| HubSpot, Inc. (HUBS.N)                          | O (03/21/2023) | \$241.65 |
| Klaviyo, Inc (KVYO.N)                           | O (04/29/2026) | \$18.20  |
| Liveramp Holdings Inc (RAMP.N)                  | E (01/13/2025) | \$37.71  |
| monday.com Ltd (MNDY.O)                         | O (08/12/2025) | \$91.01  |
| NICE Ltd. (NICE.O)                              | E (07/21/2026) | \$100.93 |
| RingCentral Inc (RNG.N)                         | E (08/08/2023) | \$65.94  |
| Sprinklr Inc (CXM.N)                            | E (07/19/2021) | \$7.13   |
| Sprout Social Inc (SPT.O)                       | E (11/17/2020) | \$10.03  |
| Toast, Inc. (TOST.N)                            | O (12/16/2021) | \$35.16  |
| Twilio Inc (TWLO.N)                             | O (02/24/2025) | \$221.87 |
| Wix.Com Ltd (WIX.O)                             | E (07/21/2026) | \$77.89  |
| Zeta Global Holdings Corp (ZETA.N)              | E (08/01/2024) | \$28.79  |
| Zoom Communications (ZM.O)                      | E (10/11/2022) | \$107.47 |
| ZoomInfo Technologies Inc (GTM.O)               | E (02/01/2024) | \$3.82   |
| <b>Meryl R Thomas, CFA</b>                      |                |          |
| Lightspeed Commerce Inc. (LSPD.N)               | E (02/18/2021) | \$10.41  |
| <b>Meta A Marshall</b>                          |                |          |
| Check Point Software Technologies Ltd. (CHKP.O) | E (10/16/2023) | \$132.35 |
| CrowdStrike Holdings Inc (CRWD.O)               | O (03/10/2026) | \$201.63 |
| Fortinet Inc. (FTNT.O)                          | E (07/21/2026) | \$152.85 |
| Gen Digital Inc. (GEN.O)                        | E (06/07/2024) | \$27.54  |
| Netskope, Inc. (NTSK.O)                         | O (10/13/2025) | \$14.70  |
| Okta, Inc. (OKTA.O)                             | O (12/02/2024) | \$141.21 |
| Palo Alto Networks Inc (PANW.O)                 | O (10/10/2017) | \$359.76 |
| Qualys Inc (QLYS.O)                             | U (02/09/2021) | \$186.68 |
| Rapid7 Inc (RPD.O)                              | U (07/21/2026) | \$12.21  |
| SailPoint Inc (SAIL.O)                          | O (09/02/2025) | \$19.19  |
| SentinelOne, Inc. (S.N)                         | E (12/02/2024) | \$21.32  |
| Tenable Holdings Inc (TENB.O)                   | E (12/02/2024) | \$35.28  |
| Varonis Systems, Inc. (VRNS.O)                  | E (01/26/2026) | \$42.28  |
| Zscaler Inc (ZS.O)                              | E (04/22/2026) | \$184.60 |
| <b>Ryan Lountzis</b>                            |                |          |
| CoreWeave (CRWV.O)                              | E (04/22/2025) | \$90.87  |
| Nebius Group NV (NBIS.O)                        | E (01/15/2026) | \$223.90 |
| <b>Sanjit K Singh</b>                           |                |          |
| Akamai Technologies, Inc. (AKAM.O)              | O (01/12/2026) | \$112.84 |
| Appian Corp (APPN.O)                            | E (04/30/2026) | \$37.53  |
| C3.ai (AI.N)                                    | U (01/04/2021) | \$10.38  |
| Cloudflare Inc (NET.N)                          | O (12/02/2024) | \$291.88 |
| Datadog, Inc. (DDOG.O)                          | O (01/12/2026) | \$233.52 |
| DigitalOcean Holdings Inc (DOCN.N)              | O (01/16/2025) | \$116.66 |
| Dynatrace Inc (DT.N)                            | E (02/13/2024) | \$49.60  |
| Elastic NV (ESTC.N)                             | E (07/21/2026) | \$87.34  |
| GitLab Inc (GTLB.O)                             | E (01/12/2026) | \$42.60  |
| JFrog Ltd. (FROG.O)                             | E (07/21/2026) | \$91.21  |
| MongoDB Inc (MDB.O)                             | O (04/12/2023) | \$440.53 |
| Oracle Corporation (ORCL.N)                     | E (01/15/2019) | \$143.81 |

|                                     |                |          |
|-------------------------------------|----------------|----------|
| PagerDuty, Inc. (PD.N)              | U (07/21/2026) | \$12.17  |
| Palantir Technologies Inc. (PLTR.O) | E (02/04/2025) | \$175.19 |
| Snowflake Inc. (SNOW.N)             | O (06/24/2025) | \$325.01 |
| UiPath Inc (PATH.N)                 | E (09/07/2022) | \$15.78  |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Semiconductors

| COMPANY (TICKER)                      | RATING (AS OF)  | PRICE* (08/19/2026) |
|---------------------------------------|-----------------|---------------------|
| <b>Joseph Moore</b>                   |                 |                     |
| Advanced Micro Devices (AMD.O)        | E (06/09/2024)  | \$466.42            |
| Aeva Technologies Inc (AEVA.O)        | E (07/19/2021)  | \$19.78             |
| Allegro Microsystems Inc (ALGM.O)     | O (02/13/2026)  | \$38.35             |
| Ambarella Inc (AMBA.O)                | O (03/29/2016)  | \$74.59             |
| Amkor Technology Inc (AMKR.O)         | E (11/08/2023)  | \$50.89             |
| Analog Devices Inc. (ADI.O)           | O (11/16/2023)  | \$373.26            |
| Astera Labs Inc (ALAB.O)              | O (05/11/2025)  | \$289.02            |
| Broadcom Inc. (AVGO.O)                | O (06/09/2024)  | \$362.48            |
| Cerebras Systems (CBRS.O)             | O (06/08/2026)  | \$215.69            |
| GlobalFoundries Inc (GFS.O)           | E (10/28/2024)  | \$48.58             |
| Intel Corporation (INTC.O)            | E (02/22/2023)  | \$92.80             |
| IonQ Inc (IONQ.N)                     | E (04/25/2023)  | \$43.36             |
| Marvell Technology Group Ltd (MRVL.O) | E (09/14/2015)  | \$237.27            |
| Microchip Technology Inc. (MCHP.O)    | E (07/10/2024)  | \$77.08             |
| Micron Technology Inc. (MU.O)         | O (10/06/2025)  | \$937.11            |
| Navitas Semiconductor Corp (NVT.S.O)  | U (04/06/2025)  | \$12.83             |
| NVIDIA Corp. (NVDA.O)                 | O (03/16/2023)  | \$217.56            |
| NXP Semiconductor NV (NXPI.O)         | O (02/11/2025)  | \$226.02            |
| ON Semiconductor Corp. (ON.O)         | ++              | \$76.58             |
| Qorvo Inc (QRVO.O)                    | E (10/28/2025)  | \$96.81             |
| Qualcomm Inc. (QCOM.O)                | E (06/24/2026)  | \$161.91            |
| Quantinuum (QNT.O)                    | E (06/29/2026)  | \$56.00             |
| SanDisk Corporation. (SNDK.O)         | O (03/03/2025)  | \$1,568.87          |
| Semtech Corp. (SMTC.O)                | E (04/06/2025)  | \$125.36            |
| Silicon Laboratories Inc. (SLAB.O)    | E (01/19/2021)  | \$218.92            |
| Skyworks Solutions Inc (SWKS.O)       | E (11/28/2018)  | \$68.48             |
| Texas Instruments (TXN.O)             | U (04/13/2020)  | \$267.45            |
| Wolfspeed, INC (WOLF.N)               | NR (04/06/2025) | \$29.09             |
| <b>Lee Simpson</b>                    |                 |                     |
| Arm Holdings plc (ARM.O)              | E (04/07/2026)  | \$249.34            |
| Cadence Design Systems Inc (CDNS.O)   | O (02/14/2024)  | \$314.98            |
| Synopsys Inc. (SNPS.O)                | E (02/27/2026)  | \$401.24            |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Technology - Software & Services

| COMPANY (TICKER)                  | RATING (AS OF) | PRICE* (08/19/2026) |
|-----------------------------------|----------------|---------------------|
| <b>Adam Wood</b>                  |                |                     |
| SAP SE (SAP.N)                    | O (02/20/2026) | \$215.60            |
| SAP SE (SAPG.DE)                  | O (03/20/2020) | €185.90             |
| <b>George W Webb</b>              |                |                     |
| Amadeus IT Holdings S.A. (AMA.MC) | O (04/10/2026) | €56.82              |
| Capgemini (CAP.PA)                | E (02/19/2026) | €110.35             |

|                                      |                |            |
|--------------------------------------|----------------|------------|
| Dassault Systemes SA (DAST.PA)       | E (03/17/2026) | €22.24     |
| Hexagon AB (HEXAb.ST)                |                | SKr 99.00  |
| IONOS Group SE (IOSn.DE)             | O (08/07/2026) | €32.54     |
| Nemetschek SE (NEKG.DE)              | E (07/13/2022) | €62.45     |
| Netcompany Group A/S (NETCG.CO)      | U (01/12/2026) | DKr 329.40 |
| Sage (SGE.L)                         | O (12/08/2021) | 1,073p     |
| <b>Mark Hyatt</b>                    |                |            |
| HBX Group International PLC (HBX.MC) | E (04/17/2026) | €7.78      |
| Temenos Group AG (TEMN.S)            | U (12/15/2017) | SFr 73.10  |
| Tieto Oyj (Tieto.HE)                 | U (01/12/2026) | €19.18     |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.

## INDUSTRY COVERAGE: Greater China Technology Semiconductors

| COMPANY (TICKER)                                     | RATING (AS OF) | PRICE* (08/19/2026) |
|------------------------------------------------------|----------------|---------------------|
| <b>Charlie Chan</b>                                  |                |                     |
| ACM Research Inc (ACMR.O)                            | O (03/07/2023) | \$77.59             |
| Advanced Micro-Fabrication Equipment Inc (688012.SS) | O (11/06/2023) | Rmb372.00           |
| Advanced Wireless Semiconductor Co (8086.TWO)        | U (07/14/2025) | NT\$115.50          |
| Alchip Technologies Ltd (3661.TW)                    | O (05/14/2021) | NT\$3,630.00        |
| ASE Technology Holding Co. Ltd. (3711.TW)            | O (09/15/2024) | NT\$588.00          |
| Cambricon Technology Corporation (688256.SS)         | O (04/27/2026) | Rmb1,050.49         |
| Global Unichip Corp (3443.TW)                        | O (06/24/2026) | NT\$5,295.00        |
| GlobalWafers Co Ltd (6488.TWO)                       | E (05/19/2026) | NT\$983.00          |
| Gudeng Precision (3680.TWO)                          | O (11/25/2025) | NT\$461.00          |
| Hua Hong Semiconductor Ltd (1347.HK)                 | E (03/12/2026) | HK\$113.70          |
| Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)     | O (04/27/2026) | HK\$368.00          |
| King Yuan Electronics Co Ltd (2449.TW)               | O (03/03/2023) | NT\$235.00          |
| Maxscend Microelectronics Co Ltd (300782.SZ)         | U (01/11/2021) | Rmb76.14            |
| MediaTek (2454.TW)                                   | O (11/28/2025) | NT\$3,845.00        |
| MetaX Integrated Circuits (688802.SS)                | E (04/27/2026) | Rmb729.00           |
| Nanya Technology Corp. (2408.TW)                     | O (05/28/2026) | NT\$481.00          |
| NAURA Technology Group Co Ltd (002371.SZ)            | O (11/06/2023) | Rmb712.95           |
| OmniVision Integrated Circuits Group Inc (603501.SS) | E (11/17/2025) | Rmb86.90            |
| Phison Electronics Corp (8299.TWO)                   | E (02/25/2026) | NT\$1,965.00        |
| SG Micro Corp. (300661.SZ)                           | E (11/03/2025) | Rmb106.49           |
| Silergy Corp. (6415.TW)                              | U (05/19/2026) | NT\$440.00          |
| SMIC (0981.HK)                                       | O (10/21/2025) | HK\$73.00           |
| TSMC (2330.TW)                                       | O (02/07/2022) | NT\$2,350.00        |
| UMC (2303.TW)                                        | O (05/19/2026) | NT\$115.50          |
| Vanguard International Semiconductor (5347.TWO)      | E (01/14/2026) | NT\$151.50          |
| WIN Semiconductors Corp (3105.TWO)                   | U (07/14/2025) | NT\$363.50          |
| <b>Daisy Dai, CFA</b>                                |                |                     |
| ASMPT Ltd (0522.HK)                                  | O (07/24/2025) | HK\$158.50          |
| China Resources Microelectronics Limited (688396.SS) | U (03/02/2026) | Rmb61.59            |
| Empyrean Technology Co Ltd (301269.SZ)               | E (01/17/2025) | Rmb97.01            |
| Hangzhou Silan Microelectronics Co. Ltd. (600460.SS) | U (08/25/2025) | Rmb34.49            |
| Hygon Information Technology Co., Ltd. (688041.SS)   | O (07/03/2026) | Rmb259.74           |
| Innoscence (2577.HK)                                 | E (10/13/2025) | HK\$43.78           |
| JCET Group Co Ltd (600584.SS)                        | E (01/16/2026) | Rmb77.74            |
| Shanghai Fudan Microelectronics (1385.HK)            | O (03/07/2025) | HK\$25.38           |
| SICC Co Ltd (688234.SS)                              | O (03/20/2026) | Rmb112.63           |
| StarPower Semiconductor Ltd (603290.SS)              | E (05/14/2026) | Rmb100.31           |
| Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)  | U (01/10/2023) | Rmb62.27            |

|                                                    |                |               |
|----------------------------------------------------|----------------|---------------|
| Universal Scientific Ind. (Shanghai) (601231.SS)   | O (11/05/2025) | Rmb26.87      |
| Yangjie Technology (300373.SZ)                     | O (06/10/2022) | Rmb101.49     |
| <b>Daniel Yen, CFA</b>                             |                |               |
| AP Memory Technology Corp (6531.TW)                | O (07/11/2025) | NT\$837.00    |
| ASMedia Technology Inc (5269.TW)                   | U (10/03/2025) | NT\$1,460.00  |
| Aspeed Technology (5274.TWO)                       | O (06/09/2025) | NT\$15,830.00 |
| Egis Technology Inc (6462.TWO)                     | E (01/28/2026) | NT\$97.00     |
| Elan Microelectronics Corp (2458.TW)               | O (10/03/2025) | NT\$145.50    |
| eMemory Technology Inc (3529.TWO)                  | O (08/18/2026) | NT\$2,260.00  |
| Espressif Systems (688018.SS)                      | O (05/15/2023) | Rmb110.83     |
| GigaDevice Semiconductor Beijing Inc (603986.SS)   | O (05/15/2025) | Rmb403.71     |
| Macronix International Co Ltd (2337.TW)            | O (09/18/2025) | NT\$119.00    |
| Montage Technology Co Ltd (6809.HK)                | O (03/18/2026) | HK\$270.40    |
| Montage Technology Co Ltd (688008.SS)              | O (03/18/2026) | Rmb200.00     |
| Novatek (3034.TW)                                  | U (02/04/2026) | NT\$534.00    |
| Nuvoton Technology Corporation (4919.TW)           | U (11/10/2025) | NT\$116.50    |
| Parade Technologies Ltd (4966.TWO)                 | O (05/27/2026) | NT\$582.00    |
| Powerchip Semiconductor Manufacturing Co (6770.TW) | O (10/27/2025) | NT\$66.60     |
| Realtek Semiconductor (2379.TW)                    | E (01/30/2026) | NT\$738.00    |
| Shenzhen Goodix Technology Co Ltd (603160.SS)      | U (07/14/2025) | Rmb49.13      |
| Winbond Electronics Corp (2344.TW)                 | O (05/28/2026) | NT\$168.00    |
| WPG Holdings (3702.TW)                             | O (03/16/2026) | NT\$106.50    |
| WT Microelectronics Co. Ltd. (3036.TW)             | O (01/27/2026) | NT\$201.50    |
| <b>Duan Liu</b>                                    |                |               |
| Dosilicon Co Ltd (688110.SS)                       | U (09/06/2024) | Rmb114.51     |
| Shenzhen Longsys Electronics Co Ltd (301308.SZ)    | E (02/25/2026) | Rmb378.98     |
| <b>Tiffany Yeh</b>                                 |                |               |
| AllRing Tech Co. (6187.TWO)                        | O (09/23/2025) | NT\$1,305.00  |
| FOCI Fiber Optic Communications Inc (3363.TWO)     | O (01/15/2025) | NT\$584.00    |
| Himax Technologies Inc (HIMX.O)                    | E (02/04/2026) | \$13.90       |
| Hon Precision (7769.TW)                            | O (04/17/2026) | NT\$6,445.00  |
| MPI Corporation (6223.TWO)                         | O (04/17/2026) | NT\$5,255.00  |
| Silicon Motion (SIMO.O)                            | O (05/06/2024) | \$242.80      |
| WinWay Technology Co Ltd (6515.TW)                 | O (04/17/2026) | NT\$6,370.00  |

Stock Ratings are subject to change. Please see latest research for each company.

\* Historical prices are not split adjusted.